Logic

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Chapter 1

Atomes

1.1 Basic tools

we start the book by building logic step by step.block by block in first it might seem a little annoying but I advice you to be patient.book will show its power soon. like euclidian geometry this book has undefined concepts. I tried to explain them for you to understand.but they don't have any definition.

Undefined 1.1.1. ***Urelements*** *are the most fundamental entities in a logical. They cannot be constructed, defined, or described in terms of other things. Their existence is primitive—accepted without internal structure or relational properties. Urelements simply are.*

In this book, we will denote individual urelements using lowercase letters u, v, w,...

Undefined 1.1.2. ***Type*** *parallel lines used to show that an object exists (you will undrestnd what is object soon.) (we will show you how to use them).*

1.	u		
2.		v	
3.			w

Rule 1.1.1. ***fusion*** *it's the Rule that explain how sets made. (this rule is schema.it means it works everywhere for every object.)*

$m.$	u_0		
	$\vdots$		
$m+n.$	u_n		
$o.$		$\{u_0, u_1 ... u_n\}$	

Undefined 1.1.3. ***Set*** *we can't say what exactly sets are but we know sets can be made by fusion (if "a" is a set we write "set(a)" from now we denote sets with uppercase letters like A, B, C, ...).*
definition 1.1.1. ***Object*** *urelements and sets.(we write obj instead of object). (from now we denote objects with lowercase letters like a, b, c, ...).*
Rule 1.1.2. ***fission*** *it show us how to break an obj. (this rule is scheme it means it works everywhere for every obj.)*

$o.$		$\{a_0, a_1 ... a_n\}$	
$m.$	a_0		
	$\vdots$		
$m+n.$	a_n		

Example 1.1.1. $\{a, b, c\} / \therefore \{a, b\}$ *(prove if* $\{a, b, c\}$ *exists then* $\{a, c\}$ *exists)*

1.		$\{a, b, c\}$	
2.	a		*fi1*
3.	b		*fi1*
4.	c		*fi1*
5.		$\{a, c\}$	*fu2,3,4*

Example 1.1.2. $\{a, b\} / \therefore \{b, a\}$

1.		$\{a, b\}$	
2.	a		*fi1*
3.	b		*fi1*
4.		$\{b, a\}$	*fu2,3*

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Chapter 2

propositional logic (zeroth-order logic)

2.1 What is proposition?

definition 2.1.1. ***predicate*** *"set()" and identity "=" and membership "$\in$" symbols are predicates. "set()" is unary and identity and membership are binary.*

capital letter "(P, Q, R)" showes **predicates**.
definition 2.1.2. ***Atomic proposition*** *we define atomic propositions like this.*

1. *if "a" is an obj then "set(a)" is an atomic proposition.*
2. *if "a, b" are obj then "$a = b$" is an atomic proposition.*
3. *if "a, b" are obj then "$a \in b$" is an atomic proposition.*

we denote atomic propositions with "p_i".
Undefined 2.1.1. ***logical operators*** *$(\neg, \rightarrow, \wedge, \vee)$ tool that used between atoms to make more complex propositions (all of the opratores are binary except "$\neg$" that is unary).*

you can easily see that formula are made up by atomic propositions.in general we say formula is an atomic formula or combination of them.
definition 2.1.3. ***Propositions*** *these are propositions.*

1. *atomic propositions "(p_i)".*
2. *if "p" is a formula then "$\neg p$" is a formula too.*
3. *if "p, q" are two propositions then "$p \bigcirc q$" is a formula too.($\bigcirc$ can be "$\rightarrow, \wedge, \vee$").*

for example "$(p_0 \rightarrow p_1, \neg p_3)$"
we denote propositions with "$p, q, r, \ldots$".

2.2 Rules of propositions.

definition 2.2.1. ***proposition schema(variable proposition)*** *a proposition that is unknown for us and it works as variable.(we show them with "$\varphi, \psi, \theta, \ldots$")*
Undefined 2.2.1. ***Assumption for conditional proof(acp)*** *is a tool for assuming a proposition. (any proposition can't get out but it can get in. right line must be shoter than left one.(it can't fall out). and every hypothesis that opens must be closed.)*

1
2
3
4
5
6

Rule 2.2.1. ***rules of logical operations*** ($\to I, \to E, \wedge I, \wedge E, \vee I, \vee E, \neg I, \neg E, \bot I, \bot E$) *this rule is scheme it means it works everywhere for every propositions. (thats why we use φ, ψ instead of p, q)*

φ $\vdots$ ψ $\therefore \varphi \to \psi$ $\quad \to I$	$\varphi \to \psi$ φ $\therefore \psi$ $\quad \to E$

Example 2.2.1. *for "p, q, r" we can prove these.*

1. $p \to q, q \to r / \therefore p \to r$

2. $q / \therefore p \to q$

1	$p \to q$	ass
2	$q \to r$	ass
3	p	acp
4	q	$\to E1, 3$
5	r	$\to E2, 4$
6	$p \to r$	$\to I$

1	q	ass
2	p	acp
3	q	ass
4	$p \to q$	$\to I$

$\begin{array}{l}\varphi \\ \psi \\ \hline \therefore \varphi \wedge \psi \end{array} \qquad \wedge I$	$\begin{array}{l}\varphi \wedge \psi \\ \hline \therefore \varphi \end{array} \qquad \wedge E \qquad \begin{array}{l}\varphi \wedge \psi \\ \hline \therefore \psi \end{array} \qquad \wedge E$

Example 2.2.2. *for "p, q, r" we can prove these.*

1. $p \wedge q / \therefore q \wedge p$
2. $(p \wedge q) \wedge r / \therefore p \wedge (q \wedge r)$
3. $p \to q / \therefore (p \wedge r) \to q$
4. $(p \wedge q) \to r / \therefore p \to (q \to r)$
5. $p \to (q \to r) / \therefore (p \wedge r) \to q$

1	$p \wedge q$	ass
2	p	$\wedge E, 1$
3	q	$\wedge E, 1$
4	$q \wedge p$	$\wedge I, 2, 3$

1	$(p \wedge q) \wedge r$	ass
2	$p \wedge q$	$\wedge E, 1$
3	r	$\wedge E, 1$
4	p	$\wedge E, 2$
5	q	$\wedge E, 2$
6	$q \wedge r$	$\wedge I, 3, 5$
7	$p \wedge (q \wedge r)$	$\wedge I, 4, 6$

1	$p \to q$	ass
2	$p \wedge r$	acp
3	p	$\wedge E, 2$
4	q	$\to E, 1, 3$
5	$(p \wedge r) \to q$	$\to I$

1	$(p \wedge q) \to r$	ass
2	p	acp
3	q	acp
4	$p \wedge q$	$\wedge I 2, 3$
5	r	$\to E 1, 4$
6	$q \to r$	$\to I$
7	$p \to (q \to r)$	$\to I$

1	$p \to (q \to r)$	ass
2	$p \wedge q$	acp
3	p	$\wedge E, 2$
4	$q \to r$	$\to E, 1, 3$
5	q	$\wedge E, 2$
6	r	$\to E, 4, 5$
7	$(p \wedge q) \to r$	$\to I$

$\dfrac{\varphi}{\therefore \varphi \vee \psi} \quad \vee I \qquad \dfrac{\varphi}{\therefore \psi \vee \varphi} \quad \vee I$	$\begin{array}{ll} \varphi \vee \psi & \\ \quad \mid \varphi & \\ \quad \mid \vdots & \\ \quad \mid \theta & \\ \quad \mid \psi & \\ \quad \mid \vdots & \\ \quad \mid \theta & \\ \therefore \theta & \vee E \end{array}$

Example 2.2.3. *for "p, q, r" we can prove these.*

1. $p \vee q / \therefore q \vee p$
2. $(p \vee q) \vee r / \therefore p \vee (q \vee r)$
3. $p \wedge (q \vee r) / \therefore (p \wedge q) \vee (p \wedge r)$
4. $(p \to q) \wedge (r \to s) / \therefore (p \vee r) \to (q \vee s)$

$$\begin{array}{lll} 1 & p \vee q & ass \\ 2 & \quad \mid p & acp \\ 3 & \quad \mid p \vee q & \vee I, 2 \\ 4 & \quad \mid q & acp \\ 5 & \quad \mid q \vee p & \vee I, 4 \\ 6 & q \vee p & \vee E, 1 \end{array}$$

$$\begin{array}{lll} 1 & (p \vee q) \vee r & ass \\ 2 & \quad \mid p \vee q & acp \\ 3 & \quad \mid \quad \mid p & acp \\ 4 & \quad \mid \quad \mid p \vee (q \vee r) & \vee I, 3 \\ 5 & \quad \mid \quad \mid q & acp \\ 6 & \quad \mid \quad \mid q \vee r & \vee I, 5 \\ 7 & \quad \mid \quad \mid p \vee (q \vee r) & \vee I, 6 \\ 8 & \quad \mid p \vee (q \vee r) & \vee E, 2 \\ 9 & \quad \mid r & acp \\ 10 & \quad \mid q \vee r & \vee I, 9 \\ 11 & \quad \mid p \vee (q \vee r) & \vee I, 10 \\ 12 & p \vee (q \vee r) & \vee E, 1 \end{array}$$

1	$p \wedge (q \vee r)$	*ass*
2	$\quad\mid\ p$	$\wedge E1$
3	$\quad\mid\ q \vee r$	$\wedge E1$
4	$\quad\mid\quad\mid\ q$	*acp*
5	$\quad\mid\quad\mid\ p \wedge q$	$\wedge E2,4$
6	$\quad\mid\quad\mid\ (p \wedge q) \vee (p \wedge r)$	$\vee I5$
7	$\quad\mid\quad\mid\ r$	*acp*
8	$\quad\mid\quad\mid\ p \wedge r$	*I∧ 2,7*
9	$\quad\mid\quad\mid\ (p \wedge q) \vee (p \wedge r)$	$\vee I8$
10	$\quad\mid\ (p \wedge q) \vee (p \wedge r)$	$\vee I8$
11	$(p \wedge q) \vee (p \wedge r)$	$\vee E3$

1	$(p \to q) \wedge (r \to s)$	*ass*
2	$(p \to q)$	$\wedge E1$
3	$(r \to s)$	$\wedge E1$
4	$\quad\mid\ p \vee r$	*acp*
5	$\quad\mid\quad\mid\ p$	*acp*
6	$\quad\mid\quad\mid\ q$	$\to E2,5$
7	$\quad\mid\quad\mid\ q \vee s$	$\vee I,6$
8	$\quad\mid\quad\mid\ r$	*acp*
9	$\quad\mid\quad\mid\ s$	$\to E3,8$
10	$\quad\mid\quad\mid\ q \vee s$	$\vee I9$
11	$\quad\mid\ q \vee s$	$\vee E,4$
12	$(p \vee r) \to (q \vee s)$	$\to I$

definition 2.2.2. $\bot : \varphi \wedge \neg\varphi$

$\mid\ \varphi$ $\mid\ \vdots$ $\mid\ \bot$ $\therefore \neg\varphi \qquad \neg I$	$\mid\ \neg\varphi$ $\mid\ \vdots$ $\mid\ \bot$ $\therefore \varphi \qquad \neg E$

definition 2.2.3. ***schema of prove*** *you've got knowed by now what is schema of a rule. if you want to prove a theorem that it's schema form.you can't prove it directly. here we use a technique calls* ***"schema of prove"****. it let us prove for all situations at the same time.*

Theorem 2.2.1. (*schema*) $(\bot I, \bot E)$

$\dfrac{\varphi \quad \neg\varphi}{\therefore \bot} \qquad \bot I$	$\dfrac{\bot}{\therefore \varphi} \qquad \bot E$

1	φ	*ass*
2	$\neg\varphi$	*ass*
3	$\varphi \wedge \neg\varphi$	$\wedge I$
4	$\bot$	$def.3$

1	$\bot$	*ass*
2	$\quad\mid\ \neg\varphi$	*acp*
3	$\quad\mid\ \vdots$	
4	$\quad\mid\ \bot$	$rp1$
5	φ	$\neg E$

Attention 2.2.1. *Intuitionists can skip the "¬E" rule that is the "RAA" rule in old notation and use "⊥E" as a rule instead.*

Example 2.2.4. *for "p, q, r" we can prove these.*

1. $p \to q, \neg q / \therefore \neg p$

2. $p \to q / \therefore \neg p \vee q$

$$\begin{array}{lll}
1 & p \to q & ass \\
2 & \neg q & ass \\
3 & \quad \mid\; p & acp \\
4 & \quad \mid\; q & \to E, 1, 3 \\
5 & \quad \mid\; \bot & \bot I, 2, 4 \\
6 & \neg p & \neg I
\end{array}$$

$$\begin{array}{lll}
1 & p \to q & ass \\
2 & \quad \mid\; \neg(\neg p \vee q) & acp \\
3 & \quad \mid\; \quad \mid\; p & acp \\
4 & \quad \mid\; \quad \mid\; q & \to E, 1, 3 \\
5 & \quad \mid\; \quad \mid\; \neg p \vee q & \vee I, 4 \\
6 & \quad \mid\; \quad \mid\; \bot & \bot I, 2, 5 \\
7 & \quad \mid\; \neg p & \neg I \\
8 & \quad \mid\; \neg p \vee q & \vee I, 7 \\
9 & \quad \mid\; \bot & \bot I, 2, 8 \\
10 & \neg p \vee q & \neg I
\end{array}$$

definition 2.2.4. $\varphi \leftrightarrow \psi : (\varphi \to \psi) \wedge (\psi \to \varphi)$

Theorem 2.2.2. ($schema$) ($\leftrightarrow I, \leftrightarrow E$)

$$\begin{array}{ll}
\mid\; \varphi & \\
\mid\; \vdots & \\
\mid\; \psi & \\
\mid\; \psi & \\
\mid\; \vdots & \\
\mid\; \varphi & \\
\therefore \varphi \leftrightarrow \psi & \leftrightarrow I
\end{array}$$

$$\frac{\varphi \leftrightarrow \psi \quad \varphi}{\therefore \psi} \leftrightarrow E \qquad \frac{\varphi \leftrightarrow \psi \quad \psi}{\therefore \varphi} \leftrightarrow E$$

1	φ	acp
	$\vdots$	
n	ψ	
$n+1$	$\varphi \to \psi$	$\to, I$
$n+2$	ψ	acp
	$\vdots$	
m	φ	
$m+1$	$\psi \to \varphi$	$\to, I$
$m+2$	$(\varphi \to \psi) \wedge (\psi \to \varphi)$	$\wedge I$
$m+3$	$\varphi \leftrightarrow \psi$	$def., m+3$

1	$\varphi \leftrightarrow \psi$	ass
2	φ	ass
3	$(\varphi \to \psi) \wedge (\psi \to \varphi)$	$def.1$
4	$\varphi \to \psi$	$\wedge E3$
5	ψ	$\to E1, 4$

Exercise 2.2.1. *for "p, q, r" prove.*

1. $p / \therefore p$
2. $/ \therefore p \vee \neg p$
3. $\neg p, p \vee q / \therefore q$
4. $\neg p / \therefore p \to q$
5. $p \to q \therefore / \therefore \neg q \to \neg p$
6. $(p \vee q) \wedge (p \vee r) \therefore / \therefore (p \vee (q \wedge r))$
7. $\neg p \wedge \neg q \therefore / \therefore \neg(p \vee q)$ *(De morgan)*
8. $\neg p \vee \neg q \therefore / \therefore \neg(p \wedge q)$ *(De morgan)*
9. $(\neg p \to q) / \therefore ((p \to q) \to q)$
10. $/ \therefore (p \wedge q \to r) \wedge (p \wedge s \to r) \to (p \wedge (q \vee s) \to r)$
11. $/ \therefore (p \wedge q \to r) \vee (p \wedge s \to r) \to (p \wedge (q \wedge s) \to r)$
12. $/ \therefore (p \to (q \to r)) \to ((p \to q) \to (p \to r))$
13. $/ \therefore (p \to q) \to ((q \to r) \to (p \to r))$

Attention 2.2.2. *we can rewrite* ***Example*** *and* ***Exercise*** *propositions and proofs with proposition schemas and schema of proves.so they are all rules.*

Chapter 3

first-order logic

3.1 what is formula?

obj are two types.we know them and we know what exactly are we talking about and unknowns are mystery.all the obj we have used until now were known ones we call them **"constants"**.
Undefined 3.1.1. ***variables*** *A variable is a symbol (usually a letter like* $x, y, or z$*) that represents an unknown obj.*
definition 3.1.1. ***Atomic formula*** *we define atomic formulas like this.*

1. *if "*x*" is an obj(constant or variable) then "*$Set(x)$*" is an atomic formula.*
2. *if "*x, y*" are obj(constant or variable) then "*$x = y$*" is an atomic formula.*
3. *if "*x, y*" are obj(constant or variable) then "*$x \in y$*" is an atomic formula.*

we denote atomic formulas with "A_i*".*
definition 3.1.2. ***well formed formula(WWF)*** *these are formulas.*

1. *atomic formulas (*A_i*).*
2. *if* A *is a formula then* $\neg A$ *is a formula too.*
3. *if* A, B *are two formulas then* $A \bigcirc B$ *is a formula too.(*$\bigcirc$ *can be* $\rightarrow, \wedge, \vee$*).*

for example $(A_0 \rightarrow A_1, \neg A_3)$
we easily call WWF "formula" and also we denote WWF with $A, B, C, ...$
definition 3.1.3. ***open formula*** *formulas that variables used in one of its atomic formulas.*

the formulas that are not open call **"closed formulas"** or **"sentence"**. Now question is "are very closed formulas proposition?" (propositions in classic logic are sentences that are true or false?) the answer is unfortunately **No**.
definition 3.1.4. ***Substitution for obj*** *we define Substitution an obj with variable like this.*

1. *for variables if* $x = y$ *then* $y[a/x] := a$ *and if* $x \neq y$ *then* $y[a/x] := y$*.*
2. *for constants* $c[a/x] := c$*.*

definition 3.1.5. ***Substitution for formulas*** *we define Substitution for a formula like this.*

1. *for atomic formulas we define it like this. if it's a single object we do it as we explain in "**Substitution for obj**". if it's like $x = y$ or $x \in y$ (x, y can be both constants and variables). we define it like this.*

$$Set(y)[a/x] := Set(y[a/x])$$

$$(x = y)[a/z] := (x[a/z] = y[a/z])$$

$$(x \in y)[a/z] := (x[a/z] \in y[a/z])$$

2. *if formula A is complex we define it like this.*

$$(B \bigcirc C)[a/x] := (B[a/x] \bigcirc C[a/x])$$

($\bigcirc$ can be $\rightarrow, \wedge, \vee$).

$$(\neg B)[a/x] := (\neg B[a/x])$$

simply means in a formula put an obj like a instead of a variable like x.we denote it by $A[a/x]$ or $A[\frac{a}{x}]$

definition 3.1.6. ***Open proposition*** *assume formula "A" with variables "$x_0, x_1, .., x_n$" in it and a set "U". we say formula "A"is a "open proposition in U" when for every obj "$a_0, a_1, .., a_n$" in "U"*

$$A([a_0/x_0][a_1/x_1]...[a_n/x_n])$$

is a proposition we call it open proposition and denote it with "$(P(x), Q(x, y), ..)$"

definition 3.1.7. ***Big "AND" and "OR"*** *assume a set like "$\{a_0, a_1, a_2, ..., a_n\}$" we define **big "AND"** like this.*

$$\bigwedge_{i=0}^{n} P(a_i) : P(a_0) \wedge P(a_1) \wedge P(a_2) \wedge ... \wedge P(a_n)$$

*and **big "OR"***

$$\bigvee_{i=0}^{n} P(a_i) : P(a_0) \vee P(a_1) \vee P(a_2) \vee ... \vee P(a_n)$$

definition 3.1.8. ***Universe of discourse*** *we specify a set like "$\{a_0, a_1, a_2, ..., a_n\}$" as universe of a Logic world.*

Rule 3.1.1. ***use of open propositions*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$"*

$c \quad P(x)$

means :

$c_0 \quad P[a_0/x]$

$c_1 \quad P[a_1/x]$

$c_2 \quad P[a_2/x]$

$\vdots$

$c_n \quad P[a_n/x]$

if Px was been assemed then like this

$$\begin{array}{l|l} c & P(x) \\ & \vdots \end{array}$$

means :

$$\begin{array}{l|l} c_0 & P[a_0/x] \\ & \vdots \end{array}$$

$$\begin{array}{l|l} c_1 & P[a_1/x] \\ & \vdots \end{array}$$

$$\begin{array}{l|l} c_2 & P[a_2/x] \\ & \vdots \end{array}$$

$$\begin{array}{l|l} c_n & P[a_n/x] \\ & \vdots \end{array}$$

definition 3.1.9. ***predicate schema(variable predicate)*** *a predicate that is unknown for us and it works as variable.(we show them with $\phi, \psi, \theta, ...$)*
definition 3.1.10. ***Universal quantification*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$" we define "$\forall x P(x)$"*

$$\forall x \phi(x) : \bigwedge_{i=0}^{n} \phi[a_i/x]$$

definition 3.1.11. ***Existential quantification*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$" we define "$\exists x P x$"*

$$\exists x \phi(x) : \bigvee_{i=0}^{n} \phi[a_i/x]$$

as you can see "$\forall x \phi(x)$" and "$\exists x \phi(x)$" are sentences.
definition 3.1.12. ***Free variables*** *variables that are not bound by a quantifier. (you can easily see that every formula with no free variables are sentences) (previously we sayed that every sentence is not necessary a proposition)*

3.2 Rules of first order logic.

Theorem 3.2.1. ***(*schema*)*** $(\forall I, \forall E, \exists I, \exists E)$

Assume that the universe of discourse is

$$D = \{a_0, a_1, a_2, \ldots, a_n\}.$$

Rules	*Terms and Conditions*
$\dfrac{\phi(x)}{\therefore \forall x\, \phi(x)}$ $\quad \forall I$	*1. x is a variable.* *2. x is arbitrary.* *3. x does not occur free in any assumption on which $\phi(x)$ depends.*
$\dfrac{\forall x\, \phi(x)}{\therefore \phi[a/x]}$ $\quad \forall E$	*1. a is an object in the universe of discourse.* *2. a is free for x in $\phi(x)$.*
$\dfrac{\phi[a/x]}{\therefore \exists x\, \phi(x)}$ $\quad \exists I$	*1. a is an object in the universe of discourse.* *2. a is free for x in $\phi(x)$.*
$\exists x\, \phi(x)$ $\quad \begin{array}{\|l} \phi(x) \\ \vdots \\ \theta \end{array}$ $\therefore \theta \qquad \exists E$	*1. x does not occur free in any assumption on which θ depends.* *2. x does not occur free in θ.* *3. $\phi(x)$ is assumed only inside the subproof.*

$$\begin{array}{lll}
1 & \phi(x) & ass \\
2 & \phi[a_0/x] & def.1 \\
 & \vdots & \\
n & \phi[a_n/x] & def.1 \\
n+1 & \bigwedge\limits_{i=0}^{n} \phi[a_i/x] & \wedge I, 2, 3, ..., n+2 \\
n+2 & \forall x \phi(x) & def.n+3
\end{array}$$

$$\begin{array}{lll}
1 & \forall x \phi(x) & ass \\
2 & \bigwedge\limits_{i=0}^{n} \phi[a_i/x] & def.1 \\
3 & \phi[a/x] & \wedge E2
\end{array}$$

$$\begin{array}{lll}
1 & \phi[a/x] & \\
2 & \bigvee\limits_{i=0}^{n} \phi[a_i/x] & \vee I1 \\
3 & \exists x \phi(x) & def2
\end{array}$$

$$\begin{array}{lll}
1 & \exists x \phi(x) & \\
2 & \bigvee\limits_{i=0}^{n} \phi[a_i/x] & \\
3 & \quad | \ \phi[a_0/x] & acp \\
 & \quad | \ \vdots & \\
n & \quad | \ \theta & \\
n+1 & \quad | \ \phi[a_1/x] & acp \\
 & \quad | \ \vdots & \\
m & \quad | \ \theta & \\
 & \vdots & \\
l & \quad | \ \phi[a_2/x] & acp \\
 & \quad | \ \vdots & \\
r & \quad | \ \theta & \\
r+1 & \theta & \exists E, 3
\end{array}$$

Rule 3.2.1. *(schema)* $(=I, =E)$

$\therefore a = a \qquad = I$	$\dfrac{a = b \quad \phi(a)}{\therefore \phi(b)} \quad = E \qquad \dfrac{a = b \quad \phi(a)}{\therefore \phi(a)} \quad = E$

definition 3.2.1. ***Uniqueness*** *we define "$\exists! x\phi(x)$"*

$$\exists! x\phi(x) : \exists x(\phi(x) \wedge \forall y(\phi(y) \rightarrow x = y))$$

Example 3.2.1. *for P,Q,R we can prove these.Proof of last ones Is Left As An Exercise To The Reader.*😀

1. $\forall x(P(x) \wedge Q(x)) \therefore / \therefore \forall x P(x) \wedge \forall x Q(x)$ *(D F $\forall\wedge$)(Distribution and Factorization of $\forall\wedge$)*
2. $\exists x(P(x) \vee Q(x)) \therefore / \therefore \exists x P(x) \vee \exists x Q(x)$ *(D F $\exists\vee$)*
3. $\forall x P(x) \vee \forall x Q(x) / \therefore \forall x(P(x) \vee Q(x))$ *(F $\forall\vee$)*
4. $\exists x(P(x) \wedge Q(x)) / \therefore \exists x P(x) \wedge \exists x Q(x)$ *(D $\exists\wedge$)*

5. $\forall x(P(x) \rightarrow Q(x)) / \therefore \forall x P(x) \rightarrow \forall x Q(x)$ *(D* $\forall \rightarrow$*)*

6. $\exists x P(x) \rightarrow \exists x Q(x) / \therefore \exists x(P(x) \rightarrow Q(x))$ *(F* $\exists \rightarrow$*)*

1	$\forall x(P(x) \wedge Q(x))$	ass
2	$P(x) \wedge Q(x)$	acp
3	$P(x)$	$\wedge E$
4	$Q(x)$	$\wedge E$
5	$\forall x P(x)$	$\forall I$
6	$\forall x Q(x)$	$\forall I$
7	$\forall x P(x) \wedge \forall x Q(x)$	$\wedge I$

1	$\forall x P(x) \wedge \forall x Q(x)$	ass
2	$\forall x P(x)$	$\wedge E$
3	$\forall x Q(x)$	$\wedge E$
4	$P(x)$	$\forall E$
5	$Q(x)$	$\forall E$
6	$P(x) \wedge Q(x)$	$\wedge I$
7	$\forall x(P(x) \wedge Q(x))$	$\forall I$

1	$\exists x(P(x) \vee Q(x))$	ass
2	$P(x) \vee Q(x)$	acp
3	$P(x)$	acp
4	$\exists x P(x)$	$\exists I$
5	$\exists x P(x) \vee \exists x Q(x)$	$\vee I$
6	$Q(x)$	acp
7	$\exists x Q(x)$	$\exists I$
8	$\exists x P(x) \vee \exists x Q(x)$	$\vee I$
9	$\exists x P(x) \vee \exists x Q(x)$	$\vee E$
10	$\exists x P(x) \vee \exists x Q(x)$	$\exists E$

1	$\exists x P(x) \vee \exists x Q(x)$	ass
2	$\exists x P(x)$	acp
3	$P(x)$	acp
4	$P(x) \vee Q(x)$	$\vee I$
5	$\exists x(P(x) \vee Q(x))$	$\exists I$
6	$\exists x(P(x) \vee Q(x))$	$\exists E$
7	$\exists x Q(x)$	acp
8	$Q(x)$	acp
9	$P(x) \vee Q(x)$	$\vee I$
10	$\exists x(P(x) \vee Q(x))$	$\exists I$
11	$\exists x(P(x) \vee Q(x))$	$\exists E$
12	$\exists x(P(x) \vee Q(x))$	$\vee E$

1	$\forall x P(x) \vee \forall x Q(x)$	*ass*
2	$\forall x P(x)$	*acp*
3	$P(x)$	$\forall E$
4	$P(x) \vee Q(x)$	$\vee I$
5	$\forall x(P(x) \vee Q(x))$	$\forall I$
6	$\forall x Q(x)$	*acp*
7	$Q(x)$	$\forall E$
8	$P(x) \vee Q(x)$	$\vee I$
9	$\forall x(P(x) \vee Q(x))$	$\forall I$
10	$\forall x(P(x) \vee Q(x))$	$\vee E$

1	$\exists x(P(x) \wedge Q(x))$	*ass*
2	$P(x) \wedge Q(x)$	*acp*
3	$P(x)$	$\wedge E$
4	$Q(x)$	$\wedge E$
5	$\exists x P(x)$	$\exists I$
6	$\exists x Q(x)$	$\exists I$
7	$\exists x P(x)$	$\exists E$
8	$\exists x Q(x)$	$\exists E$
9	$\exists x P(x) \wedge \exists x Q(x)$	$\wedge I$

1	$\forall x(P(x) \to Q(x))$	*ass*
2	$\forall x P(x)$	*acp*
3	$P(x)$	$\forall E$
4	$P(x) \to Q(x)$	$\forall E$
5	$Q(x)$	$\to E$
6	$\forall x Q(x)$	$\forall I$
7	$\forall x P(x) \to \forall x Q(x)$	$\to I$

1	$\exists x P(x) \to \exists x Q(x)$	$\to I$
2	$\exists x P(x) \vee \neg(\exists x P(x))$	*ex.*
3	$\exists x P(x)$	*acp*
4	$\exists Q(x)$	$\to I$
5	$Q(x)$	*acp*
6	$P(x) \to Q(x)$	*ex.*
7	$\exists x(P(x) \to Q(x))$	$\exists I$
8	$\exists x(P(x) \to Q(x))$	$\exists E$
9	$\neg(\exists x P(x))$	*acp*
10	$\forall x \neg P(x)$	*De morgan*
11	$\neg P(x)$	$\forall E$
12	$P(x) \to Q(x)$	*ex.*
13	$\exists x(P(x) \to Q(x))$	$\exists I$
14	$\exists x(P(x) \to Q(x))$	$\vee E$

Exercise 3.2.1. *for P,Q,R prove.*

1. $\forall x P(x) \wedge Q \therefore / \therefore \forall x(P(x) \wedge Q)$
2. $\forall x P(x) \vee Q \therefore / \therefore \forall x(P(x) \vee Q)$
3. $\exists x P(x) \wedge Q \therefore / \therefore \exists x(P(x) \wedge Q)$
4. $\exists x P(x) \vee Q \therefore / \therefore \exists x(P(x) \vee Q)$
5. $Q \to \forall x P(x) \therefore / \therefore \forall x(Q \to P(x))$

6. $Q \to \exists x P(x) \therefore / \therefore \exists x(Q \to P(x))$
7. $\forall x P(x) \to Q \therefore / \therefore \exists x(P(x) \to Q)$
8. $\exists x P(x) \to Q \therefore / \therefore \forall x(P(x) \to Q)$
9. $\exists x \neg P(x) \therefore / \therefore \neg \forall x P(x)$ *(De morgan)*
10. $\forall x \neg P(x) \therefore / \therefore \neg \exists x P(x)$ *(De morgan)*
11. $\exists x\Big(P(x) \land \exists y Q(y) \to \forall z R(z)\Big) / \therefore \forall y \forall z \Big(\forall x P(x) \land Q(y) \to R(z)\Big)$
12. $\forall x \exists y \Big(P(x) \land Q(y) \to \forall z R(z)\Big) / \therefore \forall z \Big(\exists x P(x) \land \forall y Q(y) \to R(z)\Big)$
13. $\forall x P(x) \to \exists y Q(y) / \therefore \exists x \exists y \Big(P(x) \to Q(y)\Big)$
14. $\forall x \forall z \exists y \exists w \Big(P(y,z) \lor Q(z) \to R(x,w)\Big) / \therefore \Big(\exists z \forall y P(y,z) \lor \exists z Q(z)\Big) \to \forall x \exists w R(x,w)$
15. $\exists x P(x) \lor \exists x \Big(Q(x) \land R(x)\Big) / \therefore \exists x \Big(P(x) \lor Q(x)\Big) \land \exists x \Big(P(x) \lor R(x)\Big)$
16. $\exists x \exists z \forall y \Big(P(x,y) \to Q(z) \lor R(x,y)\Big) / \therefore \forall x \exists y P(x,y) \to \Big(\exists z Q(z) \lor \exists x \exists z R(x,y)\Big)$
17. $\forall x \exists y \Big(\exists z P(x,y,z) \land Q(x,y)\Big) \lor \forall x \exists y \exists z \Big(P(x,y,z) \land R(x,y)\Big) / \therefore \forall x \exists y \exists z \Big(P(x,y,z) \land (Q(x,y) \lor R(x,y))\Big)$

Attention 3.2.1. *we can rewrite **Example** and **Exercise** propositions and proofs with proposition schemas and schema of proves.so they are all rules.*

Chapter 4

The beginning of set theory

In language of set theory if we are talking about “$\forall X(P(X))$” it means “$\forall x(Set(x) \to P(x))$” and when we are talking about “$\exists X(P(X))$” it means “$\exists x(Set(x) \wedge P(x))$”. similarly if we are talking about “$\forall x \in A \quad P(x)$” it means “$\forall x(x \in A \to P(x))$” and when we are talking about “$\exists x \in A \quad P(x)$” it means “$\exists x(x \in A \wedge P(x))$”.

and we have a definitions.

$$Type_0(X) : \exists Y \quad X \in Y$$

$$Type_1(X) : \neg\exists Y \quad X \in Y$$

4.1 Axioms of set theory

we emphasis before that set has no definition.but we can describe what set is with rules and axioms.

Axiom 4.1.1. ***Existence of universe of set theory*** *There exists a set V_2 (we denote it as V_2.it simply means the set of all sets), known as the Von Neumann Universe, which contains every set. (this axiom can't be formolized in first order logic). (there must be another axiom let us use logic for set theory but we ignore that). (from now on when we are talking about proposition “ϕ” in “set theory” we mean proposition “ϕ” in “V_2”).*

definition 4.1.1. ***Subset*** *a subset is a set where every element is also an element of another, larger set called the superset*

$$A \subseteq B : \forall x(x \in A \to x \in B)$$

or i.e.

$$\forall x \in A \quad x \in B$$

Theorem 4.1.1. (*schema*) ($\subseteq I, \subseteq E$)

$\quad\vert\; x \in A$ $\quad\vert\; \vdots$ $\quad\vert\; x \in B$ $\therefore A \subseteq B \qquad \subseteq I$	$A \subseteq B$ $\underline{a \in A}$ $\therefore a \in B \qquad \subseteq E$

1	$\vert\; x \in A$	*acp*
	$\vert\; \vdots$	
n	$\vert\; x \in B$	
$n+1$	$x \in A \to x \in B$	$I \to, 1, n$
$n+2$	$\forall x(x \in A \to x \in B)$	$\forall I, n+1$
$n+3$	$A \subseteq B$	$def. n+2$

1	$A \subseteq B$	*ass*
2	$a \in A$	*ass*
3	$\forall x(x \in A \to x \in A)$	$def.1$
4	$a \in A \to a \in B$	$\forall E, 4$
5	$a \in B$	$E \to 3, 5$

Example 4.1.1. $A \subseteq B, B \subseteq C / \therefore A \subseteq C$

1	$A \subseteq B$	*ass*
2	$B \subseteq C$	*ass*
3	$\vert\; x \in A$	*acp*
4	$\vert\; x \in B$	$\subseteq E, 1, 3$
5	$\vert\; x \in C$	$\subseteq E, 2, 4$
6	$A \subseteq C$	$\subseteq I, 3, 5$

Axiom 4.1.2. ***Extensionality*** *Two sets are equal (are the same set) if they have the same elements.*

$$\forall X, Y(\forall x(x \in X \leftrightarrow x \in Y) \to X = Y)$$

so we can rewrite axiom of Extensionality like this.

$$\forall X, Y((X \subseteq Y \wedge Y \subseteq X) \to X = Y)$$

Theorem 4.1.2. (*schema*) $(= I, = E)$

$\begin{array}{l} \left\lvert \begin{array}{l} x \in A \\ \vdots \\ x \in B \end{array} \right. \\ \left\lvert \begin{array}{l} x \in B \\ \vdots \\ x \in A \end{array} \right. \\ \therefore A = B \qquad = I \end{array}$	$\dfrac{A = B \quad a \in A}{\therefore a \in B} \; = E \qquad \dfrac{A = B \quad a \in B}{\therefore a \in A} \; = E$

1	$\mid x \in A$	acp
	$\mid \vdots$	
n	$\mid x \in B$	
$n+1$	$A \subseteq B$	$\subseteq I, 1, n$
$n+2$	$\mid x \in B$	acp
	$\mid \vdots$	
m	$\mid x \in A$	
$m+1$	$B \subseteq A$	$\subseteq I, n+2, m$
$m+2$	$\forall X, Y((X \subseteq Y \wedge Y \subseteq X) \rightarrow X = Y)$	*Extensionality*
$m+3$	$(A \subseteq B \wedge B \subseteq A) \rightarrow A = B$	$\forall E, m+2$
$m+4$	$A \subseteq B \wedge B \subseteq A$	$\wedge I, n+1, m+1$
$m+5$	$A = B$	$E \rightarrow, m+3, m+4$

Axiom 4.1.3. ***Weak comprehension*** *(schema) this axioms says for every open proposition like "$\phi(x, x_0, x_1, ..., x_n)$" there is a set like "$X$" such that every "$x$" with "$Type_0$" is a member of that set if and only if it holds "$\phi(x, x_0, x_1, ..., x_n)$".*

$$\forall x_0, x_1, ..., x_n \exists X \forall x(Type_0(x) \wedge \phi(x, x_0, x_1, ..., x_n) \leftrightarrow x \in X)$$

by axiom of extentionality we can show that X is unique. so we denote it as "$\{x \mid \phi(x, x_0, x_1, ..., x_n)\}$". sometimes instead of "$\{x \mid x \in A \wedge \phi(x, x_0, x_1, ..., x_n)\}$" we write "$\{x \in A \mid \phi(x, x_0, x_1, ..., x_n)\}$".

Let's talk about some axioms in set theoty.These daxioms will come in handy later.

Axiom 4.1.4. ***Existence*** *there is a "Set" with no member i.e..*

$$\exists x \quad Type_0(x)$$

definition 4.1.2. ***Empty set*** *also known as the null set, is the unique mathematical set containing no elements at all. (uniqueness can be proven.)*

$$\varnothing := \{x \mid \bot\}$$

therefore.

$$\forall x(x \in \varnothing \leftrightarrow \bot \wedge Type_0(x))$$

therefore.

$$\forall x(x \notin \varnothing)$$

Example 4.1.2. $/ \therefore \forall X(\varnothing \subseteq X)$

1	$\quad\mid x \in \varnothing$	*acp*
2	$\quad\mid \forall x(x \notin \varnothing)$	*def*
3	$\quad\mid x \notin \varnothing$	$\forall E, 2$
4	$\quad\mid \bot$	$\bot I, 1, 3$
5	$\quad\mid x \in X$	$\bot E, 4$
6	$\varnothing \subseteq X$	$\subseteq I, 1, 5$
7	$\forall X(\varnothing \subseteq X)$	$\forall I, 6$

Lemma 4.1.1. $A = \varnothing \therefore / \therefore \forall y(y \notin A)$

1	$\forall y(y \notin A)$	*ass*
2	$\quad\mid y \in A$	*acp*
3	$\quad\mid y \notin A$	$\forall E, 1$
4	$\quad\mid \bot$	$\bot I, 2, 3$
5	$\quad\mid y \in \varnothing$	$\bot E, 4$
6	$\quad\mid y \in \varnothing$	*acp*
7	$\quad\mid \forall y(y \notin \varnothing)$	*def.*
8	$\quad\mid y \notin \varnothing$	$\forall E, 7$
9	$\quad\mid \bot$	$\bot I, 6, 8$
10	$\quad\mid y \in A$	$\bot E, 9$
11	$A = \varnothing$	$I =$

1	$A = \varnothing$	*ass*
2	$\forall y(y \notin \varnothing)$	*def.*
3	$\forall y(y \notin A)$	$= E, 1, 2$

Theorem 4.1.3. (*schema*) $(\varnothing \neq I, \varnothing \neq E, \varnothing = I, \varnothing = E)$

$\dfrac{a \in A}{\therefore A \neq \varnothing}$ $\quad \varnothing \neq I$	$\begin{array}{l} A \neq \varnothing \\ \quad \left\vert \begin{array}{l} x \in A \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \end{array}$ $\quad \varnothing \neq E$
$\begin{array}{l} \quad \left\vert \begin{array}{l} x \in A \\ \vdots \\ \bot \end{array} \right. \\ \therefore A = \varnothing \end{array}$ $\quad \varnothing = I$	$\dfrac{A = \varnothing}{\therefore a \notin A}$ $\quad \varnothing = E$

1	$a \in A$	*ass*
2	$\exists x(x \in A)$	$\exists I, 1$
3	$\neg(\forall x(x \notin A))$	*De Morgan* 2
4	$A \neq \varnothing$	*lemma* 4.4.1, 3

1	$\quad\vert\ x \in A$	*acp*
	$\quad\vert\ \vdots$	
n	$\quad\vert\ \bot$	
$n+1$	$x \notin A$	$\neg I$
$n+2$	$\forall x(x \notin A)$	$\forall I, n+1$
$n+3$	$A = \varnothing$	*lemma* 4.4.1, 1

1	$A = \varnothing$	*ass*
2	$\forall x(x \notin A)$	*lemma* 4.4.1, 1
3	$a \notin A$	$\forall E2$

Axiom 4.1.5. ***Existence***

$$Type_0(\varnothing)$$

definition 4.1.3. ***Pairing set***

$$\{a_0, a_1, a_2, ..., a_n\} := \{x \mid x = a_0 \vee x = a_1 \vee x = a_2 \vee ... \vee x = a_n\}$$

therefore.

$$\forall x(x \in \{a_0, a_1, a_2, ..., a_n\} \leftrightarrow (Type_0(x) \wedge x = a_0 \vee x = a_1 \vee x = a_2 \vee ... \vee x = a_n))$$

Theorem 4.1.4. (*schema*) $(\{\}I, \{\}E)$

<table>
<tr>
<td>$$\frac{a = a_i}{\therefore a \in \{a_0, a_1, a_2, ..., a_n\}} \qquad \{\}I$$</td>
<td>$$\begin{array}{l} a \in \{a_0, a_1, a_2, ..., a_n\} \\ \quad \left| \begin{array}{l} a = a_0 \\ \vdots \\ \theta \end{array} \right. \\ \vdots \\ \quad \left| \begin{array}{l} a = a_n \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \qquad\qquad\qquad\qquad \{\}E \end{array}$$</td>
</tr>
</table>

Proof Is Left As An Exercise To The Reader.😃

Axiom 4.1.6. ***Pairing*** (*schema*)

$$Type_0(\{a_0, a_1, a_2, ..., a_n\})$$

definition 4.1.4. ***Union and Intersection***

Intersection:

$$A \cap B := \{x \mid x \in A \wedge x \in B\}$$

therefore.

$$\forall x(x \in A \cap B \leftrightarrow (x \in A \wedge x \in B))$$

Union:

$$A \cup B := \{x \mid x \in A \vee x \in B\}$$

therefore.

$$\forall x(x \in A \cup B \leftrightarrow (x \in A \vee x \in B))$$

Theorem 4.1.5. (*schema*) $(\cap I, \cap E, \cup I, \cup E)$

$\begin{array}{l} a \in A \\ a \in B \\ \hline \therefore a \in A \cap B \end{array} \quad \cap I$	$\begin{array}{l} a \in A \cap B \\ \hline \therefore a \in A \end{array} \quad \cap E \qquad \begin{array}{l} a \in A \cap B \\ \hline \therefore a \in B \end{array} \quad \cap E$
$\begin{array}{l} a \in A \\ \hline \therefore a \in A \cup B \end{array} \quad \cup I$ $\begin{array}{l} a \in A \\ \hline \therefore a \in B \cup A \end{array} \quad \cup I$	$\begin{array}{l} a \in A \cup B \\ \quad \begin{array}{\|l} a \in A \\ \vdots \\ \theta \end{array} \\ \quad \begin{array}{\|l} a \in B \\ \vdots \\ \theta \end{array} \\ \therefore \theta \end{array} \quad \cup E$

Proof Is Left As An Exercise To The Reader.😃

sometimes all members of a set is also set we call it “set of sets”.
definition 4.1.5. ***Big Union and Intersection*** *if “A” is “set of sets” and “$A \neq \varnothing$” we define Big Intersection:*

$$\bigcap A := \{x \mid \forall X(X \in A \to x \in X)\}$$

i.e.

$$\bigcap A := \{x \mid \forall X \in A \quad x \in X\}$$

therefore.

$$\forall x\Big(x \in \bigcap A \leftrightarrow (Type_0(x) \wedge \forall X(X \in A \to x \in X))\Big)$$

Big Union:

$$\bigcup A := \{x \mid \exists X(X \in A \wedge x \in X)\}$$

i.e.

$$\bigcup A := \{x \mid \exists X \in A \quad x \in X\}$$

therefore.

$$\forall x\Big(x \in \bigcup A \leftrightarrow (Type_0(x) \wedge \exists X(X \in A \wedge x \in X))\Big)$$

Theorem 4.1.6. (*schema*) ($\bigcap I, \bigcap E, \bigcup I, \bigcup E$)

$\quad\vert\; X \in A$ $\quad\vert\; \vdots$ $\quad\vert\; a \in X$ $\therefore a \in \bigcap A \qquad \bigcap I$	$a \in \bigcap A$ $\underline{B \in A}$ $\therefore a \in B \qquad \bigcap E$
$a \in B$ $\underline{B \in A}$ $\therefore a \in \bigcup A \qquad \bigcup I$	$a \in \bigcup A$ $\quad\vert\; a \in X$ $\quad\vert\; X \in A$ $\quad\vert\; \vdots$ $\quad\vert\; \theta$ $\therefore \theta \qquad \bigcup E$

Proof Is Left As An Exercise To The Reader.😃

definition 4.1.6. ***Complement***

$$A^c := \{x \mid x \notin A\}$$

therefore.

$$\forall x(x \in A^c \leftrightarrow Type_0(x) \land x \notin A)$$

Theorem 4.1.7. (*schema*) $(()^c I, ()^c E)$

$\quad\vert\; a \in A$ $\quad\vert\; \vdots$ $\quad\vert\; \bot$ $\therefore a \notin A \qquad ()^c I$	$\quad\vert\; a \notin A$ $\quad\vert\; \vdots$ $\quad\vert\; \bot$ $\therefore a \in A \qquad ()^c E$

Proof Is Left As An Exercise To The Reader.😃

Observation 4.1.1. *we can show Universal set with...*

$$V = \{x \mid \top\}$$

therefore.

$$\forall x(x \in V \leftrightarrow Type_0(x) \land \top)$$

therefore.

$$\forall x(x \in V)$$

definition 4.1.7. ***Subtraction***

$$A - B := \{x \mid x \in A \land x \notin B\}$$

therefore.

$$\forall x\Big(x \in A - B \leftrightarrow (Type_0(x) \land x \in A \land x \notin B)\Big)$$

Theorem 4.1.8. (*schema*) $(-I, -E)$

$$\begin{array}{ll} a \in A & \\ a \notin B & \\ \hline \therefore a \in A - B & \quad -I \end{array}$$	$$\begin{array}{ll} a \in A - B & \\ \hline \therefore a \in A & \quad -E \end{array} \qquad \begin{array}{ll} a \in A - B & \\ \hline \therefore a \notin B & \quad -E \end{array}$$

Proof Is Left As An Exercise To The Reader.😃

Lemma 4.1.2. $/ \therefore A - B = A \cap B^c$

Proof Is Left As An Exercise To The Reader.😃

Axiom 4.1.7. ***Separation***

$$\forall X, Y \quad Type_0(X) \to Type_0(X \cap Y)$$

therefore.

$$Type_0(A) \to Type_0(A \cap B)$$

therefore.

$$Type_0(\bigcap A)$$

therefore.

$$Type_0(A) \to Type_0(A - B)$$

Axiom 4.1.8. ***union***

$$\forall X \quad Type_0(X) \to Type_0(\bigcup X)$$

therefore.

$$Type_0(A) \wedge Type_0(B) \to Type_0(A \cup B)$$

definition 4.1.8. ***Powerset***

$$\mathcal{P}(A) := \{X \mid X \subseteq A\}$$

therefore.

$$\forall X\Big(X \in \mathcal{P}(A) \leftrightarrow (Type_0(X) \wedge X \subseteq A)\Big)$$

Theorem 4.1.9. (*schema*) $(\mathcal{P}I, \mathcal{P}E)$

$$\begin{array}{ll} A \subseteq B & \\ \hline \therefore A \in \mathcal{P}(B) & \quad \mathcal{P}I \end{array}$$	$$\begin{array}{ll} A \in \mathcal{P}(B) & \\ \hline \therefore A \subseteq B & \quad \mathcal{P}E \end{array}$$

Proof Is Left As An Exercise To The Reader.😃

Axiom 4.1.9. ***Powerset***

$$\forall X \quad Type_0(X) \to Type_0(P(X))$$

from now on we don't get in to all details of proofs because it will be too long and hard to read. so we will take it easy from now and don't write all details like reasons of lines and all lines we left it to reader to understand what happened there.😃

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Chapter 5

Relations and Functions

5.1 Ordered pair and Cartesian product

definition 5.1.1. ***Ordered pair***

$$(x, y) := \{\{x\}, \{x, y\}\}.$$

so $\forall x, y \quad Type_0((x, y))$
definition 5.1.2. ***Tuples****(n is a arbitrary number.)(this is a temporary definition, we will give a better one in the next chapter.)*

$$(x_1, x_2, ..., x_n) := ((x_1, x_2, ..., x_{n-1}), x_n)$$

Lemma 5.1.1. $/ \therefore (x_0, y_0) = (x_1, y_1) \leftrightarrow ((x_0 = x_1) \wedge (y_0 = y_1))$

Proof Is Left As An Exercise To The Reader.😀
definition 5.1.3. ***Cartesian product***

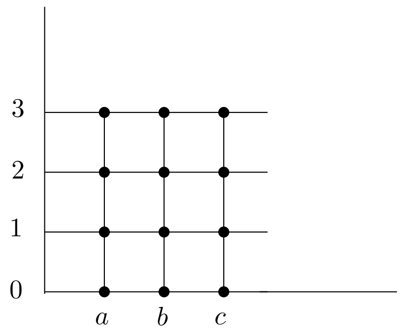
$$A \times B := \{u \mid \exists x, y\Big((x, y) = u \wedge x \in A \wedge y \in B\Big)\}.$$

or i.e.

$$A \times B := \{(x, y) \mid x \in A \wedge y \in B\}.$$

Example 5.1.1. *if* $A = \{a, b, c\}$ *and* $B = \{0, 1, 2, 3\}$ *then*

$$A \times B = \{(a, 0), (a, 1), (a, 2), (a, 3), (b, 0), (b, 1), (b, 2), (b, 3), (c, 0), (c, 1), (c, 2), (c, 3)\}$$



so $\forall X, Y \quad Type_0(X) \wedge Type_0(Y) \rightarrow Type_0(X \times Y)$

Theorem 5.1.1. *(schema)* $(\times I, \times E)$

$t \in A$ $s \in B$ $\overline{\therefore (t, s) \in A \times B} \qquad \times I$	$\therefore t \in A \times B$ $\quad \vert\ t = (x, y)$ $\quad \vert\ x \in A$ $\quad \vert\ y \in B$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\theta \qquad\qquad \times E$

Proof Is Left As An Exercise To The Reader. 😃

Theorem 5.1.2. *(schema)*

$(t, s) \in A \times B$ $\overline{\therefore t \in A}$	$(t, s) \in A \times B$ $\overline{\therefore s \in B}$

Proof Is Left As An Exercise To The Reader. 😃

Exercise 5.1.1. *for sets A, B, C, prove.*

1. $/ \therefore A \times (B \cap C) = (A \times B) \cap (A \times C)$
2. $/ \therefore A \times (B \cup C) = (A \times B) \cup (A \times C)$
3. $/ \therefore A \times (B - C) = (A \times B) - (A \times C)$

5.2 Relations

definition 5.2.1. ***Relation*** *we call a set like* $\boldsymbol{R}$ *a* ***Relation*** *when there are two sets like X and Y s.t. $R \subseteq X \times Y$.*

$$Rel(R) : \exists X, Y (R \subseteq X \times Y).$$

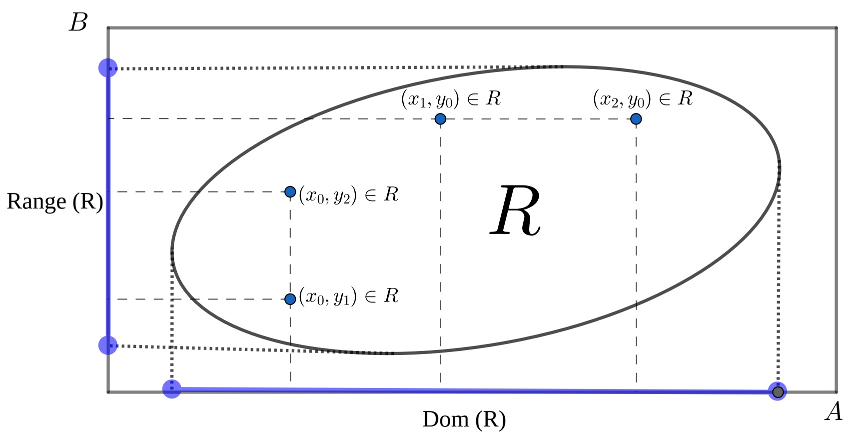
sometimes we use different notations in this book. you have to be smart and recognize it by yourself. sometimes instead of "$x \in R$" we denote it by "$R(x)$". and instead of "$(x_0, x_1, ..., x_n) \in R$" we denote it by "$R(x_0, x_1, ..., x_n)$". if tuple is binary, instead of "$(x, y) \in R$" we denote it by "xRy". or "$R(x, y)$" instead of "$R\big((x, y)\big)$"
so if we have a definition like "$R(x) : \phi(x)$" s.t. "$\phi(x)$" is a first order proposition. "weak comprehension" show us there is a straight relatioon between "propositions" and "sets" so you can see it as "$R := \{x \mid \phi(x)\}$". and sometimes we might show a formula in natural language for example below instead of "$Rel(X)$" we can say "X is relation". if it is confusing so forget it. 😃

definition 5.2.2. ***Domain and Range(also called Image)***
Domain of a set "R" is all "x" that there exists a "y" s.t. "xRy"

$$Dom(R) := \{x \mid \exists y \quad xRy\}$$

Range(Image) of a set "R" is all "y" that there exists a "x" s.t. "xRy"

$$Range(R) := \{y \mid \exists x \quad xRy\}.$$



Theorem 5.2.1. (*schema*) ($DomI, DomE$)

$\dfrac{tRs}{\therefore t \in Dom(R)}$ $\quad DomI$	$t \in Dom(R)$ $\quad$ tRy $\quad$ $\vdots$ $\quad$ θ $\therefore \theta$ $\quad DomE$

Proof Is Left As An Exercise To The Reader.😀

Theorem 5.2.2. (*schema*) ($RangeI, RangeE$)

$\dfrac{tRs}{\therefore s \in Range(R)}$ $\quad RangeI$	$s \in Range(R)$ $\quad$ xRs $\quad$ $\vdots$ $\quad$ θ $\therefore \theta$ $\quad RangeE$

Proof Is Left As An Exercise To The Reader.😃

definition 5.2.3. ***Inverse of a Relation***

$$R^{-1} = \{u \mid \exists x, y\Big(u = (x, y) \wedge yRx\Big)\}$$

or i.e.

$$R^{-1} = \{(x, y) \mid yRx\}$$

therefore.

$$\forall x, y\Big(xR^{-1}y \leftrightarrow yRx\Big)$$

Theorem 5.2.3. *(schema)* $(()^{-1}I, ()^{-1}E)$

$\dfrac{tRs}{\therefore sR^{-1}t}$ $\quad ()^{-1}I$	$t \in R^{-1}$ $\quad \vert\ t = (x, y)$ $\quad \vert\ yR^{-1}x$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\therefore \theta \qquad ()^{-1}E$

Proof Is Left As An Exercise To The Reader.😃

definition 5.2.4. ***Combination of a Relations***

$$S \circ R = \{u \mid \exists x, y\Big(u = (x, y) \wedge \exists z\big(xRz \wedge zSy\big)\Big)\}$$

or i.e.

$$S \circ R = \{(x, y) \mid \exists z\Big(xRz \wedge zSy\Big)\}$$

therefore.

$$\forall x, y\Big((x, y) \in S \circ R \leftrightarrow \exists z\big(xRz \wedge zSy\big)\Big)$$

Theorem 5.2.4. *(schema)* $(\circ I, \circ E)$

$$\begin{array}{ll} tSs & \\ \underline{sRr} & \\ \therefore sR \circ Sr & \circ I \end{array}$$	$$\begin{array}{ll} t \in R \circ S & \\ \quad \begin{array}{\|l} t = (x,y) \\ xSz \\ zRy \\ \vdots \\ \theta \end{array} & \\ \therefore \theta & \circ E \end{array}$$

Proof Is Left As An Exercise To The Reader.😃
definition 5.2.5. ***Image***

$$R[A] := \{y \mid \exists x \in A \quad xRy\}$$

you can olso see by definition of "inverse of a Relation" and "image" this equation holds.

$$R^{-1}[B] = \{x \mid \exists y \in B \quad xRy\}$$

*we cal this "**Pre-image**"*
Theorem 5.2.5. (*schema*) $(R[]I, R[]E, R^{-1}[]I, R^{-1}[]E)$

$$\begin{array}{ll} t \in A & \\ \underline{tRs} & \\ \therefore s \in R[A] & R[]I \end{array}$$	$$\begin{array}{ll} s \in R[A] & \\ \quad \begin{array}{\|l} x \in A \\ xRs \\ \vdots \\ \theta \end{array} & \\ \therefore \theta & R[]E \end{array}$$
$$\begin{array}{ll} s \in B & \\ \underline{tRs} & \\ \therefore t \in R^{-1}[B] & R^{-1}[]I \end{array}$$	$$\begin{array}{ll} t \in R^{-1}[B] & \\ \quad \begin{array}{\|l} y \in B \\ tRy \\ \vdots \\ \theta \end{array} & \\ \therefore \theta & R^{-1}[]E \end{array}$$

Proof Is Left As An Exercise To The Reader.😃

5.3 Functions

definition 5.3.1. ***Function*** *we call a relation "**f**" a "**function**" when every input has exactly one output i.e.*

$$Func(f) : Rel(f) \wedge \forall x, y_1, y_2 \Big((x, y_1) \in f \wedge (x, y_2) \in f \to y_1 = y_2 \Big)$$

Theorem 5.3.1. *(schema)* $(FuncI, FuncE)$

$\quad xfy$ $\quad xfz$ $\quad \vdots$ $\quad y = z$ $\therefore Func(f) \qquad FuncI$	$Func(f)$ tfs tfr $\therefore s = r \qquad FuncE$

Proof Is Left As An Exercise To The Reader.😃

definition 5.3.2. ***Restriction of a function*** *to a set "X" (usually a subset of "dom(f)") is the function*

$$f \restriction X := \{(x, y) \in f \mid x \in X\}$$

definition 5.3.3. ***Output of a function*** *assume a function* ***f*** *and* $(x, y) \in f$*, we define "*$f(x) := y$*" and we call "*$f(x)$*" a "**output of a function**". and the reverse is also true, if we use "*$f(x)$*", we have a function "**f**" and we can say "*$x \in Dom(f)$*". "output of a function" is object too.*

Theorem 5.3.2. *(schema)*

$Func(f)$ $(x, y) \in f$ $\therefore f(x) = y \qquad f()E$	$f(x) = y$ $\therefore Func(f) \qquad f()E$ $\therefore (x, y) \in f \qquad f()E$

Attention 5.3.1. *input of a function can be a ordered pair itself, so we can have* $f((x, y))$ *and it is a "**output of a function**". we define "*$f(x, y) := f((x, y))$*" for simplicity. this is true for tuples as well, so we can have "*$f((x_1, x_2, ..., x_n))$*" and we define "*$f(x_1, x_2, ..., x_n) := f((x_1, x_2, ..., x_n))$*" for simplicity.*

definition 5.3.4. ***Term*** *we define term like this.*

1. *every obj is a term.*
2. *if* $t_0, ...t_n$ *are terms and "*$Func(f)$*" and "*$(t_0, ...t_n) \in Dom(f)$*". then "*$f(t_0, ...t_n)$*" is also a term.*

we denote terms with small letters like $t, s, r, ...$.

definition 5.3.5. ***Substitution for term*** *we define Substitution an term with variable like this.*

1. *for variables if* $x = y$ *then* $y[t/x] := t$ *and if* $x \neq y$ *then* $y[t/x] := y$.
2. *for constants* $c[t/x] := c$.

3. *for terms like* $f(t_0, ...t_n)$

$$f(t_0, ...t_n)[t/x] := f(t_0[t/x], ...t_n[t/x])$$

definition 5.3.6. ***Substitution for formulas*** *we define Substitution for a formula like this.*

1. *for atomic formulas we define it like this. if it's a single term we do it as we explain in "**Substitution for term**". if it's like* $t_0 = t_1$ *or* $t_0 \in t_1$ *(*t_0, t_1 *are terms). we define it like this.*

$$Set(y)[t/x] := Set(y[t/x])$$

$$(t_0 = t_1)[t/x] := (t_0[t/x] = t_1[t/x])$$

$$(t_0 \in t_1)[t/x] := (t_0[t/x] \in t_1[t/x])$$

2. *if formula* A *is complex we define it like this.*

$$(B \bigcirc C)[t/x] := (B[t/x] \bigcirc C[t/x])$$

($\bigcirc$ *can be* $\to, \wedge, \vee$*).*

$$(\neg B)[t/x] := (\neg B[t/x])$$

Theorem 5.3.3. ***(schema)*** $(\forall I, \forall E, \exists I, \exists E)$

Rules	*Terms and Conditions*
$\dfrac{\phi(x)}{\therefore \forall x \phi(x)}$ $\quad \forall I$	1. x *is a variable.* 2. *"*$\phi(x)$*" must not be an assumption.* 3. x *must not occur free in any assumption on which* $\phi(x)$ *depends.*
$\dfrac{\forall x \phi(x)}{\therefore \phi[t/x]}$ $\quad \forall E$	1. t *is a term.* 2. t *is free for* x *in* $\phi(x)$*.*
$\dfrac{\phi[t/x]}{\therefore \exists x \phi(x)}$ $\quad \exists I$	1. t *is a term.* 2. t *is free for* x *in* $\phi(x)$*.*
$\exists x \phi(x)$ $\quad \vert\ \phi(x)$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\therefore \theta$ $\quad \exists E$	1. x *is a variable.* 2. *"*$\phi(x)$*" is assumed only within the subproof.* 3. x *does not occur free in* θ*.* 4. x *does not occur free in any assumption on which* θ *depends.*

$(\forall I, \exists E)$ *is the same as* $(\forall I, \exists E)$ *in chapter three so we have to prove* $(\forall E, \exists I)$*. so assume* $t = f(x)$ *so to it's easy to see* $Func(f)$ *and* $x \in Dom(f)$*.*

1	$\forall x\phi(x)$	ass
2	$Func(f)$	1
3	$x \in Dom(f)$	1
4	$\quad\vert\ (x,y) \in f$	acp
5	$\quad\vert\ f(x) = y$	def
6	$\quad\vert\ \phi(y)$	$\forall E$
7	$\quad\vert\ \phi(f(x))$	$= E$
8	$\phi(f(x))$	$\exists E$

1	$\phi(f(x))$	ass
2	$Func(f)$	ass
3	$x \in Dom(f)$	ass
4	$\quad\vert\ (x,y) \in f$	acp
5	$\quad\vert\ f(x) = y$	def
6	$\quad\vert\ \phi(y)$	$= E$
7	$\quad\vert\ \exists x\phi(x)$	$\exists I$
8	$\exists x\phi(x)$	$\exists E$

Attention 5.3.2. *in all rules and theorems you can replace all constants like $a, b, c, ...$ with terms like $t, s, r, ...$ and rewrite them.*

definition 5.3.7. ***Compatible Function***

1. *$Compatible(f, g) : \forall x \in dom(f) \cap dom(g) \quad f(x) = g(x)$.*
2. *$Compatible(F) : \forall f, g \in F \quad Compatible(f, g)$.*

Lemma 5.3.1. *.*

1. *$Compatible(f, g) \therefore / \therefore Func(f \cup g)$.*
2. *$Compatible(f, g) \therefore / \therefore f \restriction (dom(f) \cap dom(g)) = g \restriction (dom(f) \cap dom(g))$.*

Proof Is Left As An Exercise To The Reader.😃

Theorem 5.3.4. *$Compatible(F) / \therefore (1), (2), (3)$*

1. *$Func(\bigcup F)$.*
2. *$dom(\bigcup F) = \bigcup\{dom(f) \mid f \in F\}$.*
3. *$\forall f \in F \quad Compatible(f, \bigcup F)$.*

Proof Is Left As An Exercise To The Reader.😃

definition 5.3.8. ***Function from set to set*** *we call a function **f** from **A** to **B** when it's domain is **A** and it's range is subset of **B** i.e.*

$$(f : A \to B) : Func(f) \wedge Dom(f) = A \wedge Range(f) \subseteq B$$

definition 5.3.9. ***Set of functions from set to set*** *we denote set of all "$f : A \to B$" with "B^A" i.e.*

$$B^A : \{f \mid f : A \to B\}$$

definition 5.3.10. ***Injective function(Relation)*** *we call a function(Relation) **f** injective when every output has at most one input i.e.*

$$Injective(f) : \forall y, x_1, x_2\Big((x_1, y) \in f \wedge (x_2, y) \in f \to x_1 = x_2\Big).$$

*we call a function **f** injective from **A** to **B** when it's function from **A** to **B** and it's injective i.e.*

$$(f : A \xrightarrow{inj} B) : (f : A \to B) \wedge Injective(f)$$

Theorem 5.3.5. *(schema)* $(InjectiveI, InjectiveE)$

$\begin{array}{l\|l} & xfz \\ & yfz \\ & \vdots \\ & x = y \end{array}$ $\therefore Injective(f) \qquad InjectiveI$	$Injective(f)$ tfr $\underline{sfr}$ $\therefore s = r \qquad InjectiveE$

Proof Is Left As An Exercise To The Reader.😃

definition 5.3.11. ***Surjective function*** *we call a function* ***f*** *from* ***A*** *to* ***B*** *when it's domain is* ***A*** *and it's range is equal to* ***B*** *i.e.*

$$(f : A \xrightarrow[surj]{} B) : (f : A \to B) \wedge Range(f) = B$$

or i.e.

$$(f : A \xrightarrow[surj]{} B) : (f : A \to B) \wedge (\forall y \in B \quad \exists x \in A \quad (x, y) \in f)$$

definition 5.3.12. ***Bijection function*** *we call a function* ***f*** *from* ***A*** *to* ***B*** *when it's domain is* ***A*** *and it's range is equal to* ***B*** *i.e.*

$$(f : A \xrightarrow[surj]{inj} B) : (f : A \xrightarrow{inj} B) \wedge (f : A \xrightarrow[surj]{} B)$$

5.4 Family of sets

Attention 5.4.1. ***Indexes*** *are objects (like numbers) allows us to give a unique "tag" to every object. we use $i, j, k, ...$ for variable indexes and we use $m, n, l, ...$ for constant indexes.*

definition 5.4.1. ***Family of sets*** *assume $A : I \to \mathcal{F}$ is a function from a nonempty set of indices I to a family of sets $\mathcal{F}$, then we call $(Range[A])$ an indexed family of sets. we can write A_i instead of $A(i)$ and we can write $(A_i)_{i\in I}$ instead of "A" and $\{A_i\}_{i\in I}$ instead of "$Range[A]$" or "$A[I]$".*

$$(A_i)_{i\in I} := A$$

and

$$\{A_i\}_{i\in I} := A[I]$$

definition 5.4.2. ***Intersrction and union of an family of sets***

$$\bigcap_{i\in I} A_i := \bigcap \{A_i\}_{i\in I}$$

$$\bigcup_{i\in I} A_i := \bigcup \{A_i\}_{i\in I}$$

Attention 5.4.2. *we can write $\forall i \in I \quad P(i)$ instead of $\forall i(i \in I \to P(i))$ and we can write $\exists i \in I \quad P(i)$ instead of $\exists i(i \in I \wedge P(i))$.*

Lemma 5.4.1.

$$/ \therefore \bigcap_{i \in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$$

1	$x \in \bigcap_{i \in I} A_i$	acp
2	$x \in \bigcap \{A_i\}_{i \in I}$	$def.1$
3	$x \in \bigcap A[I]$	$def.1$
4	$\forall Y \in A[I] \quad x \in Y$	$def \bigcap$
5	$\forall Y ((\exists i \in I \quad A_i = Y) \to x \in Y)$	$def image$
6	$(\exists i \in I \quad A_i = A_{i'}) \to x \in A_{i'}$	$\forall E$
7	$\quad i' \in I$	acp
8	$\quad A_{i'} = A_{i'}$	$= I$
9	$\quad i' \in I \wedge A_{i'} = A_{i'}$	$\wedge I$
10	$\quad \exists i \in I \quad A_i = A_{i'}$	$\exists I$
11	$\quad x \in A_{i'}$	$\to E$
12	$i' \in I \to x \in A_{i'}$	$\to I$
13	$\forall i \in I \quad x \in A_i$	$\forall I$
14	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	def

1	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	*acp*
2	$\forall i \in I \quad x \in A_i$	*def*
3	$\exists i \in I \quad A_i = Y$	*acp*
4	$i \in I$	*acp*
5	$x \in A_i$	$\forall E$
6	$A_i = Y$	$\wedge E$
7	$x \in Y$	$= E$
8	$x \in Y$	$\exists E$
9	$(\exists i \in I \quad A_i = Y) \to x \in Y$	$\to I$
10	$\forall Y((\exists i \in I \quad A_i = Y) \to x \in Y)$	$\forall I$
11	$\forall Y \in A[I] \quad x \in Y$	*def*
12	$x \in \bigcap A[I]$	*def*
13	$x \in \bigcap \{A_i\}_{i \in I}$	*def*
14	$x \in \bigcap_{i \in I} A_i$	*def*
15	$\bigcap_{i \in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$	$= I$

$$/ \therefore \bigcup_{i \in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$$

1	$\quad x \in \bigcup_{i\in I} A_i$	*acp*
2	$\quad x \in \bigcup \{A_i\}_{i\in I}$	$def.1$
3	$\quad x \in \bigcup A[I]$	$def.1$
4	$\quad \exists Y \in A[I] \quad x \in Y$	$def \bigcup$
5	$\quad \exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$def.image$
6	$\quad\quad x \in Y$	*acp*
7	$\quad\quad \exists i \in I \quad A_i = Y$	*acp*
8	$\quad\quad\quad i \in I$	*acp*
9	$\quad\quad\quad A_i = Y$	*acp*
10	$\quad\quad\quad x \in A_i$	$=E$
11	$\quad\quad\quad \exists i \in I \quad x \in A_i$	$\exists I$
12	$\quad\quad \exists i \in I \quad x \in A_i$	$\exists E$
13	$\quad \exists i \in I \quad x \in A_i$	$\exists E$
14	$\quad x \in \{x \mid \exists i \in I \quad x \in A_i\}$	def

1	$\quad x \in \{x \mid \exists i \in I \quad x \in A_i\}$	*acp*
2	$\quad \exists i \in I \quad x \in A_i$	def
3	$\quad\quad i' \in I$	*acp*
4	$\quad\quad x \in A_{i'}$	*acp*
5	$\quad\quad A_{i'} = A_{i'}$	$=I$
6	$\quad\quad \exists i \in I \quad A_i = A_{i'}$	$\exists I$
7	$\quad\quad (\exists i \in I \quad A_i = A_{i'}) \wedge x \in A_{i'}$	$\wedge I$
8	$\quad\quad \exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$\exists I$
9	$\quad \exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$\exists E$
10	$\quad \exists Y \in A[I] \quad x \in Y$	def
11	$\quad x \in \bigcup A[I]$	def
12	$\quad x \in \bigcup \{A_i\}_{i\in I}$	def
13	$\quad x \in \bigcup_{i\in I} A_i$	def
14	$\bigcup_{i\in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$	$=I$

Theorem 5.4.1. (*schema*) ($\bigcap_I I, \bigcap_I E, \bigcup_I I, \bigcup_I E$)

$\begin{array}{\|l} i \in I \\ \vdots \\ t \in A_i \end{array}$ $\therefore t \in \bigcap_{i \in I} A_i \qquad \bigcap_I I$	$t \in \bigcap_{i \in I} A_i$ $n \in I$ $\overline{\therefore t \in A_n \qquad \bigcap_I E}$
$t \in A_n$ $n \in I$ $\overline{\therefore t \in \bigcup_{i \in I} A_i \qquad \bigcup_I I}$	$t \in \bigcup_{i \in I} A_i$ $\begin{array}{\|l} i \in I \\ t \in A_i \\ \vdots \\ \theta \end{array}$ $\therefore \theta \qquad \bigcup_I E$

Proof Is Left As An Exercise To The Reader.😃

5.5 Structures and morphisms

definition 5.5.1. ***Structure*** *"$\mathfrak{M}$" is a tuple defined like this.*

$$\mathfrak{M} := (M, f_0, f_1, ..., f_n, R_0, R_1, ..., R_m, c_0, c_1, ..., c_k)$$

*"M" is a **nonempty** set shows our universe "$\forall i \in \{0, ..., n\} \quad f_i : M^{n'} \to M$" are functions and " $\forall i \in \{0, ..., m\} \quad R_i \subseteq M^{m'}$" are relations and "$\forall i \in \{0, ..., k\} \quad c_i \in M$" are constants. ($M^n$ means $\overbrace{M \times M \times ... \times M}^{n \ times}$ we will define it later) and for the case we have arbitrary family of functions and relations and constants.*

$$\mathfrak{M} := (M, \{f_i\}_{i \in I_f}, \{R_i\}_{i \in I_R}, \{c_i\}_{i \in I})$$

definition 5.5.2. ***Homorphism*** *if "$\mathfrak{M} := (M, \{f_{i,m}\}_{i \in I_f}, \{R_{i,m}\}_{i \in I_R}, \{c_{i,m}\}_{i \in I})$" and "$\mathfrak{N} := (N, \{f_{i,n}\}_{i \in I_f}, \{R_{i,n}\}_{i \in I_R}, \{c_{i,n}\}_{i \in I})$" are two structures. function "$h : M \to N$"is called "(weak) Homorphism" when this conditions holds*

1. *for eveery relation "$R_{i,m}$"*

$$\forall x_0, x_1, x_2, ..., x_k \in M \quad R_{i,m}(x_0, x_1, x_2, ..., x_k) \to R_{i,n}(h(x_0), h(x_1), h(x_2), ..., h(x_k))$$

.

2. *for eveery function "$f_{i,m}$"*

$$\forall x_0, x_1, x_2, ..., x_k \in M \quad h(f_{i,m}(x_0, x_1, x_2, ..., x_k)) = f_{i,n}(h(x_0), h(x_1), h(x_2), ..., h(x_k))$$

.

3. *for eveery constant "$c_{i,m}$"*
$$h(c_{i,m}) = c_{i,n}$$
.

in condition (1) if "$\leftrightarrow$" be instead of "$\rightarrow$" then it's called "(strong) Homorphism"

few notices: when we are talking about "Homorphism" we usually talk about "(strong) Homorphism" .
in the case (in definition "(strong) Homorphism") we denote it by "$Hom(h, \mathfrak{M}, \mathfrak{N})$". if exists function "$h$" "$Hom(h, \mathfrak{M}, \mathfrak{N})$" we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are homomorphic. we denote it by "$Hom(\mathfrak{M}, \mathfrak{N})$".
if "$h : M \xrightarrow{inj} N$" we call it "Monomorphism" or "Embedding".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are momomorphic.
if "$h : M \xrightarrow[surj]{} N$" we call it "Epimorphism".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are Eomomorphic.
if "$h : M \xrightarrow[surj]{inj} N$" we call it "Isomorphism".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are Isomomorphic. and denote it by "$\mathfrak{M} \cong \mathfrak{N}$".
if "$k : M \rightarrow M$" is a function and "$Hom(k, \mathfrak{M}, \mathfrak{M})$". we call "$k$" "Indomorphism".
if "$k : M \xrightarrow[surj]{inj} M$" is a bijection and "$Hom(k, \mathfrak{M}, \mathfrak{M})$". we call "$k$" "Automorphism".

5.6 Equivalences and Partitions

definition 5.6.1. ***Equivalence relation*** *assume a structure "$\mathfrak{A} := (A, \sim)$"("$\sim$" is a relation). we say "$\sim$" is a "Equivalence relation" in "$\mathfrak{A}$" if it's reflexive,symmetric, and transitive.*

$$(reflexive) : \forall x \in A \quad x \sim x$$

$$(symmetric) : \forall x, y \in A \quad x \sim y \rightarrow y \sim x$$

$$(transitive) : \forall x, y, z \in A \quad x \sim y \wedge y \sim z \rightarrow x \sim z$$

Lemma 5.6.1. *"$\cong$" is a equivalence relation in "$(Structure, \cong)$"*
Proof Is Left As An Exercise To The Reader.😃
definition 5.6.2. ***Equivalence class of "x"*** *Let "$\sim$" be an equivalence relation on "$\mathfrak{A}$". For every "$x \in A$"*

$$[x]_\sim := \{y \in A \mid y \sim x\}$$

you can also see it as a function $[\_]_\sim : A \rightarrow V$.
Theorem 5.6.1. *(schema)* $([]_\sim I, []_\sim E)$

$\dfrac{t \sim s}{\therefore t \in [s]_\sim} \quad []_\sim I$	$\dfrac{t \in [s]_\sim}{\therefore t \sim s} \quad []_\sim E$

Proof Is Left As An Exercise To The Reader.😃
Lemma 5.6.2. *For every "$a, b \in A$"*

1. $a \sim b \therefore / \therefore [a]_\sim = [b]_\sim$
2. $a \nsim b \therefore / \therefore [a]_\sim \cap [b]_\sim = \varnothing$

prove.1

$/ \therefore$

1	$a \sim b$	*ass*
2	$\quad\vert\ x \in [a]_\sim$	*acp*
3	$\quad\vert\ x \sim a$	$[]_\sim E$
4	$\quad\vert\ x \sim b$	*trans*
5	$\quad\vert\ x \in [b]_\sim$	$[]_\sim I$
6	$\quad\vert\ x \in [b]_\sim$	*acp*
7	$\quad\vert\ x \sim b$	$[]_\sim E$
8	$\quad\vert\ x \sim a$	*trans*
9	$\quad\vert\ x \in [a]_\sim$	$[]_\sim I$
10	$[a]_\sim = [b]_\sim$	

$\therefore /$

1	$[a]_\sim = [b]_\sim$	*ass*
2	$a \sim a$	*reflex*
3	$a \in [a]_\sim$	$[]_\sim I$
4	$a \in [b]_\sim$	$= E$
5	$a \sim b$	$[]_\sim E$

prove.2

$/ \therefore$

1	$a \nsim b$	*ass*
2	$\quad\vert\ [a]_\sim \cap [b]_\sim \neq \varnothing$	*acp*
3	$\quad\vert\quad\vert\ x \in [a]_\sim \cap [b]_\sim$	*acp*
4	$\quad\vert\quad\vert\ x \in [a]_\sim$	$\cap E$
5	$\quad\vert\quad\vert\ x \in [b]_\sim$	$\cap E$
6	$\quad\vert\quad\vert\ x \sim a$	$[]_\sim E$
7	$\quad\vert\quad\vert\ x \sim b$	$[]_\sim E$
8	$\quad\vert\quad\vert\ a \sim b$	*trans*
9	$\quad\vert\quad\vert\ \perp$	
10	$\quad\vert\ \perp$	
11	$[a]_\sim \cap [b]_\sim = \varnothing$	

$\therefore /$

1	$[a]_\sim \cap [b]_\sim = \varnothing$	
2	$\quad\vert\ a \sim b$	*acp*
3	$\quad\vert\ a \sim a$	*reflex*
4	$\quad\vert\ a \in [a]_\sim$	$[]_\sim I$
5	$\quad\vert\ a \in [b]_\sim$	$[]_\sim I$
6	$\quad\vert\ a \in [a]_\sim \cap [b]_\sim$	$\cap I$
7	$\quad\vert\ \perp$	
8	$a \nsim b$	

definition 5.6.3. ***Partition*** *is a "set of sets" like "P" of a set like "A" with this conditions.*

1. $\forall X \in P \quad X \neq \varnothing$
2. $\forall X, Y \in P \quad X \neq Y \to X \cap Y = \varnothing$
3. $\bigcup P = A$

if these condition holds then we say "$Part(P, A)$".

definition 5.6.4. ***Set of Equivalence classes***

$$A/\sim := \{[x]_\sim \mid x \in A\}$$

Theorem 5.6.2. *if "$\sim$" is a equivalence class in "$(A, \sim)$" then "$A/\sim$" is a partition for "A" i.e. "$Part(A/\sim, A)$".*

1	$\quad\vert\ [x]_\sim \in A/\sim$	*acp*
2	$\quad\vert\ x \sim x$	*reflex*
3	$\quad\vert\ x \in [x]_\sim$	$[]_\sim I$
4	$\quad\vert\ [x]_\sim \neq \varnothing$	$\neq \varnothing I$
5	$\forall [x]_\sim \in A/\sim \quad [x]_\sim \neq \varnothing$	

1	$\quad\vert\ [x]_\sim \neq [y]_\sim$	*acp*
2	$\quad\vert\ x \nsim y$	*Lemma*5.6.2.1
3	$\quad\vert\ [x]_\sim \cap [y]_\sim = \varnothing$	*Lemma*5.6.2.2
4	$\forall [x]_\sim, [y]_\sim \in A/\sim \quad [x]_\sim \neq [y]_\sim \to [x]_\sim \cap [y]_\sim = \varnothing$	

1	$\quad\vert\ x \in \bigcup(A/\sim)$	*acp*
2	$\quad\vert\quad\vert\ [y]_\sim \in A/\sim$	*acp*
3	$\quad\vert\quad\vert\ x \in [y]_\sim$	*acp*
4	$\quad\vert\quad\vert\ [y]_\sim \subseteq A$	
5	$\quad\vert\quad\vert\ x \in A$	$\subseteq E$
6	$\quad\vert\ x \in A$	$\bigcup E$
7	$\quad\vert\ x \in A$	
8	$\quad\vert\ x \in [x]_\sim$	
9	$\quad\vert\ [x]_\sim \in (A/\sim)$	
10	$\quad\vert\ x \in \bigcup(A/\sim)$	$\bigcup I$
11	$A = \bigcup(A/\sim)$	$= I$

Theorem 5.6.3. *if "P" is a partition for "A" we define*

$$\sim_P := \{(x, y) \mid \exists Y \in P \quad (x \in Y \land y \in Y)\}$$

then "$\sim_P$" is a equivalence class in "$(A, \sim_P)$".

1	$\quad \mid x \in A$	*acp*
2	$\quad \mid x \in \bigcup P$	*Def*
3	$\quad \mid \quad \mid x \in Y$	*acp*
4	$\quad \mid \quad \mid Y \in P$	*acp*
5	$\quad \mid \quad \mid \exists Y \in P \quad (x \in Y \wedge x \in Y)$	
6	$\quad \mid \quad \mid x \sim_P x$	
7	$\quad \mid x \sim_P x$	
8	$\forall x \in A \quad x \sim_P x$	

1	$\quad \mid x, y \in A$	*acp*
2	$\quad \mid \quad \mid x \sim_P y$	*acp*
3	$\quad \mid \quad \mid \exists Y \in P \quad (x \in Y \wedge y \in Y)$	*Def*
4	$\quad \mid \quad \mid \quad \mid Y \in P \wedge x \in Y \wedge y \in Y$	*acp*
5	$\quad \mid \quad \mid \quad \mid Y \in P \wedge y \in Y \wedge x \in Y$	*commutative*$\wedge$
6	$\quad \mid \quad \mid \quad \mid \exists Y \in P \quad (y \in Y \wedge x \in Y)$	
7	$\quad \mid \quad \mid \quad \mid y \sim_P x$	
8	$\quad \mid \quad \mid y \sim_P x$	
9	$\quad \mid x \sim_P y \rightarrow y \sim_P x$	
10	$\forall x, y \in A \quad x \sim_P y \rightarrow y \sim_P x$	

1	$x, y, z \in A$	*acp*
2	$\quad x \sim_P y$	*acp*
3	$\quad y \sim_P z$	*acp*
4	$\quad \exists Y \in P \quad (x \in Y \wedge y \in Y)$	
5	$\quad \exists Y \in P \quad (y \in Y \wedge z \in Y)$	
6	$\quad\quad Y \in P \wedge x \in Y \wedge y \in Y$	
7	$\quad\quad\quad Y' \in P \wedge y \in Y' \wedge z \in Y'$	
8	$\quad\quad\quad Y \cap Y' \neq \varnothing$	
9	$\quad\quad\quad Y = Y'$	
10	$\quad\quad\quad Y \in P \wedge x \in Y \wedge z \in Y$	
11	$\quad\quad\quad \exists Y \in P \quad (x \in Y \wedge z \in Y)$	
12	$\quad\quad\quad x \sim_P z$	
13	$\quad\quad x \sim_P z$	
14	$\quad x \sim_P z$	
15	$x \sim_P y \wedge y \sim_p z \rightarrow x \sim_P z$	
16	$\forall x, y, z \in A \quad x \sim_P y \wedge y \sim_p z \rightarrow x \sim_P z$	

5.7 Orderings

definition 5.7.1. ***(partial) Order relation*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a relation). we say "$\preceq$" is a "(partial) Order relation" in "$\mathfrak{A}$" if it's reflexive,antisymmetric, and transitive.*

$$(reflexive) : \forall x \in A \quad x \preceq x$$

$$(antisymmetric) : \forall x, y \in A \quad x \preceq y \wedge y \preceq x \rightarrow x = y$$

$$(transitive) : \forall x, y, z \in A \quad x \preceq y \wedge y \preceq z \rightarrow x \preceq z$$

definition 5.7.2. ***(partial) Strict order relation*** *assume a structure "$\mathfrak{A} := (A, \prec)$"("$\prec$" is a relation). we say "$\prec$" is a "(partial) Strict order relation" in "$\mathfrak{A}$" if it's irreflexive, and transitive.*

$$(irreflexive) : \forall x \in A \quad x \nprec x$$

$$(transitive) : \forall x, y, z \in A \quad x \prec y \wedge y \prec z \rightarrow x \prec z$$

by two conditions bellow we can proof asymmetric too.

$$(asymmetric) : \forall x, y \in A \quad x \prec y \rightarrow y \nprec x$$

now we will show you relation between "(partial) Order relation" and "(partial) Strict order relation".

Theorem 5.7.1. .

1. *if "$\preceq$" is a "(partial) Order relation" in "$\mathfrak{A}$" define "$a \prec b : a \preceq b \wedge a \neq b$" prove that "$\prec$" is "(partial) Strict order relation".

1	$\mid x \in A$	*acp*
2	$\mid\quad\mid x \prec x$	*acp*
3	$\mid\quad\mid x \neq x$	$\wedge E$
4	$\mid\quad\mid x = x$	$= I$
5	$\mid\quad\mid \perp$	$\perp I$
6	$\mid x \not\prec x$	$\neg I$
7	$\forall x \in A \quad x \not\prec x$	$\to I, \forall I$

1	$\mid x, y, z \in A$	*acp*
2	$\mid\quad\mid x \prec y$	*acp*
3	$\mid\quad\mid y \prec z$	*acp*
4	$\mid\quad\mid x \preceq y$	$\wedge E$
5	$\mid\quad\mid y \preceq z$	$\wedge E$
6	$\mid\quad\mid x \preceq z$	*trans*
7	$\mid\quad\mid x \neq y$	$\wedge E$
8	$\mid\quad\mid y \neq z$	$\wedge E$
9	$\mid\quad\mid\quad\mid x = z$	*acp*
10	$\mid\quad\mid\quad\mid y \preceq x$	$= E$
11	$\mid\quad\mid\quad\mid x = y$	*antisym*
12	$\mid\quad\mid\quad\mid \perp$	$\perp I$
13	$\mid\quad\mid x \neq z$	$\neg I$
14	$\mid\quad\mid x \prec z$	$\wedge I$
15	$\mid x \prec y \wedge y \prec z \to x \prec z$	$\to I$
16	$\forall x, y, z \in A \quad x \prec y \wedge y \prec z \to x \prec z$	$\to I, \forall I$

2. *if "$\prec$" is a "(partial) Strict Order relation" in "$\mathfrak{A}$" define "$a \preceq b : a \prec b \vee a = b$" prove that "$\preceq$" is "(partial) order relation".

1	$\quad x \in A$	acp
2	$\quad x = x$	$= I$
3	$\quad x \preceq x$	$\vee I$
4	$\forall x \in A \quad x \preceq x$	$\rightarrow I, \forall I$

1	$\quad x, y \in A$	acp
2	$\quad\quad x \preceq y \wedge y \preceq x$	acp
3	$\quad\quad\quad x \prec y \wedge y \prec x$	acp
4	$\quad\quad\quad \bot$	$asym$
5	$\quad\quad\quad x = y$	
6	$\quad\quad\quad x \prec y \wedge y = x$	acp
7	$\quad\quad\quad x = y$	
8	$\quad\quad\quad x = y \wedge y \prec x$	acp
9	$\quad\quad\quad x = y$	
10	$\quad\quad\quad x = y \wedge y = x$	acp
11	$\quad\quad\quad x = y$	
12	$\quad\quad x = y$	$\vee E$
13	$\quad x \preceq y \wedge y \preceq x \rightarrow x = y$	$\rightarrow I$
14	$\forall x, y \in A \quad x \preceq y \wedge y \preceq x \rightarrow x = y$	$\rightarrow I, \forall I$

1	$x, y, z \in A$	acp
2	$\quad x \preceq y \wedge y \preceq z$	acp
3	$\quad\quad x \prec y \wedge y \prec z$	acp
4	$\quad\quad x \prec z$	$trans$
5	$\quad\quad x \preceq z$	$\vee I$
6	$\quad\quad x \prec y \wedge y = z$	acp
7	$\quad\quad x \prec z$	$= E$
8	$\quad\quad x \preceq z$	$\vee I$
9	$\quad\quad x = y \wedge y \prec z$	acp
10	$\quad\quad x \prec z$	$= E$
11	$\quad\quad x \preceq z$	$\vee I$
12	$\quad\quad x = y \wedge y = z$	acp
13	$\quad\quad x = z$	$= E$
14	$\quad\quad x \preceq z$	$\vee I$
15	$\quad x \preceq z$	$\vee E$
16	$x \preceq y \wedge y \prec z \rightarrow x \preceq z$	$\rightarrow I$
17	$\forall x, y, z \in A \quad x \preceq y \wedge y \prec z \rightarrow x \preceq z$	$\rightarrow I, \forall I$

definition 5.7.3. ***(Total-Linear) Order relation*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). we say "$\preceq$" is a "(Total-Linear) Order relation" in "$\mathfrak{A}$" if every two elements are comparable.*

$$(comparablity) : \forall x, y \in A \quad x \preceq y \vee y \preceq x$$

definition 5.7.4. ***(Total-Linear) Strict order relation*** *assume a structure "$\mathfrak{A} := (A, \prec)$"("$\prec$" is a Strict order relation). we say "$\prec$" is a "(Total-Linear) Strict order relation" in "$\mathfrak{A}$" if every two elements are comparable.*

$$(comparablity) : \forall x, y \in A \quad x \prec y \vee x = y \vee y \prec x$$

Theorem 5.7.2. *.*

1. *if "$\preceq$" is a "(Total-Linear) Order relation" in "$\mathfrak{A}$" define "$a \prec b : a \preceq b \wedge a \neq b$" prove that "$\prec$" is "(Total-Linear) Strict order relation".*
2. *if "$\prec$" is a "(Total-Linear) Strict Order relation" in "$\mathfrak{A}$" define "$a \preceq b : a \prec b \vee a = b$" prove that "$\preceq$" is "(Total-Linear) order relation".*

1	$x, y \in A$	*acp*
2	$x \preceq y \lor y \preceq x$	*linearity of $\preceq$ (1)*
3	$\quad x = y$	*acp*
4	$\quad x \prec y \lor x = y \lor y \prec x$	$\lor$*I (3)*
5	$\quad x \neq y$	*acp*
6	$\quad\quad x \preceq y$	*acp*
7	$\quad\quad x \preceq y \land x \neq y$	$\land$*I (7, 6)*
8	$\quad\quad x \prec y$	*def $\prec$ (8)*
9	$\quad\quad x \prec y \lor x = y \lor y \prec x$	$\lor$*I (9)*
10	$\quad\quad y \preceq x$	*acp*
11	$\quad\quad y \neq x$	*symmetry of $\neq$ (6)*
12	$\quad\quad y \preceq x \land y \neq x$	$\land$*I (12, 13)*
13	$\quad\quad y \prec x$	*def $\prec$ (14)*
14	$\quad\quad x \prec y \lor x = y \lor y \prec x$	$\lor$*I (15)*
15	$\quad x \prec y \lor x = y \lor y \prec x$	$\lor$*E (2, 11, 17)*
16	$x \prec y \lor x = y \lor y \prec x$	$\lor$*E (20, 5, 19)*
17	$\forall x, y \in A \quad x \prec y \lor x = y \lor y \prec x$	$\forall$*I (1-21)*

1	$x, y \in A$	*acp*
2	$x \prec y \lor x = y \lor y \prec x$	*linearity of $\prec$*
3	$\quad x \prec y \lor x = y$	*acp (Case 1 from 2)*
4	$\quad x \preceq y$	*def $\preceq$ (3)*
5	$\quad x \preceq y \lor y \preceq x$	$\lor$*I (4)*
6	$\quad y \prec x$	*acp (Case 2 from 2)*
7	$\quad y \prec x \lor y = x$	$\lor$*I (6)*
8	$\quad y \preceq x$	*def $\preceq$ (7)*
9	$\quad x \preceq y \lor y \preceq x$	$\lor$*I (8)*
10	$x \preceq y \lor y \preceq x$	$\lor$*E (2, 3-5, 6-9)*
11	$\forall x, y \in A \quad x \preceq y \lor y \preceq x$	$\forall$*I (1-10)*

Lemma 5.7.1. *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $B, C \subseteq A$ prove*

1. *$\exists b \in B \quad \forall x \in B \quad b \preceq x / \therefore \exists! b \in B \quad \forall x \in B \quad b \preceq x$ in this case we say "B" has least element or Minimum or "$HaveLeast(B)$"*
2. *$\exists c \in C \quad \forall x \in C \quad x \preceq c / \therefore \exists! c \in C \quad \forall x \in C \quad x \preceq c$ in this case we say "C" has greatest element or has Maximum or "$HaveGreatest(C)$"*

1	$\exists b \in B\ \forall x \in B (b \preceq x)$	*ass*
2	$b_1, b_2 \in B$	*acp (assumption for uniqueness)*
3	$(\forall x \in B \quad b_1 \preceq x) \land (\forall x \in B\ b_2 \preceq x)$	*acp*
4	$\forall x \in B \quad b_1 \preceq x$	*$\land E$ (3)*
5	$\forall x \in B \quad b_2 \preceq x$	*$\land E$ (3)*
6	$b_1 \preceq b_2$	*$\forall E$ (4, $x \to b_2$)*
7	$b_2 \preceq b_1$	*$\forall E$ (5, $x \to b_1$)*
8	$b_1 \preceq b_2 \land b_2 \preceq b_1$	*$\land I$ (6, 7)*
9	$b_1 = b_2$	*antisym of $\preceq$ (8)*
10	$((\forall x \in B \quad b_1 \preceq x) \land (\forall x \in B\ b_2 \preceq x)) \to b_1 = b_2$	*$\to I$ (3-9)*
11	$\exists! b \in B \quad \forall x \in B (b \preceq x)$	*$\exists!$ def (1, 10)*

1	$\exists c \in C\ \forall x \in C (x \preceq c)$	*ass*
2	$c_1, c_2 \in C$	*acp (assumption for uniqueness)*
3	$(\forall x \in C\ x \preceq c_1) \land (\forall x \in C\ x \preceq c_2)$	*acp*
4	$\forall x \in C\ x \preceq c_1$	*$\land E$ (3)*
5	$\forall x \in C\ x \preceq c_2$	*$\land E$ (3)*
6	$c_2 \preceq c_1$	*$\forall E$ (4, $x \to c_2$)*
7	$c_1 \preceq c_2$	*$\forall E$ (5, $x \to c_1$)*
8	$c_1 \preceq c_2 \land c_2 \preceq c_1$	*$\land I$ (7, 6)*
9	$c_1 = c_2$	*antisym of $\preceq$ (8)*
10	$((\forall x \in C\ x \preceq c_1) \land (\forall x \in C\ x \preceq c_2)) \to c_1 = c_2$	*$\to I$ (3-9)*
11	$\exists! c \in C\ \forall x \in C (x \preceq c)$	*$\exists!$ def (1, 10)*

Theorem 5.7.3. ***Minimum and Maximum*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $\mathcal{B} := \{B \subseteq A \mid HaveLeast(B)\}$ and $\mathcal{C} := \{C \subseteq A \mid HaveGreatest(C)\}$ we define like this*

$$min_{\mathfrak{A}} := \{(B, b) \in \mathcal{B} \times V \mid \forall x \in B \quad b \preceq x\}$$

$$max_{\mathfrak{A}} := \{(C, c) \in \mathcal{C} \times V \mid \forall x \in C \quad x \preceq c\}$$

we must prove these two relations are two functions "$min_{\mathfrak{A}} : \mathcal{B} \to V$" and "$max_{\mathfrak{A}} : \mathcal{C} \to V$".

1	$\quad B \in \mathcal{B}$	*acp*
2	$\quad \exists b \in B \quad \forall x \in B(b \preceq x)$	*def $\mathcal{B}$ (1)*
3	$\quad\quad b_0 \in B \wedge \forall x \in B(b_0 \preceq x)$	*acp (assumption for $\exists E$)*
4	$\quad\quad (B, b_0) \in min_A$	*def min_A (3)*
5	$\quad\quad \exists b \in V((B, b) \in min_A)$	*$\exists I$ (4)*
6	$\quad \exists b \in V((B, b) \in min_A)$	*$\exists E$ (2, 3-5)*
7	$\quad\quad b_1, b_2 \in V$	*acp*
8	$\quad\quad\quad (B, b_1) \in min_A \wedge (B, b_2) \in min_A$	*acp*
9	$\quad\quad\quad (\forall x \in B \quad b_1 \preceq x) \wedge (\forall x \in B\ b_2 \preceq x)$	*def min_A, $\wedge E$ (8)*
10	$\quad\quad\quad b_1 = b_2$	*Lemma 5.7.1 (uniqueness)*
11	$\quad\quad ((B, b_1) \in min_A \wedge (B, b_2) \in min_A) \to b_1 = b_2$	*$\to I$ (8-10)*
12	$\quad \exists! b \in V((B, b) \in min_A)$	*def $\exists!$ (6, 11)*
13	$\forall B \in \mathcal{B} \quad \exists! b \in V((B, b) \in min_A)$	*$\forall I$ (1-12)*

1	$\quad C \in \mathcal{C}$	*acp*
2	$\quad \exists c \in C\ \forall x \in C(x \preceq c)$	*def $\mathcal{C}$ (1)*
3	$\quad\quad c_0 \in C \wedge \forall x \in C(x \preceq c_0)$	*acp (assumption for $\exists E$)*
4	$\quad\quad (C, c_0) \in max_A$	*def max_A (3)*
5	$\quad\quad \exists c \in V((C, c) \in max_A)$	*$\exists I$ (4)*
6	$\quad \exists c \in V((C, c) \in max_A)$	*$\exists E$ (2, 3-5)*
7	$\quad\quad c_1, c_2 \in V$	*acp*
8	$\quad\quad\quad (C, c_1) \in max_A \wedge (C, c_2) \in max_A$	*acp*
9	$\quad\quad\quad (\forall x \in C \quad x \preceq c_1) \wedge (\forall x \in C\ x \preceq c_2)$	*def max_A, $\wedge E$ (8)*
10	$\quad\quad\quad c_1 = c_2$	*Lemma 5.7.1 (uniqueness)*
11	$\quad\quad ((C, c_1) \in max_A \wedge (C, c_2) \in max_A) \to c_1 = c_2$	*$\to I$ (8-10)*
12	$\quad \exists! c \in V((C, c) \in max_A)$	*def $\exists!$ (6, 11)*
13	$\forall C \in \mathcal{C} \quad \exists! c \in V((C, c) \in max_A)$	*$\forall I$ (1-12)*

definition 5.7.5. ***Minimal and Maximal*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). we define "Minimal and Maximal" like this*

$$Minimal_{\mathfrak{A}}(b) : \forall x \in A \quad x \not\prec b$$

i.e. "b" is the minimal in "$\mathfrak{A}$".

$$Maximal_{\mathfrak{A}}(c) : \forall x \in A \quad c \not\prec x$$

i.e. "c" is the maximal in "$\mathfrak{A}$".
definition 5.7.6. ***Well ordering*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). we say "$\mathfrak{A}$ " is "Well ordered" when every nonempty subset of "A " has a "Minimum " i.e.*

$$WellOrdered(\mathfrak{A}) : \forall X \subseteq A \quad X \neq \varnothing \rightarrow HaveLeast(X)$$

definition 5.7.7. ***Lower and Upper Bound*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). $B, C \subseteq A$. we define "Lower and Upper Bound" like this*

$$Lowerb_{\mathfrak{A}}(b, B) : \forall x \in B \quad b \preceq x$$

i.e. "b" is the lowerbound of B in "$\mathfrak{A}$".

$$Upperb_{\mathfrak{A}}(c, C) : \forall x \in C \quad x \preceq c$$

i.e. "C" is the upperbound of B in "$\mathfrak{A}$".

if the structure clear to us we don't write structure index "$min(\_), Minimal(\_), Lowerb(\_, \_)$". and we don't mention it.
Theorem 5.7.4. ***Infimum and Supremum*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $\mathcal{B} := \{B \subseteq A \mid HaveGreatest\{b \in A \mid Lowerb_{\mathfrak{A}}(b, B)\}\}$ and $\mathcal{C} := \{C \subseteq A \mid HaveLeast\{c \in A \mid Upperb_{\mathfrak{A}}(c, C)\}\}$ we define "$inf_{\mathfrak{A}} : \mathcal{B} \rightarrow V$" and "$sup_{\mathfrak{A}} : \mathcal{C} \rightarrow V$" like this*

$$inf_{\mathfrak{A}}(B) := max_{\mathfrak{A}}\{b \in A \mid Lowerb_{\mathfrak{A}}(b, B)\}$$

$$sup_{\mathfrak{A}}(B) := min_{\mathfrak{A}}\{c \in A \mid Upperb_{\mathfrak{A}}(c, C)\}$$

Theorem 5.7.5. *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $B \subseteq A$ prove*

1. $min_{\mathfrak{A}}(B) = b / \therefore inf_{\mathfrak{A}}(B) = b$.

1	$min_{\mathfrak{A}}(B) = b$	*ass*
2	$b \in B \wedge \forall y \in B(b \preceq y)$	*def* $min_{\mathfrak{A}}$ *(1)*
3	$b \in B$	$\wedge E$ *(2)*
4	$\forall y \in B(b \preceq y)$	$\wedge E$ *(2)*
5	$Lowerb_{\mathfrak{A}}(b, B)$	*def* $Lowerb_{\mathfrak{A}}$ *(4)*
6	$b \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}$	*set construction (5)*
7	$z \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}$	*acp*
8	$Lowerb_{\mathfrak{A}}(z, B)$	*set membership (7)*
9	$\forall y \in B(z \preceq y)$	*def* $Lowerb_{\mathfrak{A}}$ *(8)*
10	$z \preceq b$	$\forall E$ *(9,* $y \to b$ *using 3)*
11	$\forall z \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}(z \preceq b)$	$\forall I$ *(7-10)*
12	$b \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\} \wedge \forall z \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}(z \preceq b)$	$\wedge I$ *(6, 11)*
13	$max_{\mathfrak{A}}(\{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}) = b$	*def* $max_{\mathfrak{A}}$ *(12)*
14	$inf_{\mathfrak{A}}(B) = b$	*def* $inf_{\mathfrak{A}}$ *(13)*

2. $inf_{\mathfrak{A}}(B) = b, b \in B / \therefore min_{\mathfrak{A}}(B) = b$.

1	$inf_{\mathfrak{A}}(B) = b$	*ass*
2	$b \in B$	*ass*
3	$max_{\mathfrak{A}}(\{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}) = b$	*def* $inf_{\mathfrak{A}}$ *(2)*
4	$b \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\} \wedge \forall z \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}(z \preceq b)$	*def* $max_{\mathfrak{A}}$ *(4)*
5	$b \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}$	$\wedge E$ *(5)*
6	$Lowerb_{\mathfrak{A}}(b, B)$	*set membership (6)*
7	$\forall y \in B(b \preceq y)$	*def* $Lowerb_{\mathfrak{A}}$ *(7)*
8	$b \in B \wedge \forall y \in B(b \preceq y)$	$\wedge I$ *(3, 8)*
9	$min_{\mathfrak{A}}(B) = b$	*def* $min_{\mathfrak{A}}$ *(9)*

this theorem holds for "max, sup*" too.*

Chapter 6

The beginning of Abstract Algebra

we want to introduce some functions "$\bigcirc : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$" to you. we use "$m \bigcirc n$" instead of "$\bigcirc(m, n)$".

6.1 Group theory

definition 6.1.1. *A structure* $\mathfrak{G} := (G, \cdot)$

1. *If* $\forall x, y, z \in G \quad (xy)z = x(yz)$ *(associative* $\times$*),* $\mathfrak{G}$ *is a* ***semigroup****.*
2. *If* $\mathfrak{G}$ *is a* ***semigroup*** *and* $\forall x \in G \quad \exists y \in G \quad \forall x \in G \quad xy = yx = x$ *(has identity* $\times$*),* $\mathfrak{G}$ *is a* ***monoid****. you can prove for every* ***monoid*** $\mathfrak{G}$ *identity element is unique we denote it by "e"(in* $\times$ *with "*1*" and in* $+$ *with* 0*).*
3. *If* $\mathfrak{G}$ *is a* ***monoid*** *and* $\forall x \in G \quad \exists y \in G \quad xy = yx = e$ *(has inverse* $\times$*),* $\mathfrak{G}$ *is a group.*
4. *If* $\mathfrak{G}$ *is a* ***semigroup/monoid*** *and* $\forall x, y, z \in G \quad xz = yz \to x = y$ *(cancellative* $\times$*),* $\mathfrak{G}$ *is a* ***cancellative semigroup/monoid****. (you can prove every group is cancellative as an exercise.*😃*)*
5. *If* $\mathfrak{G}$ *is a* ***semigroup/monoid/group*** *and* $\forall x, y \in G \quad xy = yx$ *(commutative(abelian)* $\times$*),* $\mathfrak{G}$ *is a* ***commutative(abelian) semigroup/monoid/group****.*

Theorem 6.1.1 (Grothendieck Group Construction)**.** *Let* $(S, +, 0)$ *be a cancellative commutative monoid.*

Then there exists an abelian group $G(S)$ *together with a monoid mbedding*

$$\eta : S \longrightarrow G(S)$$

such that for every abelian group A *and every monoid mbedding*

$$f : S \to A,$$

there exists a unique group mbedding

$$\tilde{f} : G(S) \to A$$

satisfying

$$\tilde{f} \circ \eta = f.$$

The canonical map is

$$\eta(a) = [(a, 0)].$$

Proof. **Step 0: Define relation.**

Define a relation $\sim$ on $S \times S$ by

$$(a, b) \sim (c, d) \leftrightarrow a + d = b + c.$$

Claim 0: $\sim$ is an equivalence relation.

Reflexivity

1	$(a, b) \in S \times S$	
2	$a + b = b + a$	
3	$(a, b) \sim (a, b)$	

Symmetry

1	$(a, b) \sim (c, d)$	
2	$a + d = b + c$	
3	$c + b = d + a$	
4	$(c, d) \sim (a, b)$	

Transitivity

1	$(a, b) \sim (c, d)$	
2	$(c, d) \sim (e, f)$	
3	$a + d = b + c$	
4	$c + f = d + e$	
5	$a + d + c + f = b + c + d + e$	add and combine, 3, 4
6	$a + f + (c + d) = b + e + (c + d)$	rearrange, 5
7	$a + f = b + e$	cancel $+(c + d)$, 6
8	$(a, b) \sim (e, f)$	

Step 1: Define Grothendieck group

$$G(S) := (S \times S)/\sim, \quad [(a,b)] \in G(S).$$

Step 2: Define operation

$$[(a,b)] + [(c,d)] := [(a+c, b+d)]$$

$$-[(a,b)] := [(b,a)]$$

—

Step 3: Well-definedness

(Addition and inverse are well-defined by compatibility of $\sim$ with $+$.)

—

Step 4: Group structure

(a,b)] + [(b,a)] = [(a+b, a+b)][(b,a)] = [(a+b, a+b)] (a+b,a+b)] = 01	+
0 2	=
3	every element has inverse
4	$G(S)$ is an abelian group

—

Step 5: Define extension

Let $f : S \to A$ be a monoid mbedding into an abelian group A.

$$\tilde{f}([(a,b)]) := f(a) - f(b).$$

—

Step 6: Well-definedness

1	$(a,b) \sim (c,d)$
2	$a + d = b + c$
3	$f(a) + f(d) = f(b) + f(c)$
4	$f(a) - f(b) = f(c) - f(d)$

—

Step 7: Homomorphism property

(Straightforward from group operations in A.)

—

Step 8: Factorization

$$\tilde{f}([(a,0)]) = f(a)$$

—

Step 9: Uniqueness

Any mbedding is forced by:
$$[(a,b)] = [(a,0)] - [(b,0)].$$

□

6.2 Ring theory

definition 6.2.1. *A structure* $\mathfrak{R} := (R, +, \cdot)$

1. *If* $(R,+)$ *is a abelian* ***monoid/group*** *and* $\forall x,y,z \in R \quad (xy)z = x(yz)$ *(associative* $\times$*),* $\forall x,y,z \in R \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive* $\times$*),* $\mathfrak{R}$ *is a* ***Semiring/Ring****.*

2. *If* $\mathfrak{R}$ *is a* ***Semiring/Ring*** *and* $\exists y \in R \quad \forall x \in R \quad xy = yx = x$ *(has identity* $\times$*),* $\mathfrak{R}$ *is a* ***Semiring/Ring with identity****. you can prove for every* ***Semiring/Ring with identity*** $\mathfrak{R}$ *identity element is unique we denote it by "1".*

3. *If* $\mathfrak{R}$ *is a* ***Semiring/Ring*** *and* $\forall x,y \in R \quad xy = 0 \to x = 0 \vee y = 0$ *(zero divisor free* $\times$*),* $\mathfrak{R}$ *is a* ***zero divisor free Semiring/Ring****. (you can prove every* ***division Semiring/Ring*** *is zero divisor free as an exercise.😀)*

4. *If* $\mathfrak{R}$ *is a* ***Semiring/Ring with identity*** *and* $\forall x \in R, \neq 0 \quad \exists y \in R \quad xy = yx = 1$ *(has inverse* $\times$*),* $\mathfrak{R}$ *is a* ***division Semiring/Ring*** *with.*

5. *If* $\mathfrak{R}$ *is a* ***Semiring/Ring*** *and* $\forall x,y \in R \quad xy = yx$ *(commutative(abelian)* $\times$*),* $\mathfrak{R}$ *is a* ***commutative(abelian) Semiring/Ring****.*

6. *If* $\mathfrak{R}$ *is a* ***zero divisor free Semiring/Ring*** *and* ***commutative(abelian) Semiring/Ring****.* $\mathfrak{R}$ *is a* ***Semidomain/Domain*** *.*

7. *If* $\mathfrak{R}$ *is a* ***division Semiring/Ring*** *and* ***commutative(abelian) Semiring/Ring****.* $\mathfrak{R}$ *is a* ***SemiField/Field*** *.*

Theorem 6.2.1 (Grothendieck Construction for Semirings)**.** *Let* $(S,+,\cdot)$ *be a semiring whose additive structure* $(S,+,0)$ *is a cancellative commutative monoid.*

Then there exists a ring $G(S)$ *together with a semiring mbedding*

$$\eta : S \longrightarrow G(S)$$

such that for every ring T *and every semiring mbedding*

$$f : S \to T,$$

there exists a unique ring mbedding

$$\tilde{f} : G(S) \to T$$

satisfying

$$\tilde{f} \circ \eta = f.$$

The canonical map is

$$\eta(a) = [(a, 0)].$$

Proof. **Step 0: Define the underlying relation.**

Define a relation $\sim$ on $S \times S$ by

$$(a, b) \sim (c, d) \leftrightarrow a + d = b + c.$$

Claim 0: $\sim$ is an equivalence relation.

Reflexivity

1	$(a, b) \in S \times S$
2	$a + b = b + a$
3	$(a, b) \sim (a, b)$

Symmetry

1	$(a, b) \sim (c, d)$
2	$a + d = b + c$
3	$c + b = d + a$
4	$(c, d) \sim (a, b)$

Transitivity

1	$(a, b) \sim (c, d)$	
2	$(c, d) \sim (e, f)$	
3	$a + d = b + c$	
4	$c + f = d + e$	
5	$a + d + c + f = b + c + d + e$	add and combine, 3, 4
6	$a + f + (c + d) = b + e + (c + d)$	rearrange, 5
7	$a + f = b + e$	cancel $+(c + d)$, 6
8	$(a, b) \sim (e, f)$	

(additive commutative monoid cancellation NOT required; we instead justify via rearrangement in monoid equality)

Step 1: Define the carrier set.

Define

$$G(S) := (S \times S)/\sim, \quad [(a,b)] \in G(S).$$

Step 2: Define operations.

Define

$$[(a,b)] + [(c,d)] := [(a+c,\ b+d)],$$
$$[(a,b)] \cdot [(c,d)] := [(ac+bd,\ ad+bc)].$$

Define

$$\eta(a) := [(a,0)].$$

Step 3: Well-definedness of addition.

1	$(a,b) \sim (a',b'),\ (c,d) \sim (c',d')$
2	$a+b' = a'+b,\ c+d' = c'+d$
3	$(a+c)+(b'+d') = (a'+c')+(b+d)$
4	$[(a+c, b+d)] = [(a'+c', b'+d')]$
5	$+$ is well-defined

Step 4: Well-definedness of multiplication.

1	$(a,b) \sim (a',b'),\ (c,d) \sim (c',d')$
2	$a+b' = a'+b,\ c+d' = c'+d$
3	$ac+bd+(a'd'+b'c') = a'c'+bd+ac'+b'd$
4	$[(ac+bd, ad+bc)] = [(a'c'+b'd', a'd'+b'c')]$
5	$\cdot$ is well-defined

Step 5: Ring structure.

1	$[(a,b)] + [(c,d)] = [(a+c, b+d)]$
2	associativity inherited from S
3	$[(a,b)] \cdot [(c,d)]$ associative
4	$distributivityfollowsfromsemiringlaws$
5	$(0,0)$ is additive identity
6	$[(a,a)] = 0$ in $G(S)$
7	$G(S)$ is a ring

Step 6: Define extension of a mbedding.

Let

$$f : S \to T$$

be a semiring mbedding into a ring T.

Define

$$\tilde{f}([(a,b)]) := f(a) - f(b).$$

Claim 6: $\tilde{f}$ is well-defined.

1 $(a,b) \sim (c,d)$

2 $a + d = b + c$

3 $f(a) + f(d) = f(b) + f(c)$

4 $f(a) - f(b) = f(c) - f(d)$

5 $\tilde{f}([(a,b)]) = \tilde{f}([(c,d)])$

Step 7: $\tilde{f}$ is a ring mbedding.

1 $\tilde{f}([(a,b)] + [(c,d)]) = f(a+c) - f(b+d)$

2 $= (f(a) - f(b)) + (f(c) - f(d))$

3 $= \tilde{f}([(a,b)]) + \tilde{f}([(c,d)])$

4 $\tilde{f}([(a,b)] \cdot [(c,d)]) = f(ac + bd) - f(ad + bc)$

5 $= (f(a) - f(b))(f(c) - f(d))$

6 $= \tilde{f}([(a,b)])\tilde{f}([(c,d)])$

Step 8: Factorization property.

1 $\tilde{f}(\eta(a)) = \tilde{f}([(a,0)])$

2 $= f(a)$

3 $\tilde{f} \circ \eta = f$

Step 9: Uniqueness.

1 $g : G(S) \to T$ ring mbedding

2 $g \circ \eta = f$

3 $g([(a,b)]) = g([(a,0)]) - g([(b,0)])$

4 $= f(a) - f(b)$

5 $= \tilde{f}([(a,b)])$

6 $g = \tilde{f}$

Therefore $G(S)$ satisfies the universal property of the Grothendieck construction.

$\square$

Theorem 6.2.2 (Field of Fractions)**.** *Let R be a integral domain.*

Then there exists a field $\mathrm{Frac}(R)$ *together with a ring embedding*

$$\eta : R \longrightarrow \mathrm{Frac}(R)$$

such that for every field K and every ring embedding

$$f : R \to K,$$

there exists a unique field embedding

$$\tilde{f} : \mathrm{Frac}(R) \to K$$

satisfying

$$\tilde{f} \circ \eta = f.$$

The canonical map is

$$\eta(a) = \frac{a}{1}.$$

Proof. **Step 0: Define relation.**

Define a relation $\sim$ on $R \times (R \setminus \{0\})$ by

$$(a, b) \sim (c, d) \leftrightarrow ad = bc.$$

Claim 0: $\sim$ is an equivalence relation.

Reflexivity

1	$(a, b) \in R \times (R \setminus \{0\})$
2	$ab = ba$
3	$(a, b) \sim (a, b)$

Symmetry

1	$(a, b) \sim (c, d)$
2	$ad = bc$
3	$cb = da$
4	$(c, d) \sim (a, b)$

Transitivity

1	$ad = bc$	
2	$cf = de$	
3	$adf = bcf$	multiply by f, 1
4	$bcf = bde$	substitute, 2
5	$adf = bde$	
6	$(af)d = (be)d$	rearrange
7	R is an integral domain	
8	$d \neq 0$	
9	$af = be$	cancellation (no zero divisors)
10	$(a, b) \sim (e, f)$	

—

Step 1: Define field of fractions

$$\mathrm{Frac}(R) := (R \times (R \setminus \{0\}))/\sim$$

$$[(a, b)] := \frac{a}{b}$$

—

Step 2: Define operations

$$[(a, b)] + [(c, d)] := [(ad + bc, bd)]$$

$$[(a, b)] \cdot [(c, d)] := [(ac, bd)]$$

—

Step 3: Define embedding

$$\eta(a) := [(a, 1)]$$

—

Step 4: Well-definedness

(Follows from cross-multiplication and compatibility of $\sim$.)

—

Step 5: Field structure

1	$[(a,b)] \cdot [(b,a)] = [(ab,ab)]$
2	$[(ab,ab)] = 1$
3	every nonzero element has inverse
4	$\mathrm{Frac}(R)$ is a field

—

Step 6: Extension map

Let $f : R \to K$ be a ring embedding into a field K.

$$\tilde{f}([(a,b)]) := \frac{f(a)}{f(b)}$$

—

Step 7: Well-definedness

1	$(a,b) \sim (c,d)$
2	$ad = bc$
3	$f(a)f(d) = f(b)f(c)$
4	$\frac{f(a)}{f(b)} = \frac{f(c)}{f(d)}$

—

Step 8: Homomorphism and factorization

$$\tilde{f}(\eta(a)) = \tilde{f}([(a,1)]) = f(a)$$

—

Step 9: Uniqueness

$$[(a,b)] = \frac{[(a,1)]}{[(b,1)]} \Rightarrow \tilde{f} \text{ is forced}$$

□

Chapter 7

Natural numbers

7.1 Introduction to Natural numbers

definition 7.1.1. ***Natural numbers*** *we define "$0, 1, 2, 3, \ldots$" like this.*

$$\begin{aligned}
&0 := \varnothing \\
&1 := \{0\} = \varnothing \cup \{\varnothing\} = \{\varnothing\} \\
&2 := \{0, 1\} = \{\varnothing\} \cup \big\{\{\varnothing\}\big\} = \big\{\varnothing, \{\varnothing\}\big\} \\
&3 := \{0, 1, 2\} = \Big\{\varnothing, \{\varnothing\}, \big\{\varnothing, \{\varnothing\}\big\}\Big\} \\
&\vdots
\end{aligned}$$

definition 7.1.2. ***Successor function*** *is a function "$s : V \to V$" defined like this.*

$$s(x) := x \cup \{x\}$$

you can see $\forall x \quad Type_0(x) \to Type_0(s(x))$

definition 7.1.3. *we define a set* ***"Inductive"*** *like this.*

$$Ind(X) : \varnothing \in X \wedge \forall y (y \in X \to s(y) \in X)$$

notice that "$Ind(X)$" might be empty.

Axiom 7.1.1. ***The Axiom of Infinity*** *we define a set "Inductive" like this.*

$$\exists X \quad Type_0(X) \wedge Ind(X)$$

so "$Ind(X)$" is a nonempty set. so we can define

definition 7.1.4. ***Set of Natural numbers($\mathbb{N}$)***

$$\mathbb{N} := \bigcap Ind$$

Lemma 7.1.1. *prove.*

1. $\mathbb{N} = \{x \mid \underbrace{\forall Y\Big(\big(\underbrace{\varnothing \in Y \wedge \forall z(z \in Y \to s(z) \in Y)}_{Ind(Y)}\big) \to x \in Y\Big)}_{\phi(x)}\}$

2. $Ind(\mathbb{N})$
   Proof Is Left As An Exercise To The Reader.😃

notice that now "$\phi(x)$" is a proposition. so "$\{x \mid \forall Y\Big(\big(\varnothing \in Y \wedge \forall z(z \in Y \to s(z) \in Y)\big) \to x \in Y\Big)\}$" is a set.

7.2 Properties of Natural numbers

Theorem 7.2.1. ***Induction*** $P(0), \forall n \in \mathbb{N} \quad \Big(P(n) \to P(s(n))\Big) / \therefore \forall n \in \mathbb{N} \quad P(n)$
proof.

1	$S := \{x \in \mathbb{N} \mid P(x)\}$	def
2	$P(0)$	ass
3	$\forall n \in \mathbb{N} \quad \Big(P(n) \to P(s(n))\Big)$	ass
4	$\quad n \in \mathbb{N}$	acp
5	$\quad 0 \in S$	$1, 2$
6	$\quad\quad n \in S$	acp
7	$\quad\quad P(n)$	$comprhension$
8	$\quad\quad P(n) \to P(s(n))$	$\forall E, \to E, 3, 4$
9	$\quad\quad P(s(n))$	$\to E, 7, 8$
10	$\quad\quad s(n) \in S$	$4, 9$
11	$\quad \forall n\Big(n \in S \to s(n) \in S\Big)$	$1, 3$
12	$\quad Ind(S)$	$5, 6$
13	$\quad n \in \bigcap Ind$	def
14	$\quad n \in S$	$\bigcap E$
15	$\quad P(n)$	$comprhension$
16	$\forall n \in \mathbb{N} \quad P(n)$	$\to I, \forall I$

definition 7.2.1. *A set "T" is **Transitive** if $u \in v \in T$ implies $u \in T$.*

$$Transitive(T) : \forall u, v \quad u \in v \in T \to u \in T$$

Lemma 7.2.1. .

1. $\forall n \in \mathbb{N} \quad Transitive(n)$
2. $Transitive(\mathbb{N})$ 😀

proof.

1	$\forall v \quad v \notin 0$	
2	$\forall u, v \quad u \in v \in 0 \to u \in 0$	
3	$Transitive(0)$	
4	$n \in \mathbb{N}$	*acp*
5	$Transitive(n)$	*acp*
6	$u \in v \wedge v \in s(n)$	*acp*
7	$v \in n \cup \{n\}$	$\wedge E$
8	$v \in n$	$\wedge E$
9	$u \in n$	$5, 6, 8$
10	$u \in s(n)$	$\cup I$
11	$v \in \{n\}$	$\wedge E$
12	$v = n$	
13	$u \in n$	$= E$
14	$u \in s(n)$	$\cup I$
15	$u \in s(n)$	
16	$\forall u, v \quad u \in v \in s(n) \to u \in s(n)$	
17	$Transitive(s(n))$	
18	$\forall n \in \mathbb{N} \quad Transitive(n) \to Transitive(s(n))$	
19	$\forall n \in \mathbb{N} \quad Transitive(n)$	*Induction theorem*

Proof Is Left As An Exercise To The Reader.

definition 7.2.2. ***Order in Natural numbers($\mathbb{N}$)***

$$<:= \{(x, y) \in \mathbb{N} \times \mathbb{N} \mid x \in y\}$$

Theorem 7.2.2. ***Order of Natural numbers($\mathbb{N}$)*** *prove*

1. $(\mathbb{N}, <)$ *is linearly ordered.* 😀
2. $\forall n \in \mathbb{N} \quad (n, <)$ *is Well ordered.*
3. $(\mathbb{N}, <)$ *is Well ordered.*

4. $\Big(\forall x \in \mathbb{N} \quad \exists y \in \mathbb{N} \quad x < y\Big)$ *(doesn't have right endpoint)*

proof.

1	$0 \not< 0$	$def \varnothing$
2	$\vert\ n \in \mathbb{N}$	acp
3	$\vert\ \vert\ n \not< n$	acp
4	$\vert\ \vert\ \vert\ s(n) < s(n)$	acp
5	$\vert\ \vert\ \vert\ \vert\ s(n) \in n$	acp
6	$\vert\ \vert\ \vert\ \vert\ n \in s(n)$	$\cup I$
7	$\vert\ \vert\ \vert\ \vert\ n \in n$	$Transitive(\mathbb{N})$
8	$\vert\ \vert\ \vert\ \vert\ \bot$	$3, 7$
9	$\vert\ \vert\ \vert\ \vert\ s(n) \in \{n\}$	acp
10	$\vert\ \vert\ \vert\ \vert\ s(n) = n$	
11	$\vert\ \vert\ \vert\ \vert\ s(n) \not< s(n)$	$= E$
12	$\vert\ \vert\ \vert\ \vert\ \bot$	$4, 11$
13	$\vert\ \vert\ \vert\ \bot$	$\cup E$
14	$\vert\ \vert\ s(n) \not< s(n)$	$\neg I$
15	$\forall n \in \mathbb{N} \quad n \not< n$	*Induction theorem*

1	$\forall n, m, k \in \mathbb{N} \quad n < m < k \to n < k$	*Induction theorem*

Proof Is Left As An Exercise To The Reader.

Theorem 7.2.3. ***Induction ,second version*** $\Big(\forall k < n \quad P(k)\Big) \to P(n) / \therefore \forall n \in \mathbb{N} \quad P(n)$

proof.

7.3 The Recursion theorem

definition 7.3.1. ***Tuple (new definition)-Finite sequence*** *is a function "$a : n \to V$" s.t. $n \in \mathbb{N}$ we know we can show this by "$(a_i)_{i \in n}$" we can also show it by "$(a_0, a_1, a_2, ..., a_{n-1})$"*

definition 7.3.2. ***Sequence*** *is a function "$a : \mathbb{N} \to V$" we know we can show this by "$(a_i)_{i \in \mathbb{N}}$" we can also show it by "$(a_i)_{i=0}^{\infty}$"*

now we can redefine "Cartesian product". we define finite Cartesian product "$\prod_{i \in n} A_i := \{(a_i)_{i \in n} \mid a_i \in A_i\}$" and define infinite Cartesian product "$\prod_{i \in \mathbb{N}} A_i := \{(a_i)_{i \in \mathbb{N}} \mid a_i \in A_i\}$" and define Cartesian product in general way "$\prod_{i \in I} A_i := \{(a_i)_{i \in I} \mid a_i \in A_i\}$". and you can easily see that from the definition of "B^A", "A^n" is the set of all n-length finite sequences of $a \in A$ and "$A^{\mathbb{N}}$" is the set of all infinite sequences of $a \in A$.

Theorem 7.3.1 (Recursion)**.** *For any set A, any $a \in A$, and any function $g : A \times \mathbb{N} \to A$, there is a unique function $f : \mathbb{N} \to A$ such that:*

1. $f(0) = a$.

2. $\forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)$.

Proof. Define

$$Compute_n(t_n) : t_n : s(n) \to A \ \wedge \ t_n(0) = a \ \wedge \ \forall k < n \ \big(t(s(k)) = g(t(k), k)\big).$$

$$F := \Big\{ t_n \ \Big| \ Compute_n(t_n) \Big\}.$$

Claim 0: $\forall n \in \mathbb{N} \quad \exists t_n \in F$**.**

Claim 1: $F \neq \varnothing$**.**

1	$t_0 \in F$
2	$F \neq \varnothing$

Define

$$f := \bigcup F.$$

Claim 2: f **is a function.**

1	$\vert\ t_m, t_n \in F$
2	$\vert\ \vert\ Dom(t_m) = s(m)$
3	$\vert\ \vert\ Dom(t_n) = s(n)$
4	$\vert\ \vert\ \vert\ m \leq n$
5	$\vert\ \vert\ \vert\ t_m(0) = a$
6	$\vert\ \vert\ \vert\ t_n(0) = a$
7	$\vert\ \vert\ \vert\ t_m(0) = t_n(0)$
8	$\vert\ \vert\ \vert\ \vert\ k < m$
9	$\vert\ \vert\ \vert\ \vert\ \vert\ t_m(k) = t_n(k)$
10	$\vert\ \vert\ \vert\ \vert\ \vert\ g(t_m(k), k) = g(t_n(k), k)$
11	$\vert\ \vert\ \vert\ \vert\ \vert\ t_m(s(k)) = g(t_m(k), k)$
12	$\vert\ \vert\ \vert\ \vert\ \vert\ t_n(s(k)) = g(t_n(k), k)$
13	$\vert\ \vert\ \vert\ \vert\ \vert\ t_m(s(k)) = t_n(s(k))$
14	$\vert\ \vert\ \vert\ \vert\ \forall k \leq m \quad t_m(k) = t_n(k)$
15	$\vert\ \vert\ t_m \subseteq t_n \vee t_n \subseteq t_m$
16	$\forall t_m, t_n \in F \quad (t_m \subseteq t_n \vee t_n \subseteq t_m)$
17	$Compatible(F)$
18	$Func(f)$

Claim 3: Dom$(f) = \mathbb{N}$ and Range$(f) \subseteq A$.

1 | $n \in \mathbb{N}$
2 | $\exists t_n \in F$
3 | $n \in Dom(t_n)$
4 | $n \in Dom(f)$
5 | $n \in Dom(f)$
6 | | $t_m \in F$
7 | | $n \in Dom(t_m)$
8 | | $Dom(t_m) = s(m)$
9 | | $n \in s(m)$
10 | | $n \in \mathbb{N}$
11 | $n \in \mathbb{N}$
12 $Dom(f) = \mathbb{N}$
13 | $x \in Range(f)$
14 | $(n, x) \in f$
15 | | $t_m \in F$
16 | | $(n, x) \in t$
17 | | $t_m : s(m) \to A$
18 | | $x \in A$
19 | $x \in A$
20 $Range(f) \subseteq A$

Claim 4: f satisfies the recursion equations.

1 $f(0) = a$
2 | $n \in \mathbb{N}$
3 | $\exists t_{n+1} \in F$
4 | $t_{n+1}(s(n)) = g(t_{n+1}(n), n)$
5 | $t_{n+1} \subseteq f$
6 | $f(s(n)) = g(f(n), n)$
7 $\forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)$

Claim 5 : Uniqueness.

1	$\quad\mid\; h : \mathbb{N} \to A$
2	$\quad\mid\; h(0) = a$
3	$\quad\mid\; \forall n \in \mathbb{N} \quad h(s(n)) = g(h(n), n)$
4	$\quad\mid\; f(0) = h(0)$
5	$\quad\mid\quad\mid\; f(n) = h(n)$
6	$\quad\mid\quad\mid\; g(f(n), n) = g(h(n), n)$
7	$\quad\mid\quad\mid\; f(s(n)) = g(f(n), n)$
8	$\quad\mid\quad\mid\; h(s(n)) = g(h(n), n)$
9	$\quad\mid\quad\mid\; f(s(n)) = h(s(n))$
10	$\quad\mid\; \forall n \in \mathbb{N} \quad f(n) = h(n)$
11	$\quad\mid\; f = h$
12	$\exists! f : \mathbb{N} \to A \quad \big(f(0) = a \land \forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)\big)$

Therefore, there exists a unique function $f : \mathbb{N} \to A$ such that $f(0) = a$ and $\forall n \in \mathbb{N}\ f(s(n)) = g(f(n), n)$. □

Theorem 7.3.2 (Parametric Recursion Theorem)**.** *For any sets "A, P", any function "$a : P \to A$", and any function "$g : P \times A \times \mathbb{N} \to A$", there is a unique function "$f : P \times \mathbb{N} \to A$" such that*

1.
$$\forall p \in P \quad f(p, 0) = a(p)$$

2.
$$\forall p \in P\ \forall n \in \mathbb{N} \quad f\big(p, s(n)\big) = g\big(p, f(p, n), n\big)$$

Proof. Fix $p \in P$.

Define

$$G_p(x, n) := g(p, x, n).$$

Claim 1 : $G_p : A \times \mathbb{N} \to A$.

1	$\quad\mid\; p \in P$
2	$\quad\mid\quad\mid\; (x, n) \in A \times \mathbb{N}$
3	$\quad\mid\quad\mid\; g(p, x, n) \in A$
4	$\quad\mid\quad\mid\; G(p)(x, n) \in A$
5	$\quad\mid\; G(p) : A \times \mathbb{N} \to A$
6	$\forall p \in P \quad G(p) : A \times \mathbb{N} \to A$

For every $p \in P$, by the ordinary Recursion Theorem applied to $a(p) \in A$ and $G(p) : A \times \mathbb{N} \to A$, there exists a unique function

$$F_p : \mathbb{N} \to A$$

such that

$$F_p(0) = a(p)$$

and

$$\forall n \in \mathbb{N} \quad F_p(s(n)) = G(p)(F_p(n), n).$$

Define

$$f(p, n) := F_p(n).$$

Claim 2 : $f : P \times \mathbb{N} \to A$.

1 $\qquad$ $(p, n) \in P \times \mathbb{N}$
2 $\qquad$ $F_p : \mathbb{N} \to A$
3 $\qquad$ $F_p(n) \in A$
4 $\qquad$ $f(p, n) \in A$
5 $f : P \times \mathbb{N} \to A$

Claim 3 : f satisfies the recursion equations.

1 $\qquad$ $p \in P$
2 $\qquad$ $F_p(0) = a(p)$
3 $\qquad$ $f(p, 0) = a(p)$
4 $\forall p \in P \quad f(p, 0) = a(p)$
5 $\qquad$ $p \in P$
6 $\qquad$ $n \in \mathbb{N}$
7 $\qquad$ $F_p(s(n)) = G(p)(F_p(n), n)$
8 $\qquad$ $G(p)(F_p(n), n) = g(p, F_p(n), n)$
9 $\qquad$ $f(p, s(n)) = g(p, f(p, n), n)$
10 $\forall p \in P \ \forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(p, n), n)$

Define

$$H_p(n) := h(p, n).$$

Claim 4 : Uniqueness.

1 │ $h : P \times \mathbb{N} \to A$
2 │ $\forall p \in P \quad h(p, 0) = a(p)$
3 │ $\forall p \in P\ \forall n \in \mathbb{N} \quad h(p, s(n)) = g(p, h(p, n), n)$
4 │ │ $p \in P$
5 │ │ $H_p(0) = a(p)$
6 │ │ $\forall n \in \mathbb{N} \quad H_p(s(n)) = g(p, H_p(n), n)$
7 │ │ $F_p = H_p$
8 │ │ │ $n \in \mathbb{N}$
9 │ │ │ $f(p, n) = F_p(n)$
10 │ │ │ $H_p(n) = h(p, n)$
11 │ │ │ $f(p, n) = h(p, n)$
12 │ │ $\forall n \in \mathbb{N} \quad f(p, n) = h(p, n)$
13 │ $\forall p \in P\ \forall n \in \mathbb{N} \quad f(p, n) = h(p, n)$
14 │ $f = h$
15 $\exists! f : P \times \mathbb{N} \to A \quad \Big(\forall p \in P \quad f(p, 0) = a(p) \land \forall p \in P \forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(p, n), n)\Big)$

Therefore, there exists a unique function $f : P \times \mathbb{N} \to A$ such that

$$\forall p \in P \quad f(p, 0) = a(p)$$

and

$$\forall p \in P\ \forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(p, n), n).$$

□

7.4 Arithmetic of Natural numbers

Theorem 7.4.1. ***Summation*** *there is a unique function "$+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$" s.t.*

1. $\forall m \in \mathbb{N} \quad m + 0 = m$
2. $\forall m, n \in \mathbb{N} \quad m + s(n) = s(m + n)$

Proof Is Left As An Exercise To The Reader.
Theorem 7.4.2. ***Multiplication*** *there is a unique function "$\times : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$" s.t.*

1. $\forall m \in \mathbb{N} \quad m \times 0 = 0$
2. $\forall m, n \in \mathbb{N} \quad m \times s(n) = m \times n + m$

Proof Is Left As An Exercise To The Reader.

we denote also "Multiplication" with "$m \cdot n$" and "mn" too.

7.5 Peano arithmetic

assume a structure "$\mathfrak{N} := (N, +, \cdot, s, 0)$"(here "$+, \cdot, s, 0$" are arbitrary functions and constants and doesn't have meaning we defined before). with these properties.

1. $\forall x, y \in N \quad s(x) = s(y) \to x = y$
2. $\forall x \in N \quad s(x) \neq 0$
3. $\forall x \in N \quad x + 0 = x$
4. $\forall x, y \in N \quad x + s(y) = s(x + y)$
5. $\forall x \in N \quad x \cdot 0 = 0$
6. $\forall x, y \in N \quad x \cdot s(y) = x \cdot y + x$
7. $\phi(0) \wedge \forall y \in N \quad \Big(\phi(y) \to \phi(s(y))\Big) \to \forall y \in N \quad \phi(y)$ (this is true for arbitrary "ϕ")

we call "$\mathfrak{N}$" a model of "Peano arithmetic".
Lemma 7.5.1. *prove for "$(N, +)$"*

1. $\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall x \in N \quad x + 0 = x = 0 + x$ *(identity +)*
3. $\forall x, y, z \in N \quad x + z = y + z \to x = y$ *(cancellative +)*
4. $\forall x, y \in N \quad x + y = y + x$ *(commutative(abelian) +)*

1	$\quad x, y, z \in N$	*acp*
2	$\quad (x + y) + 0 = x + y = x + (y + 0)$	*Ax. 3*
3	$\qquad (x + y) + z = x + (y + z)$	*acp*
4	$\qquad (x + y) + s(z) = s\big((x + y) + z\big) = s\big(x + (y + z)\big) = x + s(y + z) = x + (y + s(z))$	*Ax. 4* , 3
5	$\quad (x + y) + z = x + (y + z)$	*Ax. Ind*
6	$\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$	$\to I, \forall I$

1	$x \in N$	*acp*
2	$0 + 0 = 0$	*Ax. 3*
3	$0 + x = x$	*acp*
4	$0 + s(x) = s(0 + x) = s(x)$	*Ax. 3* , 4
5	$0 + x = x$	*Ax. Ind*
6	$\forall x \in N \quad x + 0 = 0 + x = x$	$\to I, \forall I$

1	$x, y \in N$	*acp*
2	$y + s(0) = s(y + 0) = s(y) = s(y) + 0$	*Ax. 3*
3	$y + s(x) = s(y) + x$	*acp*
4	$y + s(s(x)) = s(y + s(x)) = s(s(y) + x) = s(y) + s(x)$	*Ax. 4* , 3
5	$x + y = y + x$	*Ax. Ind*
6	$\forall x, y \in N \quad y + s(x) = s(y) + x$	$\to I, \forall I$

1	$x, y \in N$	*acp*
2	$x + 0 = 0 + x$	*Lemma 7.5.1.2*
3	$x + y = y + x$	*acp*
4	$x + s(y) = s(x + y) = s(y + x) = y + s(x) = s(y) + x$	*Ax. Ind*
5	$x + y = y + x$	*Ax. Ind*
6	$\forall x, y \in N \quad x + y = y + x$	$\to I, \forall I$

Proof Is Left As An Exercise To The Reader.😀

Lemma 7.5.2. *prove for "$(N, +, \cdot)$"*

1. $\forall x, y, z \in N \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall x \in N \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*
3. $\forall x, y \in N \quad xy = 0 \to x = 0 \vee y = 0$ *(zero divisor free $\times$)*
4. $\forall x, y \in N \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in N \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😀

1	$x \in N$	*acp*
2	$0 \cdot 0 = 0$	*Ax. 5*
3	$\quad 0 \cdot x = 0$	*acp*
4	$\quad 0 \cdot s(x) = 0 \cdot x + 0 = 0$	
5	$0 \cdot x = 0$	*Ax. Ind*
6	$\forall x, y \in N \quad 0 \cdot x = 0$	$\to I, \forall I$

1	$x, y \in N$	*acp*
2	$0 + y0 = s(y) \cdot 0$	*Ax. 5*
3	$\quad x + yx = s(y) \cdot x$	*acp*
4	$\quad s(x) + yx = x + s(yx) = s(x + yx) = s(s(y) \cdot x) = s(y) \cdot s(x)$	
5	$x + yx = s(y) \cdot x$	*Ax. Ind*
6	$\forall x, y \in N \quad x + yx = s(y) \cdot x$	$\to I, \forall I$

1	$x, y \in N$	*acp*
2	$x \cdot 0 = 0$	*Ax. 5*
3	$x \cdot 0 = 0 \cdot x$	*Lemma*
4	$\quad xy = yx$	*acp*
5	$\quad x \cdot s(y) = xy + x = yx + x = x + yx = s(y) \cdot x$	
6	$xy = yx$	*Ax. Ind*
7	$\forall x, y \in N \quad xy = yx$	$\to I, \forall I$

Now define "$x < y : \exists z \neq 0 \quad x + z = y$". You can prove for natural numbers

$$\forall m, n \in \mathbb{N} \quad m < n \leftrightarrow \exists k \neq 0 \quad m + k = n$$

Lemma 7.5.3. *prove for "$(N, +, \times, <)$"*

1. $\forall x, y, z \in N \quad x < y \to x + z < y + z$
2. $\forall x, y, z \in N \quad x < y \wedge 0 < z \to xz < yz$

Proof Is Left As An Exercise To The Reader.😀

as you can see "$(\mathbb{N}, +, \times, s, 0)$" is a model of peano arithmetic. so properties are true for them.

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Chapter 8

From Natural numbers to Complex numbers

8.1 Integers"$\mathbb{Z}$"

we learned about "$\mathbb{N}$" before and we constructed structure "$(\mathbb{N}, +, \cdot, <)$". if you assume "$\mathbb{N} \times \mathbb{N}$" we can define an equivalence relation "$(n_1, m_1) \sim (n_2, m_2) : n_1 + m_2 = n_2 + m_1$"(check it's an equivalence relation).

definition 8.1.1. ***Integers"$\mathbb{Z}$"*** *is a set*

$$\mathbb{Z} := \mathbb{N} \times \mathbb{N} / \sim$$

we denote it's members with "$z_0, z_1, \dots$"

in this system "$0 := [(0, 0)]$" and "$\forall n \in \mathbb{N} \quad n := [(n, 0)]$" and "$\forall n \in \mathbb{N} \quad -n := [(0, n)]$". in general"$\forall m, n \in \mathbb{N} \quad m - n := [(m, n)]$". so we can say "$\mathbb{N} \subseteq \mathbb{Z}$".

Theorem 8.1.1. ***Summation*** *there is a unique function "$+_{\mathbb{Z}} : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$" s.t.*

$$\forall m_1, m_2, n_1, n_2 \in \mathbb{N} \quad [(m_1, n_1)] +_{\mathbb{Z}} [(m_2, n_2)] = [(m_1 + m_2, n_1 + n_2)]$$

Proof Is Left As An Exercise To The Reader.

Theorem 8.1.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{Z}} : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$" s.t.*

$$\forall m_1, m_2, n_1, n_2 \in \mathbb{N} \quad [(m_1, n_1)] \times_{\mathbb{Z}} [(m_2, n_2)] = [(m_1 m_2 + n_1 n_2, m_1 n_2 + m_2 n_1)]$$

Proof Is Left As An Exercise To The Reader.

definition 8.1.2. ***Order in*** $\mathbb{Z}$

$$[(m_1, n_1)] <_{\mathbb{Z}} [(m_2, n_2)] : m_1 + n_2 < m_2 + n_1$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Z}}$", "$+$" instead of "$+_{\mathbb{Z}}$" and "$\times$" instead of "$\times_{\mathbb{Z}}$".

Lemma 8.1.1. *prove.*

1. $/ \therefore \forall m, n \in \mathbb{N} \quad m <_{\mathbb{Z}} n \leftrightarrow m < n$
2. $/ \therefore \forall m, n \in \mathbb{N} \quad m +_{\mathbb{Z}} n = m + n$
3. $/ \therefore \forall m, n \in \mathbb{N} \quad m \times_{\mathbb{Z}} n = m \times n$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$z \cdot y$" and "zy" too.
Lemma 8.1.2. *prove for "$(\mathbb{Z}, +)$"*

1. $\forall x, y, z \in \mathbb{Z} \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall z \in \mathbb{Z} \quad z + 0 = 0 + z = z$ *(identity +)*
3. $\forall z \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad z + y = y + z = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{Z} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃
Lemma 8.1.3. *prove for "$(\mathbb{Z}, +, \times)$"*

1. $\forall x, y, z \in \mathbb{Z} \quad (xy)z = x(yz)$ *(associative ×)*
2. $\forall z \in \mathbb{Z} \quad z \cdot 1 = 1 \cdot z = z$ *(identity ×)*
3. $\forall x, y \in \mathbb{Z} \quad xy = 0 \rightarrow x = 0 \vee y = 0$ *(zero divisor free ×)*
4. $\forall x, y \in \mathbb{Z} \quad xy = yx$ *(commutative(abelian) ×)*
5. $\forall x, y, z \in \mathbb{Z} \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.😃
Theorem 8.1.3. ***Order of Integers($\mathbb{Z}$)*** *prove for "$(\mathbb{Z}, +, \times, <_{\mathbb{Z}})$ "*

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y, z \in \mathbb{Z} \quad x < y \rightarrow x + z < y + z$
4. $\forall x, y, z \in \mathbb{Z} \quad x < y \wedge 0 < z \rightarrow xz < yk$

Proof Is Left As An Exercise To The Reader.😃

8.2 Rational numbers "$\mathbb{Q}$"

we learned about "$\mathbb{Z}$" before and we constructed structure "$(\mathbb{Z}, +, \cdot, <)$". if you assume "$\mathbb{Z} \times (\mathbb{Z} - \{0\})$" we can define an equivalence relation "$(y_1, x_1) \sim (y_2, x_2) : y_1 \cdot x_2 = y_2 \cdot x_1$"(check it's an equivalence relation).
definition 8.2.1. ***Rational numbers"$\mathbb{Q}$"*** *is a set*

$$\mathbb{Q} := \mathbb{Z} \times (\mathbb{Z} - \{0\}) / \sim$$

we denote it's members with "$q_0, q_1, \ldots$"

in this system "$0 := [(0,1)]$" and "$1 := [(1,1)]$" and "$\forall x \in \mathbb{Z} \quad x := [(x,1)]$" and "$\forall x \in \mathbb{Z} \quad \frac{1}{x} := [(1,x)]$". in general "$\forall x, y \in \mathbb{Z} \quad \frac{x}{y} := [(x,y)]$". so we can say "$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$".

Theorem 8.2.1. ***Summation*** *there is a unique function "$+_{\mathbb{Q}} : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$" s.t.*

$$\forall x, y, z, u \in \mathbb{Z} \quad [(x,z)] +_{\mathbb{Q}} [(y,u)] = [(xu + zy, zu)]$$

Proof Is Left As An Exercise To The Reader.

Theorem 8.2.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{Q}} : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$" s.t.*

$$\forall x, y, z, u \in \mathbb{Z} \quad [(x,z)] \times_{\mathbb{Q}} [(y,u)] = [(xy, zu)]$$

Proof Is Left As An Exercise To The Reader.

definition 8.2.2. ***Order in*** $\mathbb{Q}$ *.*
if $(z > 0, u > 0)$ or $(z < 0, u < 0)$

$$[(x,z)] <_{\mathbb{Q}} [(y,u)] : x \cdot u < y \cdot z$$

if $(z > 0, u < 0)$ or $(z < 0, u > 0)$

$$[(x,z)] <_{\mathbb{Q}} [(y,u)] : x \cdot u > y \cdot z$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Q}}$", "$+$" instead of "$+_{\mathbb{Q}}$" and "$\times$" instead of "$\times_{\mathbb{Q}}$".

Lemma 8.2.1. *prove.*

1. $/ \therefore \forall z, y \in \mathbb{Z} \quad z <_{\mathbb{Q}} y \leftrightarrow z < y$
2. $/ \therefore \forall z, y \in \mathbb{Z} \quad z +_{\mathbb{Q}} y = z + y$
3. $/ \therefore \forall z, y \in \mathbb{Z} \quad z \times_{\mathbb{Q}} y = z \times y$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$x \cdot y$" and "xy" too.

Lemma 8.2.2. *prove for "$(\mathbb{Q}, +)$"*

1. $\forall x, y, z \in \mathbb{Q} \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall x \in \mathbb{Q} \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad x + y = y + x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{Q} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😀

Lemma 8.2.3. *prove for "$(\mathbb{Q}, +, \times)$"*

1. $\forall x, y, z \in \mathbb{Q} \quad (xy)z = x(yz)$ *(associative ×)*
2. $\forall x \in \mathbb{Q} \quad x \cdot 1 = 1 \cdot x = x$ *(identity ×)*
3. $\forall x \in \mathbb{Q} - \{0\} \quad \exists y \in \mathbb{Q} \quad xy = yx = 1$ *(inverse ×)*
4. $\forall x, y \in \mathbb{Q} \quad xy = yx$ *(commutative(abelian) ×)*

5. $\forall x, y, z \in \mathbb{Q} \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 8.2.3. ***Order of Rational numbers(***$\mathbb{Q}$***)*** *prove for "*$(\mathbb{Q}, <_{\mathbb{Q}})$*"*

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{Q} \quad \exists z \in \mathbb{Q} \quad x \neq y \rightarrow x < z < y$ *(dense)*
4. $\forall x, y, z \in \mathbb{Q} \quad x < y \rightarrow x + z < y + z$
5. $\forall x, y, z \in \mathbb{Q} \quad x < y \wedge 0 < z \rightarrow xz < yk$

Proof Is Left As An Exercise To The Reader.😃

8.3 Real numbers "$\mathbb{R}$"

before we we get to lesson define function "$|\_| : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$" like this.

$$|x| := \begin{cases} x & 0 \leq x \\ -x & x < 0 \end{cases}$$

if you are not familiar to this notation it means "$|\_| := \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid (0 \leq x \rightarrow y = x) \wedge (x < 0 \rightarrow y = -x)\}$". learn this notation because we want to use later.

we learned about "$\mathbb{Q}$" before and we constructed structure "$(\mathbb{Q}, +, \cdot, <)$". now assume $\mathbb{Q}^{\mathbb{N}}$(we know what does it mean by the definition. it means all the sequences of $\mathbb{Q}$). we don't want it all we just want a subset.

definition 8.3.1. ***Cauchy sequence*** *is a sequence like "*$(a_i)_{i \in \mathbb{N}} \in \mathbb{Q}^{\mathbb{N}}$*" s.t. elements of it get infinitely close to each other i.e.*

$$Cauchy(a_i)_{i \in \mathbb{N}} : \forall \varepsilon > 0 \quad \exists N \quad \forall n, m > N \quad |a_n - a_m| < \varepsilon$$

we usually give "N" natural numbers but it's not necessary.

we define an equivalence relation "$(a_i)_{i \in \mathbb{N}} \sim (b_i)_{i \in \mathbb{N}} : \forall \varepsilon > 0 \quad \exists N \quad \forall n > N \quad |a_n - b_n| < \varepsilon$"(check it's an equivalence relation). we denote every $[(a_i)_{i \in \mathbb{N}}]$ with $[a_i]_{i \in \mathbb{N}}$.

1	$a_n - a_n = 0$	
2	$\lvert a_n - a_n \rvert < \varepsilon$	
3	$n > N \rightarrow \lvert a_n - a_n \rvert < \varepsilon$	
4	$\forall \varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - a_n \rvert < \varepsilon$	$\forall I, \exists I, \forall I$
5	$(a_i)_{i \in \mathbb{N}} \sim (a_i)_{i \in \mathbb{N}}$	$def. \sim$

1	$(a_i)_{i\in\mathbb{N}} \sim (b_i)_{i\in\mathbb{N}}$	*ass*
2	$\forall\varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - b_n\rvert < \varepsilon$	*def.* $\sim$
3	$\forall\varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert b_n - a_n\rvert < \varepsilon$	
4	$(b_i)_{i\in\mathbb{N}} \sim (a_i)_{i\in\mathbb{N}}$	*def.* $\sim$

1	$(a_i)_{i\in\mathbb{N}} \sim (b_i)_{i\in\mathbb{N}}$	*ass*
2	$(b_i)_{i\in\mathbb{N}} \sim (c_i)_{i\in\mathbb{N}}$	*ass*
3	$\forall\varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - b_n\rvert < \varepsilon$	*def.* $\sim$
4	$\forall\varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert b_n - c_n\rvert < \varepsilon$	*def.* $\sim$
5	$\exists N \quad \forall n > N \quad \lvert a_n - b_n\rvert < \frac{\varepsilon}{2}$	*def.* $\sim$
6	$\exists N \quad \forall n > N \quad \lvert b_n - c_n\rvert < \frac{\varepsilon}{2}$	*def.* $\sim$
7	$\quad \forall n > N \quad \lvert a_n - b_n\rvert < \frac{\varepsilon}{2}$	*acp*
8	$\quad\quad \forall n > N' \quad \lvert b_n - c_n\rvert < \frac{\varepsilon}{2}$	*acp*
9	$\quad\quad\quad n > max\{N, N'\}$	*acp*
10	$\quad\quad\quad n > N \to \lvert a_n - b_n\rvert < \frac{\varepsilon}{2}$	$\forall E$
11	$\quad\quad\quad n > N' \to \lvert b_n - c_n\rvert < \frac{\varepsilon}{2}$	$\forall E$
12	$\quad\quad\quad \lvert a_n - b_n\rvert < \frac{\varepsilon}{2}$	$\to E$
13	$\quad\quad\quad \lvert b_n - c_n\rvert < \frac{\varepsilon}{2}$	$\to E$
14	$\quad\quad\quad \lvert a_n - b_n\rvert + \lvert b_n - c_n\rvert < \varepsilon$	
15	$\quad\quad\quad \lvert a_n - c_n\rvert < \lvert a_n - b_n\rvert + \lvert b_n - c_n\rvert$	
16	$\quad\quad\quad \lvert a_n - c_n\rvert < \varepsilon$	
17	$\quad\quad n > max\{N, N'\} \to \lvert a_n - c_n\rvert < \varepsilon$	$\to I$
18	$\quad\quad \forall n > max\{N, N'\} \quad \lvert a_n - c_n\rvert < \varepsilon$	$\forall I$
19	$\quad\quad \exists N \quad \forall n > N \quad \lvert a_n - c_n\rvert < \varepsilon$	$\exists I$
20	$\exists N \quad \forall n > N \quad \lvert a_n - c_n\rvert < \varepsilon$	$\exists E$
21	$\forall\varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - c_n\rvert < \varepsilon$	$\forall I$
22	$(a_i)_{i\in\mathbb{N}} \sim (c_i)_{i\in\mathbb{N}}$	*def.* $\sim$

definition 8.3.2. ***Real numbers “ℝ”*** *is a set*

$$\mathbb{R} := Cauchy/\sim$$

we denote it's members with “$r_0, r_1, \ldots$”

in this system “$0 := [0]_{i\in\mathbb{N}}$” and “$1 := [1]_{i\in\mathbb{N}}$” and “$\forall q \in \mathbb{Q} \quad q := [q]_{i\in\mathbb{N}}$”. so we can say “$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$”.

Theorem 8.3.1. ***Summation*** *there is a unique function "$+_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$" s.t.*

$$[a_i]_{i\in\mathbb{N}} +_{\mathbb{R}} [b_i]_{i\in\mathbb{N}} = [a_i + b_i]_{i\in\mathbb{N}}$$

Proof Is Left As An Exercise To The Reader.

Theorem 8.3.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$" s.t.*

$$[a_i]_{i\in\mathbb{N}} \times_{\mathbb{R}} [b_i]_{i\in\mathbb{N}} = [a_i b_i]_{i\in\mathbb{N}}$$

Proof Is Left As An Exercise To The Reader.

definition 8.3.3. ***Order in*** $\mathbb{R}$ *.*

$$[a_i]_{i\in\mathbb{N}} <_{\mathbb{R}} [b_i]_{i\in\mathbb{N}} : \exists\varepsilon > 0 \quad \exists N \quad \forall n > N \quad b_n - a_n > \varepsilon$$

this lemma let us use "$<$" instead of "$<_{\mathbb{R}}$", "$+$" instead of "$+_{\mathbb{R}}$" and "$\times$" instead of "$\times_{\mathbb{R}}$".

Lemma 8.3.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{Q} \quad x <_{\mathbb{R}} y \leftrightarrow x < y$
2. $/ \therefore \forall x, y \in \mathbb{Q} \quad x +_{\mathbb{R}} y = x + y$
3. $/ \therefore \forall x, y \in \mathbb{Q} \quad x \times_{\mathbb{R}} y = x \times y$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$x \cdot y$" and "xy" too.

Lemma 8.3.2. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall x \in \mathbb{R} \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x + y = y + x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{R} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 8.3.3. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$ *(associative ×)*
2. $\forall x \in \mathbb{R} \quad x \cdot 1 = 1 \cdot x = x$ *(identity ×)*
3. $\forall x \in \mathbb{R} - \{0\} \quad \exists y \in \mathbb{R} \quad xy = yx = 1$ *(inverse ×)*
4. $\forall x, y \in \mathbb{R} \quad xy = yx$ *(commutative(abelian) ×)*
5. $\forall x, y, z \in \mathbb{R} \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 8.3.3. ***Linear order of Real numbers(***$\mathbb{R}$***)*** *prove "*$(\mathbb{R}, <_{\mathbb{R}})$

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad y < x\Big)$ *(doesn't have endpoints)*

3. $\forall x, y \in \mathbb{R} \quad \exists z \in \mathbb{R} \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall X \subseteq \mathbb{R} \quad (X \neq \varnothing \wedge \exists y \in \mathbb{R} \quad Uppreb(X, y)) \to (\exists y \in \mathbb{R} \quad sup(X) = y)$ *(complete)*
5. $\forall x, y, z \in \mathbb{R} \quad x < y \to x + z < y + z$
6. $\forall x, y, z \in \mathbb{R} \quad x < y \wedge 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.😃

8.4 Complex numbers“$\mathbb{C}$”

we learned about “$\mathbb{R}$” before and we constructed structure “$(\mathbb{R}, +, \cdot, <)$”.
definition 8.4.1. ***Complex numbers“$\mathbb{C}$”*** *is a set*

$$\mathbb{C} := \mathbb{R} \times \mathbb{R}$$

we denote it's members with “$z_0, x, ...$”

in this system “$0 := (0, 0)$” and “$\forall r \in \mathbb{R} \quad r := (r, 0)$” . in general “$\forall x, y \in \mathbb{R} \quad x + iy := (x, y)$”. so we can say “$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$”.
Theorem 8.4.1. ***Summation*** *there is a unique function “$+_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$” s.t.*

$$\forall x, y, z, u \in \mathbb{R} \quad (x, z) +_{\mathbb{C}} (y, u) = (x + y, z + u)$$

Proof Is Left As An Exercise To The Reader.
Theorem 8.4.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$” s.t.*

$$\forall x, y, z, u \in \mathbb{R} \quad (x, z) \times_{\mathbb{C}} (y, u) = (xy - zu, xu + yz)$$

Proof Is Left As An Exercise To The Reader.

this lemma let us use “$+$” instead of “$+_{\mathbb{C}}$” and “$\times$” instead of “$\times_{\mathbb{C}}$”.
Lemma 8.4.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{R} \quad x +_{\mathbb{C}} y = x + y$
2. $/ \therefore \forall x, y \in \mathbb{R} \quad x \times_{\mathbb{C}} y = x \times y$

Proof Is Left As An Exercise To The Reader.

we denote “Multiplication” with “$z \cdot y$” and “zy” too.
Lemma 8.4.2. *prove for “$(\mathbb{C}, +)$”*

1. $\forall x, y, z \in \mathbb{C} \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall z \in \mathbb{C} \quad z + 0 = 0 + z = z$ *(identity +)*
3. $\forall z \in \mathbb{C} \quad \exists y \in \mathbb{C} \quad z + y = y + z = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{C} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.
Lemma 8.4.3. *prove for “$(\mathbb{C}, +, \times)$”*

1. $\forall x, y, z \in \mathbb{C} \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall z \in \mathbb{C} \quad z \cdot 1 = 1 \cdot z = z$ *(identity $\times$)*
3. $\forall x, y \in \mathbb{C} \quad xy = yx$ *(commutative(abelian) $\times$)*
4. $\forall x, y, z \in \mathbb{C} \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.
Theorem 8.4.3. ***Exponents*** *there is a unique function "$(\_)^{(\_)} : \mathbb{C} \times \mathbb{N} \to \mathbb{C}$" s.t.*

1. $\forall x \in \mathbb{C} \quad x^0 = 1$
2. $\forall x \in \mathbb{C} \quad \forall n \in \mathbb{N} \quad x^{s(n)} = x^n \cdot x$

Proof Is Left As An Exercise To The Reader.
Lemma 8.4.4. *prove for "$(\mathbb{C}, +, \cdot, (\_)^{(\_)})$"*

1. $\forall x \in \mathbb{Z} \quad \forall m, n \in \mathbb{N} \quad x^m \cdot x^n = x^{m+n}$
2. $\forall x \in \mathbb{Z} \quad \forall m, n \in \mathbb{N} \quad (x^m)^n = x^{mn}$

Proof Is Left As An Exercise To The Reader.
Theorem 8.4.4. ***Exponents*** *there is a unique function "$(\_)^{(\_)} : \mathbb{C} \times \mathbb{Z} \to \mathbb{C}$" s.t.*

1. $\forall x \in \mathbb{C} \quad x^0 = 1$
2. $\forall x \in \mathbb{C} \quad \forall m, n \in \mathbb{N} \quad x^{m-n} = \frac{x^m}{x^n}$

Proof Is Left As An Exercise To The Reader.
Lemma 8.4.5. *prove for "$(\mathbb{C}, +, \cdot)$"*

1. $\forall x \in \mathbb{C} \quad \forall y, z \in \mathbb{Z} \quad x^y \cdot x^z = x^{y+z}$
2. $\forall x \in \mathbb{C} \quad \forall y, z \in \mathbb{Z} \quad (x^y)^z = x^{yz}$

Proof Is Left As An Exercise To The Reader.
Theorem 8.4.5. ***Big Sum*** *if "$a : \mathbb{N} \to \mathbb{C}$" there is a unique function "$\sum_{i=0}^{(\_)}(\_) : \mathbb{C} \times \mathbb{N} \to \mathbb{C}$" s.t.*

1. $\forall a \quad \sum_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in \mathbb{N} \quad \sum_{i=0}^{s(n)} a_{s(n)} = \sum_{i=0}^{n} a_n + a_{s(n)}$

Proof Is Left As An Exercise To The Reader.
Lemma 8.4.6. *prove for "$(\mathbb{C}, +, \cdot)$"*

1. $\forall n \in \mathbb{N} \quad \forall a, b \quad \sum_{i=0}^{n} a_n + \sum_{i=0}^{n} b_n = \sum_{i=0}^{n}(a_n + b_n)$
2. $\forall n \in \mathbb{N} \quad \forall a \quad k \sum_{i=0}^{n} a_n = \sum_{i=0}^{n} k a_n$

Proof Is Left As An Exercise To The Reader.
Theorem 8.4.6. ***Big Product*** *if "$a : \mathbb{N} \to \mathbb{C}$" there is a unique function "$\prod_{i=0}^{(\_)}(\_) : \mathbb{C} \times \mathbb{N} \to \mathbb{C}$" s.t.*

1. $\forall a \quad \prod_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in \mathbb{N} \quad \prod_{i=0}^{s(n)} a_{s(n)} = \prod_{i=0}^{n} a_n \cdot a_{s(n)}$

Proof Is Left As An Exercise To The Reader.

Lemma 8.4.7. *prove for “$(\mathbb{C},+,\cdot)$”*

1. $\forall n \in \mathbb{N} \quad \forall a, b \quad \prod_{i=0}^{n} a_n \prod_{i=0}^{n} b_n = \prod_{i=0}^{n}(a_n b_n)$
2. $\forall n \in \mathbb{N} \quad \forall a \quad (\prod_{i=0}^{n} a_n)^k = \prod_{i=0}^{n}(a_n)^k$

Proof Is Left As An Exercise To The Reader.

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Chapter 9

Finite and Infinite sets

9.1 Equipotent and Order in sets

we talked about "Isomorphism" before now assume two Structures $\mathfrak{A} := (A)$ and $\mathfrak{B} := (B)$. because these Structures have only a set "Isomorphism" between them only means there is a bijection function $f : A \xrightarrow[surj]{inj} B$.

definition 9.1.1. ***Equipotent*** *in situation above we say "A" and "B" are "Equipotent". we denote it with "$A \cong B$" i.e.*

$$A \cong B : \exists f(f : A \xrightarrow[surj]{inj} B)$$

Lemma 9.1.1. *"$\cong$" is a equivalence relation in "$(V, \cong)$"*

1	$id : X \xrightarrow[surj]{inj} X$	
2	$\exists f \quad f : X \xrightarrow[surj]{inj} X$	$\exists I$
3	$X \cong X$	$def.2$

1	$X \cong Y$	ass
2	$\exists f \quad f : X \xrightarrow[surj]{inj} Y$	$def.1$
3	$\quad f : X \xrightarrow[surj]{inj} Y$	acp
4	$\quad f^{-1} : Y \xrightarrow[surj]{inj} X$	$lemma$
5	$\quad \exists f \quad f : Y \xrightarrow[surj]{inj} X$	$\exists I$
6	$\quad Y \cong X$	$def.5$
7	$Y \cong X$	$\exists E$

1	$X \cong Y$	*ass*
2	$Y \cong Z$	*ass*
3	$\exists f \quad f : X \xrightarrow[surj]{inj} Y$	$def.1$
4	$\exists f \quad f : Y \xrightarrow[surj]{inj} Z$	$def.2$
5	$f : X \xrightarrow[surj]{inj} Y$	*acp*
6	$g : Y \xrightarrow[surj]{inj} Z$	*acp*
7	$g \circ f : X \xrightarrow[surj]{inj} Z$	*lemma*
8	$\exists f \quad f : X \xrightarrow[surj]{inj} Z$	
9	$X \cong Z$	
10	$X \cong Z$	

so "$V/\cong$" is a partition for "V" and every partition equals to a $[A]_{\cong}$ we denote it by $[A]$.

definition 9.1.2. ***Embedding*** *if there is a function $f : A \xrightarrow{inj} B$. in situation above we say "A" is less than or equipotent "B". we denote it with "$A \precsim B$" i.e.*

$$A \precsim B : \exists f (f : A \xrightarrow{inj} B)$$

similarly we define "$[A] \preceq [B] : \forall X \in [A], Y \in [B] \quad X \precsim Y$"

Lemma 9.1.2. $A \precsim B \therefore / \therefore [A] \preceq [B]$

1	$A \precsim B$	
2	$\quad X \in [A]$	
3	$\quad Y \in [B]$	
4	$\quad X \cong A$	
5	$\quad Y \cong B$	
6	$\quad \exists f \quad f : X \xrightarrow[surj]{inj} A$	$def.4$
7	$\quad \exists f \quad f : A \xrightarrow{inj} B$	$def.1$
8	$\quad \exists f \quad f : B \xrightarrow[surj]{inj} Y$	$def.5$
9	$\quad\quad f : X \xrightarrow[surj]{inj} A$	acp
10	$\quad\quad\quad h : A \xrightarrow{inj} B$	acp
11	$\quad\quad\quad\quad g : B \xrightarrow[surj]{inj} Y$	acp
12	$\quad\quad\quad\quad g \circ h \circ f : X \xrightarrow{inj} Y$	$lemma$
13	$\quad\quad\quad\quad \exists f \quad f : X \xrightarrow{inj} Y$	$\exists I$
14	$\quad\quad\quad\quad X \precsim Y$	$def.13$
15	$\quad X \precsim Y$	$\exists E$
16	$[A] \preceq [B]$	

Theorem 9.1.1. ***Schroeder-Bernstein*** $X \precsim Y, Y \precsim X / \therefore X \cong Y$
Proof Is Left As An Exercise To The Reader.

$$h(x) := \begin{cases} f(g(y)) & \exists i \quad y \in A_i \\ y & \exists i \quad y \in B_i \end{cases}$$

$$C_0 := Y$$
$$C_{s(n)} := f \circ g[C_n]$$

$$D_0 := f[X]$$
$$D_{s(n)} := f \circ g[D_n]$$

$$A_n := C_n - D_n$$
$$B_n := D_n - C_s(n)$$

Claim 1: $\bigcup_{i \in \mathbb{N}} A_i \cap \bigcup_{i \in \mathbb{N}} B_i \neq \varnothing$

Claim 2: $Func(h)$

Claim 3: $injective(h)$

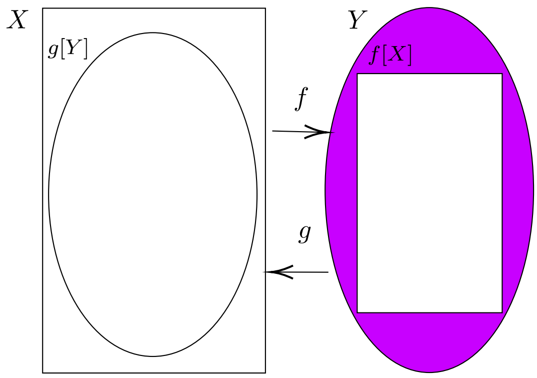
Claim 4: $h : Y \xrightarrow[surj]{inj} f[X]$

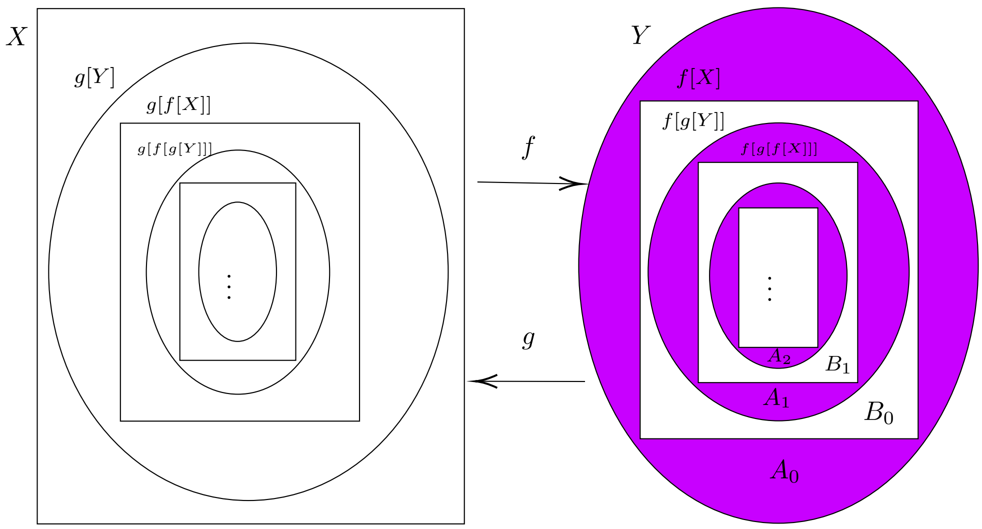
1 $\quad h : Y \xrightarrow[surj]{inj} f[X]$

2 $\quad Y \cong f[X]$

3 $\quad X \cong f[X]$

4 $\quad X \cong Y$





Lemma 9.1.3. *"$\preceq$" is a order relation in "$(V/\cong, \preceq)$"*
Proof Is Left As An Exercise To The Reader.😀

9.2 Finite sets

definition 9.2.1. *if there is a "$n \in \mathbb{N}$" s.t. $A \cong n$ we call it **"Finite"**$(Finite(A))$. if a set is not finite we call it **"Infinite"**$(Infinite(A))$.*
Lemma 9.2.1. $/ \therefore \forall n \in \mathbb{N} \quad X \subset n \to X \ncong n$

1	$\forall X \quad X \subset 0 \to X \ncong 0$	*vacuous truth*
2	$\forall X \quad X \subset n \to X \ncong n$	
3	$X \subset s(n)$	
4	$X \cong s(n)$	
5	$\exists f \quad f : s(n) \xrightarrow[surj]{inj} X$	
6	$f : s(n) \xrightarrow[surj]{inj} X$	
7	$n \notin X$	
8	$X \subset n \cup \{n\}$	
9	$X \subseteq n$	
10	$X - \{f(n)\} \subset n$	
11	$X - \{f(n)\} \subset n \to X - \{f(n)\} \ncong n$	
12	$X - \{f(n)\} \ncong n$	
13	$f \restriction n : n \xrightarrow[surj]{inj} X - \{f(n)\}$	
14	$X - \{f(n)\} \cong n$	
15	$\bot$	
16	$n \in X$	
17	$n \in Dom(f)$	
18	$n \in Range(f)$	
19	$(k, n) \in f$	
20	$(n, m) \in f$	
21	$g := f \cup \{(k, m)\} - \{(k, n), (n, m)\}$	
22	$g : n \xrightarrow[surj]{inj} X - \{n\}$	
23	$X \subset n \cup \{n\}$	
24	$X - \{n\} \subset n$	
25	$X - \{n\} \subset n \to X - \{n\} \ncong n$	
26	$X - \{n\} \ncong n$	
27	$X - \{n\} \cong n$	
28	$\bot$	
29	$\bot$	
30	$X \ncong s(n)$	
31	$\forall X \quad X \subset s(n) \to X \ncong s(n)$	

Corollary 9.2.1. .

1. $\forall m, n \in \mathbb{N} \quad m \neq n \rightarrow m \ncong n$

2. $\forall m, n \in \mathbb{N} \quad A \cong m \wedge A \cong n \rightarrow m = n$

3. $Infinite(\mathbb{N})$

4. $\forall m, n \in \mathbb{N} \quad m \prec n \leftrightarrow m < n$

Proof Is Left As An Exercise To The Reader.😃

corollary (4) let us use “$<$” instead of “$\prec$”

Lemma 9.2.2. $Finite(X), Y \subseteq X / \therefore Finite(Y)$

1 $X \cong 0 \land Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

2 $X \cong n \land Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

3 $X \cong s(n)$

4 $\exists f \quad f : s(n) \xrightarrow[surj]{inj} X$

5 $f : s(n) \xrightarrow[surj]{inj} X$

6 $Y \subseteq X$

7 $g := f - \{(n, f(n))\}$

8 $g : n \xrightarrow[surj]{inj} X - \{f(n)\}$

9 $X - \{f(n)\} \cong n$

10 $Y - \{f(n)\} \subseteq X - \{f(n)\}$

11 $X - \{f(n)\} \cong n \land Y - \{f(n)\} \subseteq X - \{f(n)\} \to \exists m \in \mathbb{N} \quad Y - \{f(n)\} \cong m$

12 $\exists m \in \mathbb{N} \quad Y - \{f(n)\} \cong m$

13 $Y - \{f(n)\} \cong m$

14 $h : m \xrightarrow[surj]{inj} Y - \{f(n)\}$

15 $f(n) \in Y$

16 $Y = (Y - \{f(n)\}) \cup \{f(n)\}$

17 $h' := h \cup \{(m, f(n))\}$

18 $h' : s(m) \xrightarrow[surj]{inj} Y$

19 $s(m) \cong Y$

20 $\exists m \in \mathbb{N} \quad Y \cong m$

21 $f(n) \notin Y$

22 $Y = Y - \{f(n)\}$

23 $h : m \xrightarrow[surj]{inj} Y$

24 $m \cong Y$

25 $\exists m \in \mathbb{N} \quad Y \cong m$

26 $\exists m \in \mathbb{N} \quad Y \cong m$

27 $X \cong s(n) \land Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

28 $\forall n \in \mathbb{N}(X \cong n \land Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m)$

29 $\exists n \in \mathbb{N} \quad X \cong n \land Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

30 $Finite(X) \land Y \subseteq X \to Finite(Y)$

9.3 Countable sets

definition 9.3.1. *set "A" is* ***"Countable"*** *if "$A \cong \mathbb{N}$". set "A" is* ***"At most countable"*** *if "$A \precsim \mathbb{N}$"*
Lemma 9.3.1. $Countable(A), Infinite(B), B \subseteq A / \therefore Countable(B)$
Proof Is Left As An Exercise To The Reade.

Proof. Let A be countable and let $B \subseteq A$ be infinite.

Then there exists a bijective enumeration

$$f : \mathbb{N} \xrightarrow[surj]{inj} A.$$

Define recursively a sequence $(k_n)_{n \in \mathbb{N}}$ in $\mathbb{N}$:

$$k_0 := \min\{k \in \mathbb{N} : f(k) \in B\},$$
$$k_{n+1} := \min\{k \in \mathbb{N} : f(k) \in B \wedge f(k) \neq f(k_i)\ \forall i \leq n\}.$$

Define

$$b_n := f(k_n).$$

1 $\quad b_n \in B$
2 $\quad\quad | \ x \in B$
3 $\quad\quad | \ \exists k \in \mathbb{N}\ f(k) = x$
4 $\quad\quad | \ \exists n \in \mathbb{N}\ b_n = x$
5 $\quad B \subseteq \{b_n : n \in \mathbb{N}\}$
6 $\quad \{b_n : n \in \mathbb{N}\} \subseteq B$
7 $\quad B = \{b_n : n \in \mathbb{N}\}$
8 $\quad \forall m < n\ (b_m \neq b_n)$
9 $\quad (b_n)_{n \in \mathbb{N}}$ injective
10 $\quad \mathbb{N} \xrightarrow[surj]{inj} B$
11 $\quad Countable(B)$

□

Corollary 9.3.1. *a set is at most countable iff it's finite or countable.*
Theorem 9.3.1. *The range of an infinite sequence $(a_n)_{n \in \mathbb{N}}$ is at most countable.*

Proof. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence in X.

We construct recursively a sequence (b_n) such that $\mathrm{Range}(b_n) = \mathrm{Range}(a_n)$ and (b_n) is injective.

Define

$$b_0 := a_0.$$

Having defined $b_0, \ldots, b_n$, define k_{n+1} as the least $k \in \mathbb{N}$ such that

$$a_k \neq b_i \quad \forall i \leq n.$$

If such k_{n+1} exists, set

$$b_{n+1} := a_{k_{n+1}}.$$

Otherwise stop the construction.

1 $b_n \in \text{Range}(a_n)$

2 | $x \in \text{Range}(a_n)$

3 | $\exists k \in \mathbb{N}\ (a_k = x)$

4 | $\exists i\ (b_i = x)$

5 $\text{Range}(a_n) \subseteq \{b_i\}$

6 $\{b_i\} \subseteq \text{Range}(a_n)$

7 $\text{Range}(b_n) = \text{Range}(a_n)$

8 $\forall i < j\ (b_i \neq b_j)$

9 (b_n) injective

10 $\text{Range}(a_n)$ is image of an injective sequence

11 $\text{Range}(a_n)$ is at most countable

□

Theorem 9.3.2. *The union of two countable sets is countable.*

Proof. Let $A = \{a_n : n \in \mathbb{N}\}$ and $B = \{b_n : n \in \mathbb{N}\}$.

Define a sequence $(c_n)_{n\in\mathbb{N}}$ by

$$c_{2k} := a_k, \qquad c_{2k+1} := b_k.$$

1 $x \in A \cup B$

2 $\quad x \in A$

3 $\quad \exists k \in \mathbb{N}\ (a_k = x)$

4 $\quad \exists k \in \mathbb{N}\ (c_{2k} = x)$

5 $\quad \exists n \in \mathbb{N}\ (c_n = x)$

6 $\quad x \in B$

7 $\quad \exists k \in \mathbb{N}\ (b_k = x)$

8 $\quad \exists k \in \mathbb{N}\ (c_{2k+1} = x)$

9 $\quad \exists n \in \mathbb{N}\ (c_n = x)$

10 $\forall x \in A \cup B\ \exists n \in \mathbb{N}\ (c_n = x)$

11 $\{c_n : n \in \mathbb{N}\} = A \cup B$

12 $c : \mathbb{N} \xrightarrow[surj]{} A \cup B$

13 $AtMostCountable(A \cup B)$

14 $Infinite(A)$

15 $A \precsim A \cup B$

16 $Infinite(A \cup B)$

17 $Countable(A \cup B)$

□

Corollary 9.3.2. *Finite union of countable sets is countable.*

Proof. Let $A_1, \ldots, A_n$ be countable.

1 $Countable(A_1)$

2 $\quad \forall k < n \quad Countable(A_k) \to Countable(\bigcup_{i \in n} A_i)$

3 $\quad\quad \forall k < s(n) \quad Countable(A_k)$

4 $\quad\quad Countable(A_n)$

5 $\quad\quad Countable(\bigcup_{i \in s(n)} A_i)$

6 $Countable(\bigcup_{i \in s(n)} A_i)$

□

Theorem 9.3.3. *If A and B are countable, then $A \times B$ is countable.*

Proof. It suffices to show $\mathbb{N} \times \mathbb{N}$ is countable.

Define

$$f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$

by

$$f(m,n) := \begin{cases} n^2 + m & \text{if } m \leq n, \\ m^2 + 2m - n & \text{if } n < m. \end{cases}$$

1 $(m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$

2 $f(m_1, n_1) = f(m_2, n_2)$

3 $m \leq n \to n^2 = f(0, n) \leq f(m, n) \leq f(n, n) \leq f(n, m) < f(0, n+1) = (n+1)^2$

4 $k = max\{m, n\} \to k^2 \leq f(m, n) < (k+1)^2$

5 $k' = max\{m', n'\} \to k'^2 \leq f(m', n') < (k'+1)^2$

6 $f(m, n) = f(m', n')$

7 $k = k'$

8 $\quad m \leq n = k \land m' \leq n' = k'$

9 $\quad n = n'$

10 $\quad n^2 + m = n'^2 + m'$

11 $\quad n^2 + m = n^2 + m'$

12 $\quad m = m'$

13 $\quad n < m = k \land n' < m' = k'$

14 $\quad m = m'$

15 $\quad m^2 + 2m - n = m'^2 + 2m' - n'$

16 $\quad m^2 + 2m - n = m^2 + 2m - n'$

17 $\quad n = n'$

18 $\quad n < m = k \land n' = k' > m'$

19 $\quad m = n'$

20 $\quad f(m, n) > f(m, m)$

21 $\quad f(n', n') > f(m', n')$

22 $\quad f(n', n') = f(m, m)$

23 $\quad f(m, n) > f(m', n')$

24 $\quad \perp$

25 $(m_1, n_1) = (m_2, n_2)$

26 $\mathrm{f} isinjective$

27 $\quad (m, n) \in \mathbb{N} \times \mathbb{N}$

28 $\quad \exists k \in \mathbb{N}\ (f(k) = (m, n))$

29 $\quad \mathbb{N} \times \mathbb{N} \cong \mathbb{N}$

30 $Countable(\mathbb{N} \times \mathbb{N})$

□

Corollary 9.3.3. *Finite Cartesian product of countable sets is countable, and A^n is countable for any countable A and $n > 0$.*

Proof. Let $A_1, \ldots, A_n$ be countable.

1 $\quad Countable(A_1)$

2 $\quad \vert \; \forall k < n \quad Countable(A_k) \to Countable(\prod_{i \in n} A_i)$

3 $\quad \vert \; \vert \; \forall k < s(n) \quad Countable(A_k)$

4 $\quad \vert \; \vert \; Countable(A_n)$

5 $\quad \vert \; \vert \; Countable(\prod_{i \in s(n)} A_i)$

6 $\quad Countable(\prod_{i \in s(n)} A_i)$

□

Lemma 9.3.2. *If "$(A_i)_{i \in \mathbb{N}}$" is family of countable sets with a family of sequences "$(a_i)_{i \in \mathbb{N}}$" s.t. "$\forall n \in \mathbb{N} \quad a_n : \mathbb{N} \xrightarrow[surj]{inj} A_n$" then "$\bigcup_{i \in \mathbb{N}} A_i$" is countable.*
Proof Is Left As An Exercise To The Reader.😀

Proof. Define

$$h : \mathbb{N} \times \mathbb{N} \to \bigcup_{i \in \mathbb{N}} A_i \quad \text{by} \quad h(i, n) = a_i(n).$$

1 $\quad x \in \bigcup_{i \in \mathbb{N}} A_i$

2 $\quad \vert \; \exists i \in \mathbb{N} \; (x \in A_i)$

3 $\quad \vert \; \exists n \in \mathbb{N} \; a_i(n) = x$

4 $\quad \vert \; h(i, n) = x$

5 $\quad h : \mathbb{N} \times \mathbb{N} \xrightarrow[surj]{inj} \bigcup_{i \in \mathbb{N}} A_i$

6 $\quad Countable(\mathbb{N} \times \mathbb{N})$

7 $\quad Countable(\bigcup_{i \in \mathbb{N}} A_i)$

□

Theorem 9.3.4. *If A is countable, then $Seq(A)$, the set of all finite sequences of elements of A, is countable.*

Proof. It suffices to prove the theorem for $A = \mathbb{N}$.

Since

$$Seq(\mathbb{N}) = \bigcup_{n \in \mathbb{N}} \mathbb{N}^n,$$

it is enough to construct, for each $n \geq 1$, an enumeration

$$a_n : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^n.$$

Let

$$g : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N} \times \mathbb{N}.$$

Define recursively:

$$a_1(i) := (i),$$

and

$$a_{n+1}(i) := (b_0, \ldots, b_{n-1}, i_2),$$

where

$$g(i) = (i_1, i_2)$$

and

$$a_n(i_1) = (b_0, \ldots, b_{n-1}).$$

1 $\quad a_1 : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}$

2 $\quad \mid\; a_n : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^n$

3 $\quad \mid\; \mid\; (b_0, \ldots, b_n) \in \mathbb{N}^{n+1}$

4 $\quad \mid\; \mid\; \exists i_1 \in \mathbb{N} \quad a_n(i_1) = (b_0, \ldots, b_{n-1})$

5 $\quad \mid\; \mid\; \exists i \in \mathbb{N} \quad g(i) = (i_1, b_n)$

6 $\quad \mid\; \mid\; a_{n+1}(i) = (b_0, \ldots, b_n)$

7 $\quad \mid\; a_{n+1} : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^{n+1}$

8 $\quad \forall n \geq 1 \quad a_n : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^n$

9 $\quad \forall n \in \mathbb{N} \quad \mathbb{N}^n$ is countable

10 $\quad \bigcup_{n \in \mathbb{N}} \mathbb{N}^n$ is countable

11 $\quad Seq(\mathbb{N})$ is countable

Since every countable set is equipotent to a subset of $\mathbb{N}$,

$$Seq(A)$$

is countable.

$\square$

Corollary 9.3.4. *The set of all finite subsets of a countable set is countable.*

Proof. Let A be countable and define

$$\mathcal{F}(A) := \{F \subseteq A : F \text{ finite}\}.$$

For each $n \in \mathbb{N}$ define

$$\mathcal{F}_n(A) := \{F \subseteq A : |F| = n\}.$$

1 A^n is countable

2 $\{(a_1, \ldots, a_n) \in A^n : a_i \text{ distinct}\}$ is countable

3 $\mathcal{F}_n(A)$ is countable

4 | $F \in \mathcal{F}(A)$

5 | $\exists n \in \mathbb{N}\ (F \in \mathcal{F}_n(A))$

6 $\mathcal{F}(A) = \bigcup_{n\in\mathbb{N}} \mathcal{F}_n(A)$

7 $\forall n\ \mathcal{F}_n(A)$ is countable

8 $\bigcup_{n\in\mathbb{N}} \mathcal{F}_n(A)$ is countable

9 $\mathcal{F}(A)$ is countable

$\square$

Corollary 9.3.5. *The set of all finite subsets of a countable set is countable.*

Proof. Let

$$\mathcal{F}(A) := \{X \subseteq A : X \text{ is finite}\}.$$

Define

$$F : Seq(A) \to \mathcal{F}(A)$$

by

$$F((a_0, \ldots, a_{n-1})) = \{a_0, \ldots, a_{n-1}\}.$$

1 $Seq(A)$ is countable

2 $\quad\mid X \in \mathcal{F}(A)$

3 $\quad\mid X = \{a_0, \ldots, a_{n-1}\}$

4 $\quad\mid F((a_0, \ldots, a_{n-1})) = X$

5 $F : Seq(A) \xrightarrow[surj]{} \mathcal{F}(A)$

6 $\mathcal{F}(A)$ is at most countable

7 $\mathcal{F}(A)$ is countable

$\square$

Theorem 9.3.5. *An equivalence relation on a countable set has at most countably many equivalence classes.*

Proof. Let $\sim$ be an equivalence relation on a countable set A.

Define

$$F : A \to A/\sim$$

by

$$F(a) := [a]_\sim$$

1 A is countable

2 $\quad\mid C \in A/\sim$

3 $\quad\mid \exists a \in A \quad C = [a]_E$

4 $\quad\mid F(a) = C$

5 $F : A \xrightarrow[surj]{} A/\sim$

6 $A/\sim$ is at most countable

$\square$

Corollary 9.3.6. *The set of all integers $\mathbb{Z}$ and the set of all rational numbers $\mathbb{Q}$ are countable.*

Proof. $\mathbb{Z}$ **is countable.**

By definition

$$\mathbb{Z} = \mathbb{N} \times \mathbb{N}/\sim .$$

1 $\mathbb{N} \times \mathbb{N}$ is countable

2 $\mathbb{Z} := (\mathbb{N} \times \mathbb{N})/\sim$

3 $\mathbb{Z}$ is at most countable

4 $\mathbb{Z}$ is infinite

5 $\mathbb{Z}$ is countable

$\mathbb{Q}$ is countable.

By definition

$$\mathbb{Q} = \mathbb{Z} \times (\mathbb{Z} - \{0\})/\sim .$$

1 $\mathbb{Z}$ is countable

2 $\mathbb{Z} - \{0\}$ is countable

3 $\mathbb{Z} \times (\mathbb{Z} - \{0\})$ is countable

4 $F : \mathbb{Z} \times (\mathbb{Z} - \{0\}) \to \mathbb{Z} \times (\mathbb{Z} - \{0\})/\sim$

5 $\mathbb{Q} = (\mathbb{Z} \times (\mathbb{Z} - \{0\}))/\sim$

6 $\mathbb{Q}$ is at most countable

7 $\mathbb{Q}$ is infinite

8 $\mathbb{Q}$ is countable

$\square$

9.4 Uncountable sets

Lemma 9.4.1. $/ \therefore \mathbb{N} \prec 2^{\mathbb{N}}$

Proof. **Claim 0:** $\mathbb{N} \precsim 2^{\mathbb{N}}$

Define

$$f : \mathbb{N} \to 2^{\mathbb{N}}$$

by

$$f(n)(k) = \begin{cases} 1 & k = n, \\ 0 & k \neq n. \end{cases}$$

1 $\quad\mid f(m) = f(n)$

2 $\quad\mid\quad\mid m \neq n$

3 $\quad\mid\quad\mid f(m)(m) = 1 \neq 0 = f(n)(m)$

4 $\quad\mid\quad\mid \bot$

5 $\quad\mid m = n$

6 $\quad f : \mathbb{N} \xrightarrow{inj} 2^{\mathbb{N}}$

7 $\quad \mathbb{N} \precsim 2^{\mathbb{N}}$

Claim 1: $\mathbb{N} \prec 2^{\mathbb{N}}$

1 $\quad\mid g : \mathbb{N} \xrightarrow[surj]{} 2^{\mathbb{N}}$

2 $\quad\mid d(n) := 1 - g(n)(n)$

3 $\quad\mid d \in 2^{\mathbb{N}}$

4 $\quad\mid \exists k \in \mathbb{N} \quad g(k) = d$

5 $\quad\mid d(k) = g(k)(k)$

6 $\quad\mid d(k) = 1 - g(k)(k)$

7 $\quad\mid g(k)(k) = 1 - g(k)(k)$

8 $\quad\mid \bot$

9 $\quad \nexists g : \mathbb{N} \xrightarrow[surj]{} 2^{\mathbb{N}}$

10 $\quad \mathbb{N} \ncong 2^{\mathbb{N}}$

11 $\quad \mathbb{N} \prec 2^{\mathbb{N}}$

$\square$

Lemma 9.4.2. $/ \therefore 2^{\mathbb{N}} \cong \mathcal{P}(\mathbb{N})$
Proof Is Left As An Exercise To The Reader.😃
Theorem 9.4.1.

$$\mathcal{P}(\mathbb{N}) \cong 2^{\mathbb{N}}$$

Proof. For each

$$S \subseteq \mathbb{N}$$

define its characteristic function

$$\chi_S : \mathbb{N} \to 2$$

by

$$\chi_S(n) := \begin{cases} 1 & n \in S, \\ 0 & n \notin S. \end{cases}$$

Define

$$\chi_{()} : \mathcal{P}(\mathbb{N}) \to 2^{\mathbb{N}}$$

1		$\chi_S = \chi_T$
2		$\forall n \in \mathbb{N} \quad (n \in S \leftrightarrow n \in T)$
3		$S = T$
4	$\chi_{()}$ is injective	
5		$f \in 2^{\mathbb{N}}$
6		$S := \{n \in \mathbb{N} : f(n) = 1\}$
7		$\chi_{(S)} = f$
8	$\chi_{()} : \mathcal{P}(\mathbb{N}) \xrightarrow[surj]{inj} 2^{\mathbb{N}}$	
9	$\mathcal{P}(\mathbb{N}) \cong 2^{\mathbb{N}}$	

□

Lemma 9.4.3. $/ \therefore 2^{\mathbb{N}} \cong \mathbb{R}$
Proof Is Left As An Exercise To The Reader.😀
Theorem 9.4.2. ***Cantor theorem*** $/ \therefore \forall X \quad Type_0(X) \to X \prec 2^X$
Proof Is Left As An Exercise To The Reader.😀
Theorem 9.4.3. $/ \therefore \forall X \quad Type_0(X) \to 2^X \cong \mathcal{P}(X)$
Proof Is Left As An Exercise To The Reader.😀

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Chapter 10

Ordinals

10.1 Well ordered sets

definition 10.1.1. *if "$(W, <)$" is well ordered and $a \in W$* ***"Initial segment"*** *is*

$$W[a] := \{x \in W \mid x < a\}$$

before get to next lemma Let's have a drfinition. assume "$(W_1, <_1)$" and "$(W_2, <_2)$" are two ordered structures function "f" is increasing when "$f : W_1 \to W_2$" is a (weak) homomorphism.
Lemma 10.1.1. *if "$(W, <)$" is well ordered and $f : W \to W$ is increasing then $\forall x \in W \quad x \leq f(x)$. Proof Is Left As An Exercise To The Reader.*😃
Corollary 10.1.1. *.*

1. *no well ordered structure is isomorph to it's initial segment of itself.*
2. *each well ordered structure has only one automorphism, the identity.*
3. *if "$(W_1, <_1)$" and "$(W_2, <_2)$" are two isomorphic well ordered structures then the isomorphism is unique.*

Proof Is Left As An Exercise To The Reader.😃
Theorem 10.1.1. *if "$(W_1, <_1)$" and "$(W_2, <_2)$" are two well ordered structures then exactly one of the following holds.*

1. $(W_1, <_1) \cong (W_2, <_2)$
2. $\exists a \in W_1 \quad (W_1[a], <_1) \cong (W_2, <_2)$
3. $\exists a \in W_2 \quad (W_1, <_1) \cong (W_2[a], <_2)$

Proof Is Left As An Exercise To The Reader.😃

Proof. Let $(W_1, <_1)$ and $(W_2, <_2)$ be well ordered sets.

Define

$$f \subseteq W_1 \times W_2 \quad \text{by} \quad (x, y) \in f \leftrightarrow W_1[x] \cong W_2[y].$$

Claim 1: f is a function.

1	$(x, y) \in f \wedge (x, y') \in f$	assumption
2	$W_2[y] \cong W_2[y']$	definition of f
3	$y = y'$	uniqueness of isomorphism of well orders

Claim 2: f is injective.

1	$(x, y) \in f \wedge (x', y) \in f$	assumption
2	$W_1[x] \cong W_1[x']$	definition of f
3	$x = x'$	uniqueness of isomorphism of well orders

Claim 3: f is increasing.

1	$x <_1 x'$	assumption
2	$W_1[x'] \cong W_2[f(x')]$	definition of f
3	$W_1[x] \cong W_2[f(x)]$	definition of f
4	$f(x) <_2 f(x')$	comparison of initial segments

Claim 4: f is isomorphism between $S = \mathrm{dom}(f)$ and $T = \mathrm{ran}(f)$.

1	$f : S \to T$ bijective	Claims 1,2
2	$x <_1 x' \Rightarrow f(x) <_2 f(x')$	Claim 3
3	f order isomorphism	definition

Claim 5: S initial segment of W_1.

1	$x \in S \wedge z <_1 x$	assumption
2	$W_1[x] \cong W_2[f(x)]$	definition of S
3	$z \in S$	restriction of initial segments

Claim 6: T initial segment of W_2 or $T = W_2$.

1	$y \in T$	definition of T
2	$T \neq W_2$	assumption of case split
3	$T = W_2[b] \;\vee\; T = W_2$	well-order trichotomy

Claim 7: exclusion of mixed case.

1	$S = W_1[a] \wedge T = W_2[b]$	assumption
2	$f : W_1[a] \cong W_2[b]$	restriction of f
3	$(a, b) \in f$	definition of domain/range
4	$a \in W_1[a]$	initial segment property
5	$a < a$	element characterization
6	$\perp$	irreflexivity

Conclusion:

$$(W_1 \cong W_2) \ \vee \ (W_1[a] \cong W_2) \ \vee \ (W_1 \cong W_2[a]).$$

□

Lemma 10.1.2. *Every element of an ordinal number is an ordinal number.*

10.2 Ordinal numbers

definition 10.2.1. ***Order in sets***

$$<_A := \{(x, y) \in A \times A \mid x \in y\}$$

definition 10.2.2. *A set "α" is a **Ordinal** if*

1. *α is transitive.*
2. *$(\alpha, <_\alpha)$ is well ordered.*

we denote it by Ord so we have the set of ordinals(Ord).
Lemma 10.2.1. *Every natural number is an ordinal.*

Proof. Let $\mathbb{N}$ be the set of natural numbers.

1	$\forall n \in \mathbb{N}\ (n \subseteq \mathbb{N})$	von Neumann construction
2	$Transitive(\mathbb{N})$	definition of transitive set
3	$<_{\mathbb{N}} = \in \cap (\mathbb{N} \times \mathbb{N})$	definition
4	$(\mathbb{N}, <_{\mathbb{N}})$ is well ordered	standard well-order of $\mathbb{N}$
5	$Ordinal(\mathbb{N})$	definition of ordinal

Hence $\mathbb{N}$ is an ordinal.

□

In ordinal numbers we show "$\mathbb{N}$" with "ω". so it's just a renaming.
Lemma 10.2.2. *if α is an ordinal then $s(\alpha)$ is an ordinal too.*
Proof Is Left As An Exercise To The Reader.😃

Proof.

$$s(\alpha) = \alpha \cup \{\alpha\}.$$

1	α transitive	ordinal assumption
2	$s(\alpha) = \alpha \cup \{\alpha\}$	definition
3	$x \in y \wedge y \in s(\alpha) \Rightarrow x \in s(\alpha)$	case split on $y \in \alpha \cup \{\alpha\}$
4	$Transitive(s(\alpha))$	definition of transitive set
5	$<_\alpha = \in \cap (\alpha \times \alpha)$	definition
6	α is greatest element of $s(\alpha)$	construction
7	$<_{s(\alpha)}$ extends $<_\alpha$	restriction
8	$(s(\alpha), <_{s(\alpha)})$ is well ordered	extension by maximum element
9	$Ordinal(s(\alpha))$	definition of ordinal

Hence $s(\alpha)$ is an ordinal number.

□

Lemma 10.2.3. *prove*

1. $Transitive(Ord)$.
2. $\forall \alpha, \beta \in Ord \quad \alpha \subset \beta \to \alpha \in \beta$.

Proof. **(1):** $Transitive(Ord)$**.**

Claim 1: x **is transitive.**

1	$u \in v \wedge v \in x$	
2	$x \in \alpha$	assumption
3	$v \in \alpha$	transitivity of α
4	$u \in \alpha$	transitivity of α
5	$u, v, x \in \alpha$	closure
6	$u <_\alpha v <_\alpha x$	definition
7	$u <_\alpha x$	linearity of $<_\alpha$

Hence $Transitive(x)$.

Claim 2: $(x, <_x)$ **is well ordered.**

1	$x \subseteq \alpha$	transitivity of α
2	$<_x = <_\alpha \restriction x$	definition
3	$<_\alpha$ well order on α	ordinal assumption
4	$<_x$ well order on x	restriction of well-order

Thus x is transitive and well ordered, hence

$$x \in Ord.$$

1	$\alpha \in Ord$
2	$x \in y \wedge y \in \alpha$
3	α transitive
4	$x \in \alpha$

Hence $Transitive(\alpha)$ for all $\alpha \in Ord$, so $Transitive(Ord)$.

(2): $\alpha \subset \beta \Rightarrow \alpha \in \beta$.

Let $\alpha, \beta \in Ord$ and $\alpha \subset \beta$.

1	$\alpha \subset \beta$	assumption
2	$\beta - \alpha \neq \varnothing$	set difference
3	$\exists \gamma \in \beta - \alpha \; \forall \delta \in \beta - \alpha \; (\gamma \leq \delta)$	well-order of β
4	$\gamma \in \beta \wedge \gamma \notin \alpha$	definition of set difference
5	$\forall \delta \in \alpha \; (\delta < \gamma)$	minimality of γ
6	$\alpha \subseteq \gamma$	transitivity of $<$
7	$\gamma \in \beta$	step 8
8	$\alpha = \gamma$	ordinal comparison lemma ($\alpha, \gamma \in Ord$)
9	$\alpha \in \beta$	since $\gamma \in \beta$ and $\alpha = \gamma$

Hence

$$\alpha \subset \beta \Rightarrow \alpha \in \beta.$$

□

Theorem 10.2.1. ***Order of Ordinals*** *prove*

1. *$(Ord, <)$ is linearly ordered.*
2. *$(Ord, <)$ is Well ordered.*

Proof Is Left As An Exercise To The Reader.😃

Theorem 10.2.2. ***Order of Ordinals.*** *prove*

1. *$(Ord, <)$ is linearly ordered.*
2. *$(Ord, <)$ is well ordered.*

Proof. **Claim 1: $(Ord, <)$ is linear.**

1	$\alpha \in \beta$	assumption
2	$\beta \in \gamma$	assumption
3	γ transitive	$\gamma \in Ord$
4	$\alpha \in \gamma$	transitivity of γ

1	$\mid$ $\alpha \in \alpha$	assumption
2	$\mid$ $(\alpha, <_\alpha)$ is linearly ordered	α is an ordinal
3	$\mid$ $x := \alpha$	definition
4	$\mid$ $x \in x$	1,3
5	$\mid$ $\perp$	asymmetry of $<_\alpha$
6	$\alpha \notin \alpha$	

Let $\alpha, \beta, \gamma \in Ord$.

1	$\alpha \subseteq \beta \wedge \beta \subseteq \gamma$	assumption
2	$\alpha \subseteq \gamma$	transitivity of $\subseteq$
3	$\alpha < \gamma$	Lemma 2.6(a)
4	$\alpha \not< \beta \wedge \beta \not< \alpha \Rightarrow \alpha = \beta$	Lemma 2.6(b),(c)
5	$\alpha < \beta \ \vee \ \alpha = \beta \ \vee \ \beta < \alpha$	Lemma 2.6(c)

Hence $(Ord, <)$ is linearly ordered.

Claim 2: $(Ord, <)$ is well ordered.

Let $X \subseteq Ord$, $X \neq \varnothing$.

1	$X \neq \varnothing$	assumption
2	$\exists \alpha \in X$	nonempty
3	$\exists \beta \in X \ \forall \gamma \in X \ (\beta \leq \gamma)$	Lemma 2.6(d)
4	β is $<$-least element of X	definition

Hence every nonempty set of ordinals has a least element.

Conclusion:

$$(Ord, <) \text{ is linearly ordered and well ordered.}$$

□

Theorem 10.2.3. *The natural numbers are exactly the finite ordinal numbers.*

Proof. **Claim 1: Every natural number is a finite ordinal.**

1	$n \in \mathbb{N}$	
2	$Ordinal(n)$	Theorem 2.3
3	$Finite(n)$	definition of $\mathbb{N}$

Claim 2: Every ordinal that is not a natural number is infinite.

1	$Ordinal(\alpha) \wedge \alpha \notin \mathbb{N}$	assumption
2	$\alpha \neq \omega$	$\alpha \notin \mathbb{N}$
3	$\alpha < \omega \ \vee \ \alpha = \omega \ \vee \ \omega < \alpha$	Theorem 2.6(c)
4	$\omega < \alpha$	Theorem 2.6(b),5,6
5	$\omega \in \alpha$	definition of $<$
6	$\omega \subseteq \alpha$	transitivity of α
7	$Infinite(\omega)$	definition
8	$Infinite(\alpha)$	$\omega \subseteq \alpha$

Hence every finite ordinal is a natural number.

□

Theorem 10.2.4. *Every well ordered structure is isomorphic to a unique ordinal.*

Proof. Let $(W, <)$ be a well ordered structure.
Define

$$A := \{a \in W \mid \exists \alpha \in Ord \ ((W[a], <) \cong (\alpha, <))\}.$$

For each $a \in A$, let α_a be the unique ordinal such that

$$(W[a], <) \cong (\alpha_a, <).$$

Define

$$S := \{\alpha_a \mid a \in A\}.$$

Claim 1: $Transitive(S)$.

1	$\alpha_a \in S$	
2	$\gamma \in \alpha_a$	
3	$h : (W[a], <) \cong (\alpha_a, <)$	definition of α_a
4	$c := h^{-1}(\gamma)$	
5	$(W[c], <) \cong (\gamma, <)$	restriction of h
6	$c \in A$	definition of A
7	$\gamma = \alpha_c$	
8	$\gamma \in S$	

Hence $Transitive(S)$.

Claim 2: S is an ordinal.

1	$S \subseteq Ord$	definition of S
2	$(S, \in)$ is well ordered	Order of Ordinals
3	$Transitive(S)$	Claim 1
4	$Ordinal(S)$	definition of ordinal

Let

$$S = \alpha.$$

Claim 3: $b < a \wedge a \in A \Rightarrow b \in A$.

1	$a \in A \wedge b < a$	
2	$h : (W[a], <) \cong (\alpha_a, <)$	definition of A
3	$h[W[b]]$ is an initial segment of α_a	
4	$\exists \beta \in \alpha_a \quad h[W[b]] = \beta$	initial segments of ordinals
5	$(W[b], <) \cong (\beta, <)$	
6	$b \in A$	definition of A
7	$\alpha_b = \beta$	
8	$\alpha_b \in \alpha_a$	

Claim 4: $A = W$ or $A = W[c]$ for some $c \in W$.

1	$A \subseteq W$	
2	$\forall a \in A\ \forall b < a\ (b \in A)$	Claim 3
3	A is an initial segment of W	
4	$A = W\ \vee\ \exists c \in W\ (A = W[c])$	Lemma 1.2

Define

$$f : A \to \alpha$$

by

$$f(a) := \alpha_a.$$

Claim 5: f is an isomorphism.

1	$a, b \in A$	
2	$a < b \Rightarrow \alpha_a \in \alpha_b$	Claim 3
3	$a < b \Rightarrow f(a) < f(b)$	
4	f injective	uniqueness of ordinals
5	$Range(f) = \alpha$	definition of $\alpha = S$
6	$f : (A, <) \cong (\alpha, <)$	

Claim 6: $A = W$.

1	$\vert\ A = W[c]$	
2	$\vert\ f : (W[c], <) \cong (\alpha, <)$	Claim 5
3	$\vert\ c \in A$	definition of A
4	$\vert\ c \in W[c]$	31
5	$\vert\ \perp$	definition of initial segment
6	$A = W$	

Since $A = W$, Claim 5 yields

$$f : (W, <) \cong (\alpha, <).$$

The ordinal α is unique since isomorphic ordinals are equal.

□

10.3 Transfinite Induction

Theorem 10.3.1 (Second Version of Transfinite Induction)**.** *Let $P(\alpha)$ be a property. Assume*

1. $P(0)$.
2. $\forall \alpha \in Ord \quad P(\alpha) \rightarrow P(s(\alpha))$.
3. $\forall \alpha \in Ord \quad \big(Limit(\alpha) \wedge \alpha \neq 0 \wedge \forall \beta < \alpha \ P(\beta)\big) \rightarrow P(\alpha)$.

Then

$$\forall \alpha \in Ord \quad P(\alpha).$$

Proof. We prove the hypothesis of Transfinite Induction.

1	$\alpha \in Ord$	
2	$\forall \beta < \alpha \quad P(\beta)$	
3	$\alpha = 0$	
4	$P(\alpha)$	assumption (a)
5	$\exists \beta < \alpha \quad \alpha = s(\beta)$	
6	$P(\beta)$	2
7	$P(\alpha)$	assumption (b)
8	$Limit(\alpha) \wedge \alpha \neq 0$	
9	$P(\alpha)$	assumption (c),2
10	$P(\alpha)$	trichotomy of ordinals
11	$\forall \alpha \in Ord\Big(\forall \beta < \alpha \ P(\beta) \rightarrow P(\alpha)\Big)$	
12	$\forall \alpha \in Ord \quad P(\alpha)$	Transfinite Induction

□

Theorem 10.3.2 (Transfinite Induction Principle)**.** *Let $P(\alpha)$ be a property.*

Assume

$$\forall \alpha \in Ord \quad \Big(\forall \beta < \alpha \quad P(\beta) \rightarrow P(\alpha)\Big).$$

Then

$$\forall \alpha \in Ord \quad P(\alpha).$$

Proof. Define

$$S := \{\beta \le \gamma \mid \neg P(\beta)\}.$$

1	$\quad\mid\ \exists\gamma \in Ord \quad \neg P(\gamma)$	assumption
2	$\quad\mid\ \gamma \in S$	definition of S
3	$\quad\mid\ S \neq \varnothing$	2
4	$\quad\mid\ \exists\alpha \in S \quad \forall\delta \in S\ (\alpha \le \delta)$	well ordering of Ord
5	$\quad\mid\ \alpha \in S$	4
6	$\quad\mid\ \neg P(\alpha)$	definition of S
7	$\quad\mid\ \forall\beta < \alpha \quad P(\beta)$	minimality of α
8	$\quad\mid\ P(\alpha)$	hypothesis
9	$\quad\mid\ \perp$	6,8
10	$\forall\alpha \in Ord \quad P(\alpha)$	contradiction

□

10.4 Transfinite Recurtion

Theorem 10.4.1 (Transfinite Recursion Theorem)**.** *Let G be an operation. Then there exists a unique operation F such that*

$$F(\alpha) = G(F \restriction \alpha) \quad \textit{for all ordinals } \alpha.$$

Proof. We say t is a **computation of length α based on G** if:

$$dom(t) = \alpha + 1 \quad \text{and} \quad \forall\beta \le \alpha\ t(\beta) = G(t \restriction \beta).$$

Define the property:

$$P(x, y)\ :\ \begin{cases} x \in Ord\ \wedge\ y = t(x) \text{ for some computation } t \text{ of length } x, \\ \text{or } x \notin Ord\ \wedge\ y = \varnothing. \end{cases}$$

We show P defines a function.

Claim 1: For every ordinal α there exists a unique computation of length α.

1	$\forall\beta < \alpha\ \exists! t_\beta$ computation of length β	induction hypothesis
2	$T := \{t_\beta \mid \beta < \alpha\}$	Replacement
3	T is a compatible system of functions	uniqueness
4	$t := \bigcup T$	definition
5	$dom(t) = \alpha$	union of domains
6	t is a function	compatibility of , **??**
7	$\forall\beta < \alpha\ t(\beta) = G(t \restriction \beta)$	compatibility
8	$t(\alpha) = G(t \restriction \alpha)$	definition extension step
9	t computation of length α	definition

Claim 2: Uniqueness of computation of length α.

1	t, s computations of length α	
2	$\forall\beta < \alpha\ t(\beta) = s(\beta)$	induction on β
3	$t = s$	extensionality

Hence P defines a function F.

Define
$$F(\alpha) := t(\alpha)$$
where t is the unique computation of length α.

Claim 3: Recursion equation.

1	$\alpha \in Ord$	
2	t computation of length α	
3	$F(\alpha) = t(\alpha)$	definition
4	$t(\alpha) = G(t \restriction \alpha)$	computation rule
5	$F(\alpha) = G(F \restriction \alpha)$	definition of , **??**

Claim 4: Uniqueness of F.

1	F, S satisfy recursion equation	
2	$\forall\beta < \alpha\ F(\beta) = S(\beta)$	induction
3	$F = S$	extensionality

□

Theorem 10.4.2 (Parametric Transfinite Recursion)**.** *Let G be an operation. Then there exists a unique operation F such that*
$$F(z, \alpha) = G\big(z, F(z) \restriction \alpha\big) \quad \textit{for all sets } z \textit{ and ordinals } \alpha.$$

Proof. Fix a set z.

We say t is a **computation of length α based on G and z** if:

$$dom(t) = \alpha + 1 \quad \wedge \quad \forall \beta \leq \alpha \; t(\beta) = G\big(z, t \restriction \beta\big).$$

Define the property

$$Q(z, x, y) : \begin{cases} x \in Ord \;\wedge\; y = t(x) \text{ for some computation } t \text{ of length } x \text{ based on } G, z, \\ \text{or } x \notin Ord \;\wedge\; y = \varnothing. \end{cases}$$

Claim 1: For every ordinal α there exists a unique computation of length α based on G and z.

1	$\forall \beta < \alpha \; \exists! t_\beta$	induction hypothesis
2	$T := \{t_\beta \mid \beta < \alpha\}$	
3	T is compatible	uniqueness
4	$t := \bigcup T$	
5	$dom(t) = \alpha$	
6	t is a function	compatibility
7	$\forall \beta < \alpha \; t(\beta) = G(z, t \restriction \beta)$	
8	$t(\alpha) = G(z, t \restriction \alpha)$	
9	t computation of length α	

Claim 2: Uniqueness.

1	t, s computations of length α
2	$\forall \beta < \alpha \; t(\beta) = s(\beta)$
3	$t = s$

Hence Q defines a function F.

Define

$$F(z, \alpha) := t(\alpha)$$

where t is the unique computation of length α based on G and z.

Claim 3: Recursion equation.

1	$\alpha \in Ord$
2	t computation of length α
3	$F(z, \alpha) = t(\alpha)$
4	$t(\alpha) = G(z, t \restriction \alpha)$
5	$F(z, \alpha) = G(z, F(z) \restriction \alpha)$

Claim 4: Uniqueness of F.

1 F, S satisfy recursion

2 $\forall \beta < \alpha \; F(z, \beta) = S(z, \beta)$

3 $F = S$

□

10.5 Arithmetic of Ordinals

Theorem 10.5.1. ***Summation*** *there is a unique function "$+ : Ord \times Ord \to Ord$" s.t.*

1. $\forall \alpha \in Ord \quad 0 + \alpha = \alpha = \alpha + 0$
2. $\forall \alpha, \beta \in Ord \quad \alpha + s(\beta) = s(\alpha + \beta) = s(\alpha) + \beta$
3. $\forall \alpha, \beta \in Ord \quad limit(\alpha) \wedge limit(\beta) \to \alpha + \beta = (\bigcup_{\beta' \in \beta} \alpha + \beta') \cup (\bigcup_{\beta' \in \beta} \beta' + \alpha)$

Proof Is Left As An Exercise To The Reader.😃

Theorem 10.5.2. ***Multiplication*** *there is a unique function "$\times : Ord \times Ord \to Ord$" s.t.*

1. $\forall \alpha \in Ord \quad \alpha \times 0 = 0$
2. $\forall \alpha, \beta \in Ord \quad \alpha \times s(\beta) = \alpha \times \beta + \alpha$
3. $\forall \alpha, \beta \in Ord \quad limit(\beta) \wedge \alpha < \beta \to \alpha \times \beta = \beta \times \alpha$
4. $\forall \alpha, \beta \in Ord \quad limit(\beta) \wedge \beta \leq \alpha \to \alpha \times \beta = \bigcup_{\beta' \in trunk(\beta)} (\alpha \times \beta') + \alpha \times branch(\beta)$

Proof Is Left As An Exercise To The Reader.😃

we denote also "Multiplication" with "$\alpha \cdot \beta$" and "$\alpha\beta$" too.

10.6 Transfinite Peano arithmetic

assume a structure "$\mathfrak{N} := (N, +, \cdot, s, 0)$"(here "$+, \cdot, s, 0$" are arbitrary functions and constants and doesn't have meaning we defined before). with "$Limit(x) : \forall y \in N \quad s(y) \neq x$ " and these properties.

1. $\forall x, y \in N \quad s(x) = s(y) \to x = y$
2. $\forall x \in N \quad s(x) \neq 0$
3. $\forall x \in N \quad x + 0 = x$
4. $\forall x \in N \quad x \cdot 0 = 0$
5. $\forall x, y \in N \quad x + s(y) = s(x + y)$
6. $\forall x, y \in N \quad x \cdot s(y) = x \cdot y + x$
7. $\forall x, y \in N \quad Limit(y) \to x + y = y + x$
8. $\forall x, y \in N \quad Limit(y) \to xy = yx$

9. $\forall x, y, z \in N \quad Limit(z) \to (x + y) + z = y + (x + z)$
10. $\forall x, y, z \in N \quad Limit(z) \to (xy)z = y(xz)$
11. $\forall x, y, z \in N \quad Limit(z) \to x(y + z) = xy + xz$
12. $\forall x, y, z \in N \quad Limit(z) \to x + z = y + z \to x = y$
13. $\forall x, y \in N \quad Limit(y) \to xy = 0 \to x = 0 \vee y = 0$
14. $\Big(\forall x \in N \quad Limit(x) \to \phi(x)\Big) \wedge \Big(\forall y \in N \quad \big(\phi(y) \to \phi(s(y))\big)\Big) \to \forall y \in N \quad \phi(y)$ (this is true for arbitrary "ϕ")

we call "$\mathfrak{N}$" a model of "Hyper Peano arithmetic".
Lemma 10.6.1. *prove for "$(N, +)$"*

1. $\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall x \in N \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x, y, z \in N \quad x + z = y + z \to x = y$ *(cancellative +)*
4. $\forall x, y \in N \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃
Lemma 10.6.2. *prove for "$(N, +, \cdot)$"*

1. $\forall x, y, z \in N \quad (xy)z = x(yz)$ *(associative ×)*
2. $\forall x \in N \quad x \cdot 1 = 1 \cdot x = x$ *(identity ×)*
3. $\forall x, y \in N \quad xy = 0 \to x = 0 \vee y = 0$ *(zero divisor free ×)*
4. $\forall x, y \in N \quad xy = yx$ *(commutative(abelian) ×)*
5. $\forall x, y, z \in N \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.😃

Now define "$x < y : \exists z \neq 0 \quad x + z = y$". You can prove for natural numbers

$$\forall \alpha, \beta \in Ord \quad \alpha < \beta \leftrightarrow \exists k \neq 0 \quad \alpha + k = \beta$$

Lemma 10.6.3. *prove for "$(N, +, \times, <)$"*

1. $\forall x, y, z \in N \quad x < y \to x + z < y + z$
2. $\forall x, y, z \in N \quad x < y \wedge 0 < z \to xz < yz$

Proof Is Left As An Exercise To The Reader.😃

as you can see "$(Ord, +, \times, s, 0)$" is a model of hyper peano arithmetic. so properties are true for them.

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Chapter 11

Cardinals

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Chapter 12

Axiom of choice

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Chapter 13

fantacy numbers($\mathbb{Z}(Ord), \mathbb{Q}(Ord), \mathbb{R}(Ord), \mathbb{C}(Ord)$)

13.1 fantacy Integers“$\mathbb{Z}(Ord)$”

we learned about “Ord” before and we constructed structure “$(Ord, +, \cdot, <)$”. if you assume “$Ord \times Ord$” we can define an equivalence relation “$(\alpha_1, \beta_1) \sim (\alpha_2, \beta_2) : \alpha_1 + \beta_2 = \alpha_2 + m_1$”(check it’s an equivalence relation).

definition 13.1.1. ***fantacy Integers“$\mathbb{Z}(Ord)$”*** *is a set*

$$\mathbb{Z}(Ord) := Ord \times Ord/ \sim$$

we denote it’s members with “$z_0, z_1, ...$”

in this system “$0 := [(0, 0)]$” and “$\forall \beta \in Ord \quad \beta := [(\beta, 0)]$” and “$\forall \beta \in Ord \quad -\beta := [(0, \beta)]$”. in general“$\forall \alpha, \beta \in Ord \quad \alpha - \beta := [(\alpha, \beta)]$”. so we can say “$Ord \subseteq \mathbb{Z}(Ord)$”.

Theorem 13.1.1. ***Summation*** *there is a unique function “$+_{\mathbb{Z}}(Ord) : \mathbb{Z}(Ord) \times \mathbb{Z}(Ord) \to \mathbb{Z}(Ord)$” s.t.*

$$\forall \alpha_1, \alpha_2, \beta_1, \beta_2 \in Ord \quad [(\alpha_1, \beta_1)] +_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] = [(\alpha_1 + \alpha_2, \beta_1 + \beta_2)]$$

Proof Is Left As An Exercise To The Reader.😃

Theorem 13.1.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{Z}(Ord)} : \mathbb{Z}(Ord) \times \mathbb{Z}(Ord) \to \mathbb{Z}(Ord)$” s.t.*

$$\forall \alpha_1, \alpha_2, \beta_1, \beta_2 \in Ord \quad [(\alpha_1, \beta_1)] \times_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] = [(\alpha_1\alpha_2 + \beta_1\beta_2, \alpha_1\beta_2 + \alpha_2\beta_1)]$$

Proof Is Left As An Exercise To The Reader.😃

definition 13.1.2. ***Order in*** *$\mathbb{Z}(Ord)$*

$$[(\alpha_1, \beta_1)] <_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] : \alpha_1 + \beta_2 < \alpha_2 + \beta_1$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Z}(Ord)}$", "$+$" instead of "$+_{\mathbb{Z}(Ord)}$" and "$\times$" instead of "$\times_{\mathbb{Z}(Ord)}$".

Lemma 13.1.1. *prove.*

1. $/ \therefore \forall \alpha, \beta \in Ord \quad \alpha <_{\mathbb{Z}(Ord)} \beta \leftrightarrow \alpha < \beta$
2. $/ \therefore \forall \alpha, \beta \in Ord \quad \alpha +_{\mathbb{Z}(Ord)} \beta = \alpha + \beta$
3. $/ \therefore \forall \alpha, \beta \in Ord \quad \alpha \times_{\mathbb{Z}(Ord)} \beta = \alpha \times \beta$

Proof Is Left As An Exercise To The Reader.😃

we denote "Multiplication" with "$z \cdot y$" and "zy" too.

Lemma 13.1.2. *prove for "$(\mathbb{Z}(Ord), +)$"*

1. $\forall x, y, z \in \mathbb{Z}(Ord) \quad (x+y)+z = x+(y+z)$ *(associative +)*
2. $\forall z \in \mathbb{Z}(Ord) \quad z+0 = 0+z = z$ *(identity +)*
3. $\forall z \in \mathbb{Z}(Ord) \quad \exists y \in \mathbb{Z}(Ord) \quad z+y = y+z = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{Z}(Ord) \quad x+y = y+x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 13.1.3. *prove for "$(\mathbb{Z}(Ord), +, \times)$"*

1. $\forall x, y, z \in \mathbb{Z}(Ord) \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall z \in \mathbb{Z}(Ord) \quad z \cdot 1 = 1 \cdot z = z$ *(identity $\times$)*
3. $\forall x, y \in \mathbb{Z}(Ord) \quad xy = 0 \to x = 0 \vee y = 0$ *(zero divisor free $\times$)*
4. $\forall x, y \in \mathbb{Z}(Ord) \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{Z}(Ord) \quad \Big(x(y+z) = xy+xz\Big) \wedge \Big((x+y)z = xz+yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 13.1.3. ***Order of fantacy Integers($\mathbb{Z}(Ord)$)*** *prove for "$(\mathbb{Z}(Ord), +, \times, <_{\mathbb{Z}(Ord)})$"*

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{Z}(Ord) \quad \exists y \in \mathbb{Z}(Ord) \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Z}(Ord) \quad \exists y \in \mathbb{Z}(Ord) \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y, z \in \mathbb{Z}(Ord) \quad x < y \to x+z < y+z$
4. $\forall x, y, z \in \mathbb{Z}(Ord) \quad x < y \wedge 0 < z \to xz < y\gamma$

Proof Is Left As An Exercise To The Reader.😃

13.2 fantacy Rational numbers "$\mathbb{Q}(Ord)$"

we learned about "$\mathbb{Z}(Ord)$" before and we constructed structure "$(\mathbb{Z}(Ord), +, \cdot, <)$". if you assume "$\mathbb{Z}(Ord) \times (\mathbb{Z}(Ord) - \{0\})$" we can define an equivalence relation "$(y_1, x_1) \sim (y_2, x_2) : y_1 \cdot x_2 = y_2 \cdot x_1$"(check it's an equivalence relation).

definition 13.2.1. ***fantacy Rational numbers*“$\mathbb{Q}(Ord)$”** *is a set*

$$\mathbb{Q}(Ord) := \mathbb{Z}(Ord) \times (\mathbb{Z}(Ord) - \{0\})/\sim$$

we denote it's members with “$q_0, q_1, ...$”

in this system “$0 := [(0,1)]$” and “$1 := [(1,1)]$” and “$\forall x \in \mathbb{Z}(Ord) \quad x := [(x,1)]$” and “$\forall x \in \mathbb{Z}(Ord) \quad \frac{1}{x} := [(1,x)]$”. in general “$\forall x, y \in \mathbb{Z}(Ord) \quad \frac{x}{y} := [(x,y)]$”. so we can say “$Ord \subseteq \mathbb{Z}(Ord) \subseteq \mathbb{Q}(Ord)$”.

Theorem 13.2.1. ***Summation*** *there is a unique function “$+_{\mathbb{Q}(Ord)} : \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \to \mathbb{Q}(Ord)$” s.t.*

$$\forall x, y, z, u \in \mathbb{Z}(Ord) \quad [(x,z)] +_{\mathbb{Q}(Ord)} [(y,u)] = [(xu + zy, zu)]$$

Proof Is Left As An Exercise To The Reader.😃

Theorem 13.2.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{Q}(Ord)} : \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \to \mathbb{Q}(Ord)$” s.t.*

$$\forall x, y, z, u \in \mathbb{Z}(Ord) \quad [(x,z)] \times_{\mathbb{Q}(Ord)} [(y,u)] = [(xy, zu)]$$

Proof Is Left As An Exercise To The Reader.😃

definition 13.2.2. ***Order in*** $\mathbb{Q}(Ord)$ *.*

if ($z > 0, u > 0$) or ($z < 0, u < 0$)

$$[(x,z)] <_{\mathbb{Q}(Ord)} [(y,u)] : x \cdot u < y \cdot z$$

if ($z > 0, u < 0$) or ($z < 0, u > 0$)

$$[(x,z)] <_{\mathbb{Q}(Ord)} [(y,u)] : x \cdot u > y \cdot z$$

this lemma let us use “$<$” instead of “$<_{\mathbb{Q}(Ord)}$”, “$+$” instead of “$+_{\mathbb{Q}(Ord)}$” and “$\times$” instead of “$\times_{\mathbb{Q}(Ord)}$”.

Lemma 13.2.1. *prove.*

1. $/ \therefore \forall z, y \in \mathbb{Z}(Ord) \quad z <_{\mathbb{Q}(Ord)} y \leftrightarrow z < y$
2. $/ \therefore \forall z, y \in \mathbb{Z}(Ord) \quad z +_{\mathbb{Q}(Ord)} y = z + y$
3. $/ \therefore \forall z, y \in \mathbb{Z}(Ord) \quad z \times_{\mathbb{Q}(Ord)} y = z \times y$

Proof Is Left As An Exercise To The Reader.😃

we denote “Multiplication” with “$x \cdot y$” and “xy” too.

Lemma 13.2.2. *prove for “$(\mathbb{Q}(Ord), +)$”*

1. $\forall x, y, z \in \mathbb{Q}(Ord) \quad (x + y) + z = x + (y + z)$ *(associative $+$)*
2. $\forall x \in \mathbb{Q}(Ord) \quad x + 0 = 0 + x = x$ *(identity $+$)*
3. $\forall x \in \mathbb{Q}(Ord) \quad \exists y \in \mathbb{Q}(Ord) \quad x + y = y + x = 0$ *(inverse $+$)*
4. $\forall x, y \in \mathbb{Q}(Ord) \quad x + y = y + x$ *(commutative(abelian) $+$)*

Proof Is Left As An Exercise To The Reader.😃

Lemma 13.2.3. *prove for "$(\mathbb{Q}(Ord), +, \times)$"*

1. $\forall x, y, z \in \mathbb{Q}(Ord) \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall x \in \mathbb{Q}(Ord) \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*
3. $\forall x \in \mathbb{Q}(Ord) - \{0\} \quad \exists y \in \mathbb{Q}(Ord) \quad xy = yx = 1$ *(inverse $\times$)*
4. $\forall x, y \in \mathbb{Q}(Ord) \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{Q}(Ord) \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃

Theorem 13.2.3. ***Order of fantacy Rational numbers(**$\mathbb{Q}(Ord)$**)*** *prove for "$(\mathbb{Q}(Ord), <_{\mathbb{Q}(Ord)})$"*

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{Q}(Ord) \quad \exists y \in \mathbb{Q}(Ord) \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Q}(Ord) \quad \exists y \in \mathbb{Q}(Ord) \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{Q}(Ord) \quad \exists z \in \mathbb{Q}(Ord) \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall x, y, z \in \mathbb{Q}(Ord) \quad x < y \to x + z < y + z$
5. $\forall x, y, z \in \mathbb{Q}(Ord) \quad x < y \wedge 0 < z \to xz < y\gamma$

Proof Is Left As An Exercise To The Reader.😃

13.3 fantacy Real numbers "$\mathbb{R}(Ord)$"

before we we get to lesson define function "$|\_| : \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \to \mathbb{Q}(Ord)$" like this.

$$|x| := \begin{cases} x & 0 \leq x \\ -x & x < 0 \end{cases}$$

if you are not familiar to this notation it means "$|\_| := \{(x, y) \in \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \mid (0 \leq x \to y = x) \wedge (x < 0 \to y = -x)\}$". learn this notation because we want to use later.

we learned about "$\mathbb{Q}(Ord)$" before and we constructed structure "$(\mathbb{Q}(Ord), +, \cdot, <)$". now assume $\mathbb{Q}(Ord)^{Ord}$(we know what does it mean by the definition. it means all the sequences of $\mathbb{Q}(Ord)$). we don't want it all we just want a subset.

definition 13.3.1. ***Cauchy sequence*** *is a sequence like "$(a_i)_{i \in Ord} \in \mathbb{Q}(Ord)^{Ord}$" s.t. elements of it get infinitely close to each other i.e.*

$$Cauchy(a_i)_{i \in Ord} : \forall \varepsilon > 0 \quad \exists N \quad \forall n, m \quad N < n, m < roof(N) \to |a_n - a_m| < \varepsilon$$

we usually give "N" natural numbers but it's not necessary.

we define an equivalence relation "$(a_i)_{i \in Ord} \sim (b_i)_{i \in Ord} : \forall \varepsilon > 0 \quad \exists N \quad \forall n \quad N < n < roof(N) \to |a_n - b_n| < \varepsilon$"(check it's an equivalence relation). we denote every $[(a_i)_{i \in Ord}]$ with $[a_i]_{i \in Ord}$.

definition 13.3.2. ***fantacy Real numbers* “$\mathbb{R}(Ord)$”** *is a set*

$$\mathbb{R}(Ord) := Cauchy/\sim$$

we denote it's members with “$r_0, r_1, \ldots$”

in this system “$0 := [0]_{i\in Ord}$” and “$1 := [1]_{i\in Ord}$” and “$\forall q \in \mathbb{Q}(Ord) \quad q := [q]_{i\in Ord}$”. so we can say “$Ord \subseteq \mathbb{Z}(Ord) \subseteq \mathbb{Q}(Ord) \subseteq \mathbb{R}(Ord)$”.

Theorem 13.3.1. ***Summation*** *there is a unique function “$+_{\mathbb{R}(Ord)} : \mathbb{R}(Ord) \times \mathbb{R}(Ord) \to \mathbb{R}(Ord)$” s.t.*

$$[a_i]_{i\in Ord} +_{\mathbb{R}(Ord)} [b_i]_{i\in Ord} = [a_i + b_i]_{i\in Ord}$$

Proof Is Left As An Exercise To The Reader.☻

Theorem 13.3.2. ***Multiplication*** *there is a unique function “$\times_{\mathbb{R}(Ord)} : \mathbb{R}(Ord) \times \mathbb{R}(Ord) \to \mathbb{R}(Ord)$” s.t.*

$$[a_i]_{i\in Ord} \times_{\mathbb{R}(Ord)} [b_i]_{i\in Ord} = [a_i b_i]_{i\in Ord}$$

Proof Is Left As An Exercise To The Reader.☻

definition 13.3.3. ***Order in*** $\mathbb{R}(Ord)$ *.*

$$[a_i]_{i\in Ord} <_{\mathbb{R}(Ord)} [b_i]_{i\in Ord} : \exists \varepsilon > 0 \quad \exists N \quad \forall n \quad N < n < roof(N) \to b_n - a_n > \varepsilon$$

this lemma let us use “$<$” instead of “$<_{\mathbb{R}(Ord)}$”, “$+$” instead of “$+_{\mathbb{R}(Ord)}$” and “$\times$” instead of “$\times_{\mathbb{R}(Ord)}$”.

Lemma 13.3.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{Q}(Ord) \quad x <_{\mathbb{R}(Ord)} y \leftrightarrow x < y$
2. $/ \therefore \forall x, y \in \mathbb{Q}(Ord) \quad x +_{\mathbb{R}(Ord)} y = x + y$
3. $/ \therefore \forall x, y \in \mathbb{Q}(Ord) \quad x \times_{\mathbb{R}(Ord)} y = x \times y$

Proof Is Left As An Exercise To The Reader.☻

we denote “Multiplication” with “$x \cdot y$” and “xy” too.

Lemma 13.3.2. *prove.*

1. $\forall x, y, z \in \mathbb{R}(Ord) \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall x \in \mathbb{R}(Ord) \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x \in \mathbb{R}(Ord) \quad \exists y \in \mathbb{R}(Ord) \quad x + y = y + x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{R}(Ord) \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.☻

Lemma 13.3.3. *prove.*

1. $\forall x, y, z \in \mathbb{R}(Ord) \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall x \in \mathbb{R}(Ord) \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*

3. $\forall x \in \mathbb{R}(Ord) - \{0\} \quad \exists y \in \mathbb{R}(Ord) \quad xy = yx = 1$ *(inverse $\times$)*
4. $\forall x, y \in \mathbb{R}(Ord) \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{R}(Ord) \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.😃
Theorem 13.3.3. ***Linear order of fantacy Real numbers(**$\mathbb{R}(Ord)$**)*** *prove "($\mathbb{R}(Ord), <_{\mathbb{R}(Ord)}$)*

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{R}(Ord) \quad \exists y \in \mathbb{R}(Ord) \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{R}(Ord) \quad \exists y \in \mathbb{R}(Ord) \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{R}(Ord) \quad \exists z \in \mathbb{R}(Ord) \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall x, y, z \in \mathbb{R}(Ord) \quad x < y \to x + z < y + z$
5. $\forall x, y, z \in \mathbb{R}(Ord) \quad x < y \wedge 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.😃

13.4 fantacy Complex numbers"$\mathbb{C}(Ord)$"

we learned about "$\mathbb{R}(Ord)$" before and we constructed structure "$(\mathbb{R}(Ord), +, \cdot, <)$".
definition 13.4.1. ***fantacy Complex numbers"**$\mathbb{C}(Ord)$**"*** *is a set*

$$\mathbb{C}(Ord) := \mathbb{R}(Ord) \times \mathbb{R}(Ord)$$

we denote it's members with "$z_0, x, \ldots$"

in this system "$0 := (0,0)$" and "$\forall n \in \mathbb{R}(Ord) \quad r := (r, 0)$" . in general "$\forall x, y \in \mathbb{R}(Ord) \quad x + iy := (x, y)$". so we can say "$Ord \subseteq \mathbb{Z}(Ord) \subseteq \mathbb{Q}(Ord) \subseteq \mathbb{R}(Ord) \subseteq \mathbb{C}(Ord)$".
Theorem 13.4.1. ***Summation*** *there is a unique function "$+_{\mathbb{C}(Ord)} : \mathbb{C}(Ord) \times \mathbb{C}(Ord) \to \mathbb{C}(Ord)$" s.t.*

$$\forall x, y, z, u \in \mathbb{R}(Ord) \quad (x, z) +_{\mathbb{C}(Ord)} (y, u) = (x + y, z + u)$$

Proof Is Left As An Exercise To The Reader.😃
Theorem 13.4.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{C}(Ord)} : \mathbb{C}(Ord) \times \mathbb{C}(Ord) \to \mathbb{C}(Ord)$" s.t.*

$$\forall x, y, z, u \in \mathbb{R}(Ord) \quad (x, z) \times_{\mathbb{C}(Ord)} (y, u) = (xy - zu, xu + yz)$$

Proof Is Left As An Exercise To The Reader.😃

this lemma let us use "$+$" instead of "$+_{\mathbb{C}(Ord)}$" and "$\times$" instead of "$\times_{\mathbb{C}(Ord)}$".
Lemma 13.4.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{R}(Ord) \quad x +_{\mathbb{C}(Ord)} y = x + y$

2. $/ \therefore \forall x, y \in \mathbb{R}(Ord) \quad x \times_{\mathbb{C}(Ord)} y = x \times y$

Proof Is Left As An Exercise To The Reader.☻

we denote "Multiplication" with "$z \cdot y$" and "zy" too.
Lemma 13.4.2. *prove for "$(\mathbb{C}(Ord), +)$"*

1. $\forall x, y, z \in \mathbb{C}(Ord) \quad (x+y)+z = x+(y+z)$ *(associative +)*
2. $\forall z \in \mathbb{C}(Ord) \quad z+0 = 0+z = z$ *(identity +)*
3. $\forall z \in \mathbb{C}(Ord) \quad \exists y \in \mathbb{C}(Ord) \quad z+y = y+z = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{C}(Ord) \quad x+y = y+x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.☻
Lemma 13.4.3. *prove for "$(\mathbb{C}(Ord), +, \times)$"*

1. $\forall x, y, z \in \mathbb{C}(Ord) \quad (xy)z = x(yz)$ *(associative ×)*
2. $\forall z \in \mathbb{C}(Ord) \quad z \cdot 1 = 1 \cdot z = z$ *(identity ×)*
3. $\forall x, y \in \mathbb{C}(Ord) \quad xy = yx$ *(commutative(abelian) ×)*
4. $\forall x, y, z \in \mathbb{C}(Ord) \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.☻
Theorem 13.4.3. ***Exponents*** *there is a unique function "$(\_)^{(\_)} : \mathbb{C}(Ord) \times Ord \to \mathbb{C}(Ord)$" s.t.*

1. $\forall x \in \mathbb{C}(Ord) \quad x^0 = 1$
2. $\forall x \in \mathbb{C}(Ord) \quad \forall n \in Ord \quad x^{s(n)} = x^n \cdot x$

Proof Is Left As An Exercise To The Reader.☻
Lemma 13.4.4. *prove for "$(\mathbb{C}(Ord), +, \cdot, (\_)^{(\_)})$"*

1. $\forall x \in \mathbb{Z}(Ord) \quad \forall m, n \in Ord \quad x^m \cdot x^n = x^{m+n}$
2. $\forall x \in \mathbb{Z}(Ord) \quad \forall m, n \in Ord \quad (x^m)^n = x^{mn}$

Proof Is Left As An Exercise To The Reader.☻
Theorem 13.4.4. ***Exponents*** *there is a unique function "$(\_)^{(\_)} : \mathbb{C}(Ord) \times \mathbb{Z}(Ord) \to \mathbb{C}(Ord)$" s.t.*

1. $\forall x \in \mathbb{C}(Ord) \quad x^0 = 1$
2. $\forall x \in \mathbb{C}(Ord) \quad \forall m, n \in Ord \quad x^{m-n} = \frac{x^m}{x^n}$

Proof Is Left As An Exercise To The Reader.☻
Lemma 13.4.5. *prove for "$(\mathbb{C}(Ord), +, \cdot)$"*

1. $\forall x \in \mathbb{C}(Ord) \quad \forall y, z \in \mathbb{Z}(Ord) \quad x^y \cdot x^z = x^{y+z}$
2. $\forall x \in \mathbb{C}(Ord) \quad \forall y, z \in \mathbb{Z}(Ord) \quad (x^y)^z = x^{yz}$

Proof Is Left As An Exercise To The Reader.☻
Theorem 13.4.5. ***Big Sum*** *if "$a : Ord \to \mathbb{C}(Ord)$" there is a unique function "$\sum_{i=0}^{(\_)}(\_) : \mathbb{C}(Ord) \times Ord \to \mathbb{C}(Ord)$" s.t.*

1. $\forall a \quad \sum_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in Ord \quad \sum_{i=0}^{s(n)} a_{s(n)} = \sum_{i=0}^{n} a_n + a_{s(n)}$

Proof Is Left As An Exercise To The Reader.😃

Lemma 13.4.6. *prove for "$(\mathbb{C}(Ord), +, \cdot)$"*

1. $\forall n \in Ord \quad \forall a, b \quad \sum_{i=0}^{n} a_n + \sum_{i=0}^{n} b_n = \sum_{i=0}^{n}(a_n + b_n)$
2. $\forall n \in Ord \quad \forall a \quad k \sum_{i=0}^{n} a_n = \sum_{i=0}^{n} k a_n$

Proof Is Left As An Exercise To The Reader.😃

Theorem 13.4.6. ***Big Product*** *if "$a : Ord \to \mathbb{C}(Ord)$" there is a unique function "$\prod_{i=0}^{(\_)}(\_) : \mathbb{C}(Ord) \times Ord \to \mathbb{C}(Ord)$" s.t.*

1. $\forall a \quad \prod_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in Ord \quad \prod_{i=0}^{s(n)} a_{s(n)} = \prod_{i=0}^{n} a_n \cdot a_{s(n)}$

Proof Is Left As An Exercise To The Reader.😃

Lemma 13.4.7. *prove for "$(\mathbb{C}(Ord), +, \cdot)$"*

1. $\forall n \in Ord \quad \forall a, b \quad \prod_{i=0}^{n} a_n \prod_{i=0}^{n} b_n = \prod_{i=0}^{n}(a_n b_n)$
2. $\forall n \in Ord \quad \forall a \quad (\prod_{i=0}^{n} a_n)^k = \prod_{i=0}^{n}(a_n)^k$

Proof Is Left As An Exercise To The Reader.😃

Chapter 14

hyperreal numbers($^*\mathbb{N}, {}^*\mathbb{Z}, {}^*\mathbb{Q}, {}^*\mathbb{R}$)

14.1 Filters

Filters are mathematical objects used to formalize the idea of "large subsets" of a set.
definition 14.1.1 (Filter)**.** *Let X be a nonempty set. A* filter *$\mathcal{F}$ on X is a collection of subsets of X such that:*

1. $\varnothing \notin \mathcal{F}$.
2. $X \in \mathcal{F}$.
3. *If $A, B \in \mathcal{F}$, then*

$$A \cap B \in \mathcal{F}.$$

4. *If $A \in \mathcal{F}$ and $A \subseteq B \subseteq X$, then*

$$B \in \mathcal{F}.$$

Intuitively, the sets in $\mathcal{F}$ are considered "large". The intersection of two large sets is still large, and every superset of a large set is also large.
Example 14.1.1 (Trivial Filter)**.** *For any set X,*

$$\mathcal{F} = \{X\}$$

is a filter on X.
Example 14.1.2 (Principal Filter)**.** *Fix $A \subseteq X$ with $A \neq \varnothing$. Define*

$$\mathcal{F}_A := \{B \subseteq X : A \subseteq B\}.$$

Then $\mathcal{F}_A$ is a filter on X. This is called the principal filter generated by *A.*

Example 14.1.3 (Fréchet Filter)**.** *Let $X = \mathbb{N}$. Define*

$$\mathcal{F} = \{A \subseteq \mathbb{N} : \mathbb{N} \setminus A \textit{ is finite}\}.$$

Then $\mathcal{F}$ is a filter. Its elements are the cofinite subsets of $\mathbb{N}$. This is called the Fréchet filter*.*

14.2 Ultrafilters

Ultrafilters are maximal filters.

definition 14.2.1 (Ultrafilter)**.** *A filter $\mathcal{U}$ on X is called an* ultrafilter *if it is maximal among filters; that is, whenever $\mathcal{F}$ is a filter on X and $\mathcal{U} \subseteq \mathcal{F}$, then*

$$\mathcal{F} = \mathcal{U}.$$

There is a very important characterization of ultrafilters.

Theorem 14.2.1 (Ultrafilter Characterization)**.** *Let $\mathcal{U}$ be a filter on X. Then $\mathcal{U}$ is an ultrafilter if and only if for every subset $A \subseteq X$, exactly one of the following holds:*

$$A \in \mathcal{U}, \quad \textit{or} \quad X \setminus A \in \mathcal{U}.$$

Proof. Suppose first that $\mathcal{U}$ is an ultrafilter. Let $A \subseteq X$. We cannot have both $A \in \mathcal{U}$ and $X \setminus A \in \mathcal{U}$, since then $\varnothing = A \cap (X \setminus A) \in \mathcal{U}$, which is impossible. Suppose that neither A nor $X \setminus A$ belongs to $\mathcal{U}$. Then the filter generated by $\mathcal{U} \cup \{A\}$ is strictly larger than $\mathcal{U}$, contradicting the maximality of $\mathcal{U}$. Hence exactly one of them belongs to $\mathcal{U}$.

Conversely, suppose that for every $A \subseteq X$, exactly one of A and $X \setminus A$ belongs to $\mathcal{U}$. Let $\mathcal{F}$ be a filter on X such that $\mathcal{U} \subseteq \mathcal{F}$. If $\mathcal{F} \neq \mathcal{U}$, choose $A \in \mathcal{F} \setminus \mathcal{U}$. By assumption, $X \setminus A \in \mathcal{U} \subseteq \mathcal{F}$. Therefore

$$A \cap (X \setminus A) = \varnothing \in \mathcal{F},$$

a contradiction. Hence $\mathcal{F} = \mathcal{U}$, so $\mathcal{U}$ is an ultrafilter.

□

Thus an ultrafilter makes a complete decision about every subset of X: exactly one of A and $X \setminus A$ is "large".

Example 14.2.1. *Fix $x \in X$. Define*

$$\mathcal{U}_x := \{A \subseteq X : x \in A\}.$$

Then $\mathcal{U}_x$ is an ultrafilter. It is called the principal ultrafilter at *x.*

Proof. First, $\varnothing \notin \mathcal{U}_x$ and $X \in \mathcal{U}_x$. If $A, B \in \mathcal{U}_x$, then $x \in A \cap B$, so $A \cap B \in \mathcal{U}_x$. If $A \in \mathcal{U}_x$ and $A \subseteq B$, then $x \in B$, so $B \in \mathcal{U}_x$. Thus $\mathcal{U}_x$ is a filter.

Moreover, for every $A \subseteq X$, either $x \in A$ or $x \notin A$. Hence either

$$A \in \mathcal{U}_x, \qquad X \setminus A \in \mathcal{U}_x.$$

By the Ultrafilter Characterization, $\mathcal{U}_x$ is an ultrafilter.

□

14.3 Nonprincipal Ultrafilters

An ultrafilter on $\mathbb{N}$ is called *nonprincipal* if it contains no finite set.

Equivalently, it extends the Fréchet filter.

Nonprincipal ultrafilters cannot be explicitly constructed in ordinary mathematics; their existence usually requires the Axiom of Choice.

definition 14.3.1 (Extension of a Filter)**.** *Let $\mathcal{F}$ and $\mathcal{G}$ be filters on a set X.*

We say that $\mathcal{G}$ extends $\mathcal{F}$ if

$$\mathcal{F} \subseteq \mathcal{G}.$$

This means every set considered "large" by $\mathcal{F}$ is also considered "large" by $\mathcal{G}$.

Theorem 14.3.1 (Ultrafilter Lemma)**.** *Every filter on a set X can be extended to an ultrafilter.*

In other words, if $\mathcal{F}$ is a filter on X, then there exists an ultrafilter $\mathcal{U}$ such that

$$\mathcal{F} \subseteq \mathcal{U}.$$

So every filter can be enlarged until it becomes maximal.

Example 14.3.1. *Let*

$$\mathcal{F} = \{A \subseteq \mathbb{N} : \mathbb{N} \setminus A \text{ is finite}\}.$$

This is the Fréchet filter. It is not an ultrafilter, because for the set of even numbers

$$E = \{0, 2, 4, 6, \dots\},$$

neither $E \in \mathcal{F}$ nor $\mathbb{N} \setminus E \in \mathcal{F}$. By the Ultrafilter Lemma, there exists an ultrafilter $\mathcal{U}$ such that $\mathcal{F} \subseteq \mathcal{U}$. Since $\mathcal{U}$ is an ultrafilter, exactly one of

$$E \in \mathcal{U}, \quad \text{or} \quad \mathbb{N} \setminus E \in \mathcal{U}$$

holds.

The Ultrafilter Lemma is usually proved using Zorn's Lemma.

Theorem 14.3.2 (Ultrafilter Lemma)**.** *Let $\mathcal{F}$ be a filter on a set X. Then there exists an ultrafilter $\mathcal{U}$ on X such that*

$$\mathcal{F} \subseteq \mathcal{U}.$$

Proof. Let

$$\mathcal{P} = \{\mathcal{G} : \mathcal{G} \text{ is a filter on } X \text{ and } \mathcal{F} \subseteq \mathcal{G}\}.$$

We partially order $\mathcal{P}$ by inclusion. Let $\mathcal{C} \subseteq \mathcal{P}$ be a chain and define

$$\mathcal{H} = \bigcup_{\mathcal{G} \in \mathcal{C}} \mathcal{G}.$$

Clearly $\varnothing \notin \mathcal{H}$ and $X \in \mathcal{H}$. If $A, B \in \mathcal{H}$, then $A \in \mathcal{G}_1$ and $B \in \mathcal{G}_2$ for some $\mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}$. Since $\mathcal{C}$ is a chain, one of $\mathcal{G}_1 \subseteq \mathcal{G}_2$ or $\mathcal{G}_2 \subseteq \mathcal{G}_1$ holds, so A, B belong to the larger filter and hence

$$A \cap B \in \mathcal{H}.$$

If $A \in \mathcal{H}$ and $A \subseteq B \subseteq X$, then $A \in \mathcal{G}$ for some $\mathcal{G} \in \mathcal{C}$, and since $\mathcal{G}$ is a filter, $B \in \mathcal{G} \subseteq \mathcal{H}$. Thus $\mathcal{H}$ is a filter. Moreover,

$$\mathcal{F} \subseteq \mathcal{H},$$

so $\mathcal{H} \in \mathcal{P}$. Therefore every chain in $\mathcal{P}$ has an upper bound, and by Zorn's Lemma, $\mathcal{P}$ has a maximal element, say $\mathcal{U}$.

We claim that $\mathcal{U}$ is an ultrafilter. Suppose $\mathcal{U} \subseteq \mathcal{V}$, where $\mathcal{V}$ is a filter on X. Then

$$\mathcal{F} \subseteq \mathcal{U} \subseteq \mathcal{V},$$

so $\mathcal{V} \in \mathcal{P}$. By the maximality of $\mathcal{U}$,

$$\mathcal{V} = \mathcal{U}.$$

Therefore $\mathcal{U}$ is an ultrafilter extending $\mathcal{F}$. □

Theorem 14.3.3. *Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$. Then*

$$\mathcal{F}_{\mathrm{Fr}} \subseteq \mathcal{U}.$$

That is, every cofinite subset of $\mathbb{N}$ belongs to $\mathcal{U}$.

Proof. Let $A \in \mathcal{F}_{\mathrm{Fr}}$. Then $\mathbb{N} \setminus A$ is finite. Suppose, toward a contradiction, that $A \notin \mathcal{U}$. Since $\mathcal{U}$ is an ultrafilter, exactly one of

$$A \in \mathcal{U}, \quad \text{or} \quad \mathbb{N} \setminus A \in \mathcal{U}$$

holds. Hence $\mathbb{N} \setminus A \in \mathcal{U}$. Thus $\mathcal{U}$ contains a finite set, say

$$F = \{n_1, \dots, n_k\} \in \mathcal{U}.$$

Suppose that $\{n_i\} \notin \mathcal{U}$ for every i. Then, since $\mathcal{U}$ is an ultrafilter,

$$\mathbb{N} \setminus \{n_i\} \in \mathcal{U}$$

for every i. By closure under finite intersections,

$$\mathbb{N} \setminus F = \bigcap_{i=1}^{k} (\mathbb{N} \setminus \{n_i\}) \in \mathcal{U}.$$

Since also $F \in \mathcal{U}$, we obtain

$$F \cap (\mathbb{N} \setminus F) = \varnothing \in \mathcal{U},$$

a contradiction. Therefore $\{n_i\} \in \mathcal{U}$ for some i, so $\mathcal{U}$ is principal, contradicting our assumption.

Hence $A \in \mathcal{U}$. Since A was arbitrary,

$$\mathcal{F}_{\mathrm{Fr}} \subseteq \mathcal{U}.$$

□

14.4 Definition of ${}^*\mathbb{N}$, ${}^*\mathbb{Z}$, ${}^*\mathbb{Q}$, ${}^*\mathbb{R}$, and ${}^*\mathbb{R}_{Ord}$

definition 14.4.1 (Ultrapower Construction)**.** *Let X be a set and let $\mathcal{U}$ be a nonprincipal ultrafilter on ω. Consider the set*

$$X^\omega$$

of all functions $a : \omega \to X$. We usually write such a function as

$$a = (a_i)_{i\in\omega},$$

where $a_i \in X$ for every $i \in \omega$.

definition 14.4.2. *Define a relation $\sim$ on X^ω by*

$$a \sim b$$

if

$$\{i \in \omega : a_i = b_i\} \in \mathcal{U}.$$

In words, two sequences are equivalent if they agree on a set of indices belonging to the ultrafilter.

Theorem 14.4.1. *The relation $\sim$ is an equivalence relation on X^ω.*

Proof. **Reflexivity.** For every $a \in X^\omega$,

$$\{i \in \omega : a_i = a_i\} = \omega \in \mathcal{U},$$

so $a \sim a$.

Symmetry. If $a \sim b$, then

$$\{i \in \omega : a_i = b_i\} = \{i \in \omega : b_i = a_i\},$$

so $b \sim a$.

Transitivity. Suppose $a \sim b$ and $b \sim c$. Then

$$A = \{i \in \omega : a_i = b_i\} \in \mathcal{U}, \qquad B = \{i \in \omega : b_i = c_i\} \in \mathcal{U}.$$

Since $\mathcal{U}$ is a filter, $A \cap B \in \mathcal{U}$. If $i \in A \cap B$, then $a_i = b_i = c_i$, so

$$A \cap B \subseteq \{i \in \omega : a_i = c_i\}.$$

Since $\mathcal{U}$ is closed upward,

$$\{i \in \omega : a_i = c_i\} \in \mathcal{U},$$

and therefore $a \sim c$. □

definition 14.4.3. *The quotient*

$$X^\omega / \sim$$

is called the ultrapower *of X modulo $\mathcal{U}$. The equivalence class of $a \in X^\omega$ is denoted by $[a]$.*

We now apply this construction to the standard number systems.
definition 14.4.4. *The set of hypernatural numbers is*

$${}^*\mathbb{N} = \mathbb{N}^\omega / \sim .$$

definition 14.4.5. *The set of hyperintegers is*

$${}^*\mathbb{Z} = \mathbb{Z}^\omega / \sim .$$

definition 14.4.6. *The set of hyperrational numbers is*

$${}^*\mathbb{Q} = \mathbb{Q}^\omega / \sim .$$

definition 14.4.7. *The set of hyperreal numbers is*

$${}^*\mathbb{R} = \mathbb{R}^\omega / \sim .$$

definition 14.4.8 (The Structure ${}^*\mathbb{R}_{Ord}$)**.** *For each ordinal α, let ${}^*\mathbb{R}_\alpha$ denote the corresponding stage of hyperreal extension. Define*

$${}^*\mathbb{R}_{Ord} := \bigcup_{\alpha \in \mathrm{Ord}} {}^*\mathbb{R}_\alpha.$$

Since Ord $\subseteq {}^*\mathbb{R}_{Ord}$*, every ordinal β is itself an element of ${}^*\mathbb{R}_{Ord}$. In particular, given $N \in {}^*\mathbb{R}_{Ord}$ and the index set $\{0, \dots, N-1\} \subseteq {}^*\mathbb{R}_{Ord}$, we say $i \in \{0, \dots, N-1\}$* ***is an ordinal*** *if $i \in$* Ord *(as opposed to i being a non-ordinal element of some ${}^*\mathbb{R}_\alpha$).*

The algebraic operations are defined componentwise. For example, if $[a], [b] \in {}^*\mathbb{R}$, then

$$[a] + [b] = [(a_i + b_i)_{i\in\omega}], \qquad [a] \cdot [b] = [(a_i b_i)_{i\in\omega}].$$

Similarly, the order relation is defined by

$$[a] < [b]$$

if

$$\{i \in \omega : a_i < b_i\} \in \mathcal{U}.$$

Theorem 14.4.2. *The standard structures $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ embed into ${}^*\mathbb{N}$, ${}^*\mathbb{Z}$, ${}^*\mathbb{Q}$, and ${}^*\mathbb{R}$, respectively.*

Proof. We prove the statement for $\mathbb{R}$. Define

$$\iota : \mathbb{R} \to {}^*\mathbb{R}, \qquad \iota(r) = [(r, r, r, \dots)].$$

Suppose $\iota(r) = \iota(s)$. Then

$$\{i \in \omega : r = s\} \in \mathcal{U}.$$

If $r \neq s$, then $\{i \in \omega : r = s\} = \varnothing$, which cannot belong to a filter. Hence $r = s$, so ι is injective. $\square$

Example 14.4.1. *Define* $a_i = i$ *for every* $i \in \omega$. *Then*

$$[a] \in {}^*\mathbb{N}$$

is larger than every standard natural number. Indeed, for every $n \in \mathbb{N}$,

$$\{i \in \omega : n < a_i\} = \{i \in \omega : n < i\}$$

is cofinite, hence belongs to $\mathcal{U}$. *Therefore*

$$n < [a].$$

14.5 The Ultrapower Construction of the Hyperreals

Let $r = (r_n)_{n\in\mathbb{N}}$ and $s = (s_n)_{n\in\mathbb{N}}$ be two sequences of real numbers. We denote the set $\{n \in \mathbb{N} : r_n = s_n\}$ by $[\![r = s]\!]$. Thus

$$[\![r = s]\!] = \{n \in \mathbb{N} : r_n = s_n\}.$$

The set $[\![r = s]\!]$ may be thought of as the interpretation, or value, of the statement "$r = s$". Instead of assigning a truth value to the statement, we assign the set of indices at which it is true.

Let $\mathcal{F}$ be a free ultrafilter on $\mathbb{N}$. We define $r \sim s$ if and only if

$$[\![r = s]\!] \in \mathcal{F}.$$

In this case we say that $r = s$ holds **almost everywhere modulo** $\mathcal{F}$.

The same idea applies to other assertions. We define

$$[\![r < s]\!] = \{n \in \mathbb{N} : r_n < s_n\}, \qquad [\![r > s]\!] = \{n \in \mathbb{N} : r_n > s_n\},$$

and

$$[\![r \le s]\!] = \{n \in \mathbb{N} : r_n \le s_n\}.$$

Thus, for example, $[\![r < s]\!] \in \mathcal{F}$ means that $r_n < s_n$ for $\mathcal{F}$-almost all n. We call these sets the **Statement Values** of the corresponding statements.

Theorem 14.5.1. *The relation* $\sim$ *is an equivalence relation on* $\mathbb{R}^{\mathbb{N}}$.

Proof. For every $r \in \mathbb{R}^{\mathbb{N}}$, $[\![r = r]\!] = \mathbb{N} \in \mathcal{F}$, so $r \sim r$.

If $r \sim s$, then $[\![r = s]\!] = [\![s = r]\!] \in \mathcal{F}$, so $s \sim r$.

Finally, suppose $r \sim s$ and $s \sim t$. Then $[\![r = s]\!], [\![s = t]\!] \in \mathcal{F}$, so

$$[\![r = s]\!] \cap [\![s = t]\!] \in \mathcal{F}.$$

Since

$$[\![r = s]\!] \cap [\![s = t]\!] \subseteq [\![r = t]\!],$$

and $\mathcal{F}$ is closed upward, $[\![r = t]\!] \in \mathcal{F}$. Hence $r \sim t$.

$\square$

definition 14.5.1. *The equivalence class of $r \in \mathbb{R}^{\mathbb{N}}$ under $\sim$ is denoted by*

$$[r] = \{s \in \mathbb{R}^{\mathbb{N}} : r \sim s\}.$$

definition 14.5.2. *For $r, s \in \mathbb{R}^{\mathbb{N}}$, define addition and multiplication on $^*\mathbb{R}$ by*

$$[r] + [s] = [r + s], \qquad [r] \cdot [s] = [r \cdot s],$$

where $(r + s)_n = r_n + s_n$ and $(r \cdot s)_n = r_n s_n$.

We define the order relation by

$$[r] < [s] \leftrightarrow [\![r < s]\!] \in \mathcal{F}.$$

These definitions are well-defined. If $r \sim r'$ and $s \sim s'$, then $r + s \sim r' + s'$ and $r \cdot s \sim r' \cdot s'$. Similarly,

$$[r] < [s] \leftrightarrow [r'] < [s'].$$

Theorem 14.5.2 (Almost-All Criterion)**.** *For sequences $r = (r_n)$, $s = (s_n)$, and $t = (t_n)$, we have*

$$[r] = [s] \leftrightarrow r_n = s_n \quad \textit{for } \mathcal{F}\textit{-almost all } n,$$

$$[r] < [s] \leftrightarrow r_n < s_n \quad \textit{for } \mathcal{F}\textit{-almost all } n,$$

$$[r] + [s] = [t] \leftrightarrow r_n + s_n = t_n \quad \textit{for } \mathcal{F}\textit{-almost all } n,$$

and

$$[r] \cdot [s] = [t] \leftrightarrow r_n s_n = t_n \quad \textit{for } \mathcal{F}\textit{-almost all } n.$$

The same principle holds for many other mathematical statements and is the basis of the ***Transfer Principle****.*

Theorem 14.5.3. *The structure*

$$({}^*\mathbb{R}, +, \cdot, <)$$

is an ordered field with zero $[0]$ *and unity* $[1]$.

Proof. The ring properties follow directly from the corresponding properties of $\mathbb{R}$. In particular, the additive inverse of $[r]$ is $-[r] = [-r]$.

It remains to show that every nonzero element has a multiplicative inverse. Suppose $[r] \neq [0]$. Then $[\![r = 0]\!] \notin \mathcal{F}$. Since $\mathcal{F}$ is an ultrafilter,

$$J = \{n \in \mathbb{N} : r_n \neq 0\} \in \mathcal{F}.$$

Define

$$s_n = \begin{cases} \dfrac{1}{r_n}, & n \in J, \\ 0, & n \notin J. \end{cases}$$

Then $[\![r \cdot s = 1]\!] = J \in \mathcal{F}$, so $[r] \cdot [s] = [1]$. Thus $[s]$ is the multiplicative inverse of $[r]$.

Finally, since $\mathcal{F}$ is an ultrafilter, for any r, s, exactly one of

$$[\![r < s]\!], \qquad [\![r = s]\!], \qquad [\![s < r]\!]$$

belongs to $\mathcal{F}$. Hence the order on ${}^*\mathbb{R}$ is linear. Therefore

$$({}^*\mathbb{R}, +, \cdot, <)$$

is an ordered field.

□

definition 14.5.3. *We identify every real number* $r \in \mathbb{R}$ *with the equivalence class of the constant sequence* $(r, r, r, \ldots)$. *Thus*

$$r = [(r, r, r, \ldots)].$$

Under this identification,

$$\mathbb{R} \subseteq {}^*\mathbb{R}.$$

For example, consider

$$\varepsilon = \left[\left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right)\right].$$

For every $n \in \mathbb{N}$, $0 < \varepsilon < 1/n$, and therefore ε is a positive infinitesimal.

Similarly,

$$\omega = [(1, 2, 3, 4, \dots)]$$

satisfies

$$\omega > n \qquad \forall n \in \mathbb{N}.$$

Therefore ω is a positive infinite hyperreal number.

definition 14.5.4. *Let $f : A \to \mathbb{R}$, where $A \subseteq \mathbb{R}$. The **Nonstandard Extension** of f is the function*

$$^*f : {}^*A \to {}^*\mathbb{R}$$

defined by

$$^*f([x_n]) = [f(x_n)],$$

where

$$^*A = \{[x_n] : x_n \in A\}.$$

The definition is well-defined. If $[x_n] = [y_n]$, then

$$\{n \in \mathbb{N} : x_n = y_n\} \in \mathcal{F}.$$

Therefore

$$\{n \in \mathbb{N} : f(x_n) = f(y_n)\} \in \mathcal{F},$$

so $[f(x_n)] = [f(y_n)]$. Thus the value of $^*f([x_n])$ does not depend on the representative chosen for $[x_n]$.

Moreover, if $x \in A$ is standard, then

$$^*f(x) = f(x).$$

Indeed,

$$^*f([(x, x, x, \ldots)]) = [(f(x), f(x), f(x), \ldots)] = f(x).$$

Therefore $^*f$ is genuinely an extension of f.

The Almost-All Criterion is the fundamental reason why this construction allows ordinary statements about real numbers to be extended to nonstandard numbers. For example, if $f(x) = x^2$ and $\omega = [(1, 2, 3, \ldots)]$, then

$$^*f(\omega) = [(1, 4, 9, \ldots)].$$

Thus expressions such as

$$^*f(a + \varepsilon)$$

are well-defined even when $a + \varepsilon \notin \mathbb{R}$.

This is the construction that will allow us to formulate limits, continuity, derivatives, and integrals using infinitesimals.

14.6 (Un)limited, Infinitesimal, and Appreciable Numbers

A hyperreal number $b \in {}^*\mathbb{R}$ is called:

- **limited** if $r < b < s$ for some $r, s \in \mathbb{R}$;
- **positive unlimited** if $r < b$ for every $r \in \mathbb{R}$;
- **negative unlimited** if $b < r$ for every $r \in \mathbb{R}$;
- **unlimited** if it is positive unlimited or negative unlimited;
- **positive infinitesimal** if $0 < b < r$ for every $r \in \mathbb{R}_{>0}$;
- **negative infinitesimal** if $r < b < 0$ for every $r \in \mathbb{R}_{<0}$;
- **infinitesimal** if it is positive infinitesimal, negative infinitesimal, or 0;
- **appreciable** if it is limited but not infinitesimal.

Thus all real numbers and all infinitesimals are limited. The only infinitesimal real number is 0, while every other real number is appreciable. An appreciable number is therefore a hyperreal number that is neither infinitely small nor infinitely large.

The above definitions can also be expressed using the natural numbers. For $b \in {}^*\mathbb{R}$,

$$b \text{ is limited} \leftrightarrow |b| < n \quad \text{for some } n \in \mathbb{N},$$

$$b \text{ is unlimited} \leftrightarrow |b| > n \quad \forall n \in \mathbb{N},$$

$$b \text{ is infinitesimal} \leftrightarrow |b| < \frac{1}{n} \quad \forall n \in \mathbb{N},$$

and

$$b \text{ is appreciable} \leftrightarrow \frac{1}{n} < |b| < m \quad \text{for some } n, m \in \mathbb{N}.$$

In particular, if $\varepsilon \neq 0$ is an infinitesimal, then

$$|\varepsilon| < \frac{1}{n} \qquad \forall n \in \mathbb{N}.$$

For example,

$$\varepsilon = \left[\left(1, \frac{1}{2}, \frac{1}{3}, \dots\right)\right]$$

is a positive infinitesimal, while

$$\omega = [(1, 2, 3, \dots)]$$

is a positive unlimited hyperreal number.

We denote the set of unlimited hypernaturals by

$$^*\mathbb{N}_\infty = {}^*\mathbb{N} - \mathbb{N}.$$

Similarly, we denote the sets of positive and negative unlimited hyperreals by

$$^*\mathbb{R}^+_\infty = \{x \in {}^*\mathbb{R} : x \text{ is positive unlimited}\}$$

and

$$^*\mathbb{R}^-_\infty = \{x \in {}^*\mathbb{R} : x \text{ is negative unlimited}\}.$$

This notation may be adapted to an arbitrary subset $X \subseteq {}^*\mathbb{R}$. For example,

$$X_\infty = \{x \in X : x \text{ is unlimited}\},$$

$$X^+ = \{x \in X : x > 0\}, \qquad X^- = \{x \in X : x < 0\}.$$

We will also use

$$\mathcal{L} = \{x \in {}^*\mathbb{R} : x \text{ is limited}\}$$

for the set of all limited hyperreal numbers, and

$$\mathcal{I} = \{x \in {}^*\mathbb{R} : x \text{ is infinitesimal}\}$$

for the set of all infinitesimals.

14.7 Arithmetic of Hyperreals

Let ε, δ be infinitesimals, let b, c be appreciable, and let H, K be unlimited hyperreal numbers. Then the following properties hold.

Sums: $\varepsilon + \delta$ is infinitesimal, $b + \varepsilon$ is appreciable, $b + c$ is limited (possibly infinitesimal), and $H + \varepsilon$ and $H + b$ are unlimited.

Opposites:

$$-\varepsilon \text{ is infinitesimal}, \qquad -b \text{ is appreciable}, \qquad -H \text{ is unlimited}.$$

Products: $\varepsilon\delta$ and εb are infinitesimal, bc is appreciable, and bH and HK are unlimited.

Reciprocals: If $\varepsilon \neq 0$, then $1/\varepsilon$ is unlimited. If $b \neq 0$, then $1/b$ is appreciable. If $H \neq 0$, then $1/H$ is infinitesimal.

Quotients: If ε, δ are nonzero infinitesimals, then ε/δ may be infinitesimal, appreciable, or unlimited. If $b \neq 0$ is appreciable and ε is infinitesimal, then ε/b is infinitesimal. If $b, c \neq 0$ are appreciable, then b/c is appreciable. If H is unlimited and $b \neq 0$ is appreciable, then H/b is unlimited. If $H \neq 0$ is unlimited and b is appreciable, then b/H is infinitesimal. If $H, K \neq 0$ are unlimited, then H/K may be infinitesimal, appreciable, or unlimited.

Roots: If $\varepsilon > 0$ is infinitesimal, then $\sqrt{\varepsilon}$ is infinitesimal. If $b > 0$ is appreciable, then $\sqrt{b}$ is appreciable. If $H > 0$ is unlimited, then $\sqrt{H}$ is unlimited.

Indeterminate Forms: The expressions

$$\frac{\varepsilon}{\delta}, \qquad \frac{H}{K}, \qquad \varepsilon H, \qquad H + K$$

do not have a determined classification from the information given alone.

For example, if $\varepsilon = 1/\omega$, then

$$\varepsilon\omega = 1,$$

which is appreciable. On the other hand, if $\varepsilon = 1/\sqrt{\omega}$, then

$$\varepsilon\omega = \sqrt{\omega},$$

which is unlimited. Thus arithmetic involving unlimited and infinitesimal numbers must be considered carefully.

In $^*\mathbb{R}$, the terminology is interpreted as follows:

- **Finite** means limited (bounded by a real number).
- **Infinite** means unlimited (larger in magnitude than every real number).

This distinction is relative to $\mathbb{R}$ inside $^*\mathbb{R}$: a number may be finite in this sense while still being nonstandard.

14.8 Halos, Galaxies, and Real Comparisons

Let $x, y \in {}^*\mathbb{R}$.
definition 14.8.1 (Infinitesimal Closeness)**.** *We say that x is* infinitely close *to y, and write*

$$x \approx y,$$

if $x - y$ is infinitesimal.
definition 14.8.2 (Halo)**.** *The* halo *of $x \in {}^*\mathbb{R}$ is the set*

$$\mathrm{hal}(x) = \{y \in {}^*\mathbb{R} : y \approx x\}.$$

definition 14.8.3 (Galaxy)**.** *The* galaxy *of $x \in {}^*\mathbb{R}$ is the set*

$$\mathrm{gal}(x) = \{y \in {}^*\mathbb{R} : x - y \textit{ is limited}\}.$$

Lemma 14.8.1. *Every halo is contained in a galaxy.*

Proof. If $y \in \mathrm{hal}(x)$, then $x - y$ is infinitesimal, hence limited. Therefore $y \in \mathrm{gal}(x)$.

□

Lemma 14.8.2. *For standard $r, s \in \mathbb{R}$,*

$$r \approx s \quad \leftrightarrow \quad r = s.$$

Proof. If $r \approx s$, then $r - s$ is a real infinitesimal, hence $r - s = 0$. Therefore $r = s$.

□

14.9 Standard Parts

Theorem 14.9.1. *Every limited hyperreal b is infinitely close to exactly one real number. This real number is called the* *****standard part***** *of b and is denoted by $\mathrm{st}(b)$. Thus*

$$b \approx \mathrm{st}(b).$$

Proof. Let
$$A = \{r \in \mathbb{R} : r < b\}.$$
Since b is limited, there exist real numbers r, s such that $r < b < s$. Therefore A is nonempty and bounded above by s. By the completeness of $\mathbb{R}$, A has a least upper bound

$$c = \sup A \in \mathbb{R}.$$

We show that $b \approx c$. Let $\epsilon > 0$ be any positive real number. Since c is an upper bound of A, we cannot have $c + \epsilon \in A$. Therefore
$$b \leq c + \epsilon.$$
On the other hand, $c - \epsilon$ cannot be an upper bound of A, since c is the least upper bound. Hence there exists some $a \in A$ such that $c - \epsilon < a < b$. Therefore

$$c - \epsilon < b.$$

Thus
$$c - \epsilon < b \leq c + \epsilon,$$
and consequently
$$|b - c| \leq \epsilon.$$
Since this holds for every positive real ϵ, we have

$$b \approx c.$$

It remains to prove uniqueness. Suppose $b \approx c'$ for some $c' \in \mathbb{R}$. Since also $b \approx c$, we obtain

$$c \approx c'.$$

But $c, c' \in \mathbb{R}$, and the only real infinitesimal is 0. Hence

$$c = c'.$$

Therefore there exists exactly one real number infinitely close to b. We define this number to be

$$\mathrm{st}(b) = c.$$

□

Theorem 14.9.2. *Let b, c be limited hyperreal numbers and let $n \in \mathbb{N}$. Then*

$$\mathrm{st}(b \pm c) = \mathrm{st}(b) \pm \mathrm{st}(c),$$

$$\mathrm{st}(bc) = \mathrm{st}(b)\,\mathrm{st}(c),$$

and
$$\mathrm{st}\left(\frac{b}{c}\right) = \frac{\mathrm{st}(b)}{\mathrm{st}(c)} \qquad \textit{if } \mathrm{st}(c) \neq 0.$$

Also,
$$\mathrm{st}(b^n) = \mathrm{st}(b)^n,$$

$$\mathrm{st}(|b|) = |\,\mathrm{st}(b)|,$$

and, if $b \geq 0$,

$$\mathrm{st}(\sqrt{b}) = \sqrt{\mathrm{st}(b)}.$$

Furthermore,

$$b \leq c \quad \rightarrow \quad \mathrm{st}(b) \leq \mathrm{st}(c).$$

Proof. Exercise.

□

These properties show that the standard-part map

$$\mathrm{st} : \mathcal{L} \rightarrow \mathbb{R},$$

where $\mathcal{L}$ denotes the set of all limited hyperreal numbers, is an order-preserving ring homomorphism onto $\mathbb{R}$. Its kernel is precisely the set of infinitesimals:

$$\ker(\mathrm{st}) = \{b \in \mathcal{L} : \mathrm{st}(b) = 0\} = \mathcal{I}.$$

Thus two limited hyperreal numbers have the same standard part exactly when they are infinitely close:

$$\mathrm{st}(b) = \mathrm{st}(c) \leftrightarrow b \approx c.$$

Equivalently, the equivalence classes of the relation $\approx$ on the limited hyperreals are precisely the sets

$$\mathrm{hal}(b) = \{c \in \mathcal{L} : c \approx b\},$$

called the ****halos**** of b.

Therefore the quotient of the ring of limited hyperreals by the ideal of infinitesimals is isomorphic to the real number field.

Theorem 14.9.3. *The quotient ring*

$$\mathcal{L}/\mathcal{I}$$

is isomorphic to $\mathbb{R}$ *by the correspondence*

$$\mathrm{hal}(b) \longmapsto \mathrm{st}(b).$$

In particular,

$$\mathcal{I}$$

is a maximal ideal of the ring $\mathcal{L}$.

14.10 Shadows and Completeness

Lemma 14.10.1. *Every limited* $x \in {}^*\mathbb{R}$ *is infinitely close to a unique real number.*

Proof. This follows from the Standard Part Theorem.

□

Chapter 15

New beginning for calculus

15.1 Limits

First we have to fix a "Main variable". It is almost all the time x.

(sometimes it is "t").

definition 15.1.1. *Let $f : A \to \mathbb{R}$ where $A \subseteq \mathbb{R}$, and let a be a limit point of A. We say that $L \in \mathbb{R}$ is the* ***Limit*** *of f as x approaches a if*

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in A \, (0 < |x - a| < \delta \to |f(x) - L| < \varepsilon) \, .$$

We write

$$\lim_{x \to a} f(x) = L.$$

Theorem 15.1.1 (Infinitesimal Characterization of Limits)**.** *Let$f : A \to \mathbb{R}$, $\quad A \subseteq \mathbb{R}$,and let a be a limit point of A.Then$\lim_{x\to a} f(x) = L$ if and only if, for every nonzero infinitesimal$\varepsilon \in \mathbb{R}(\mathrm{Ord})$ such that $a + \varepsilon \in A$, we have ${}^*f(a+\varepsilon) \approx L$. Equivalently,*

$$\lim_{x \to a} f(x) = L \leftrightarrow \mathrm{st}\left({}^*f(a+\varepsilon)\right) = L.$$

Proof. Suppose first that $\lim_{x\to a} f(x) = L$. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in A$. We prove that

$${}^*f(a+\varepsilon) \approx L.$$

Let $\varepsilon > 0$. By the definition of the limit, there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - L| < \varepsilon$. Since ε is infinitesimal, $|\varepsilon| < \delta$. Therefore $0 < |(a+\varepsilon) - a| < \delta$.Hence$|{}^*f(a+\varepsilon) - L| < \varepsilon$.Since this holds for every $\varepsilon > 0$,${}^*f(a+\varepsilon) \approx L$. Conversely, suppose that for every nonzero infinitesimal ε such that $a + \varepsilon \in {}^*A$, we have ${}^*f(a+\varepsilon) \approx L$. Suppose, toward a contradiction, that $\lim_{x\to a} f(x) \neq L$. Then there exists $\varepsilon > 0$ such that for every $\delta > 0$ there exists $x \in A$ such that $0 < |x - a| < \delta$ and $|f(x) - L| \geq \varepsilon$.

In particular, for every $n \in \mathbb{N}$ there exists $x_n \in A$ such that$0 < |x_n - a| < \frac{1}{n}$ and $|f(x_n) - L| \geq \varepsilon$.By the nonstandard extension, taking an infinite index N, we obtain$0 < |x_N - a| < \frac{1}{N}$.Since N is infinite, $1/N$ is infinitesimal. Therefore $\varepsilon := x_N - a$ is a nonzero infinitesimal. But

$$|{}^*f(a+\varepsilon) - L| = |f(x_N) - L| \geq \varepsilon,$$

so ${}^*f(a+\varepsilon) \not\approx L$, contradicting the assumption. Therefore $\lim_{x\to a} f(x) = L$. Finally,${}^*f(a+\varepsilon) \approx L$is equivalent to $\operatorname{st}\big({}^*f(a+\varepsilon)\big) = L$.

□

Theorem 15.1.2 (Uniqueness of Limit)**.** *If* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} f(x) = M$*, then* $L = M$*.*

Proof. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in {}^*A$. By the Infinitesimal Characterization of Limits,$\operatorname{st}\big({}^*f(a+\varepsilon)\big) = Land\operatorname{st}\big({}^*f(a+\varepsilon)\big) = M$.Hence$L = M$.

□

Theorem 15.1.3 (Limit Laws)**.** *Suppose* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$*. Then* $\lim_{x\to a}(f(x) + g(x)) = L + M$*,* $\lim_{x\to a}(f(x) - g(x)) = L - M$*, and* $\lim_{x\to a} f(x)g(x) = LM$*. If* $M \neq 0$*, then* $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}$*. Moreover, for every* $c \in \mathbb{R}$*,* $\lim_{x\to a} cf(x) = cL$*.*

Proof. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in A$. By the Infinitesimal Characterization of Limits,

$${}^*f(a+\varepsilon) \approx L \qquad \text{and} \qquad {}^*g(a+\varepsilon) \approx M.$$

Therefore, by the Standard Part Theorem,

$$\operatorname{st}\big({}^*f(a+\varepsilon) + {}^*g(a+\varepsilon)\big) = L + M.$$

Similarly,

$$\operatorname{st}\big({}^*f(a+\varepsilon) - {}^*g(a+\varepsilon)\big) = L - M.$$

For the product,

$$\operatorname{st}\big({}^*f(a+\varepsilon){}^*g(a+\varepsilon)\big) = \operatorname{st}({}^*f(a+\varepsilon))\operatorname{st}({}^*g(a+\varepsilon)) = LM.$$

If $M \neq 0$, then $\operatorname{st}({}^*g(a+\varepsilon)) = M \neq 0$, and hence ${}^*g(a+\varepsilon) \neq 0$. Therefore,

$$\operatorname{st}\left(\frac{{}^*f(a+\varepsilon)}{{}^*g(a+\varepsilon)}\right) = \frac{\operatorname{st}({}^*f(a+\varepsilon))}{\operatorname{st}({}^*g(a+\varepsilon))} = \frac{L}{M}.$$

Finally, for every $c \in \mathbb{R}$,

$$\mathrm{st}\big(c^*f(a+\varepsilon)\big) = c\,\mathrm{st}(^*f(a+\varepsilon)) = cL.$$

Thus all the stated limit laws hold.

$\square$

Theorem 15.1.4 (Squeeze Theorem)**.** *Suppose that for all x sufficiently close to a, $f(x) \leq g(x) \leq h(x)$. If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} h(x) = L$, then $\lim_{x\to a} g(x) = L$.*

Proof. Let ε be a nonzero infinitesimal such that $a + \varepsilon \in {}^*A$. Since the inequality holds for all x sufficiently close to a, by the Transfer Principle,${}^*f(a+\varepsilon) \leq {}^*g(a+\varepsilon) \leq {}^*h(a+\varepsilon)$.By the Infinitesimal Characterization of Limits,${}^*f(a+\varepsilon) \approx L$ and ${}^*h(a+\varepsilon) \approx L$.Hence, by the Squeeze Principle for infinitesimals,${}^*g(a+\varepsilon) \approx L$.Therefore, by the Infinitesimal Characterization of Limits,$\lim_{x\to a} g(x) = L$.

$\square$

Theorem 15.1.5 (Absolute Value)**.** *If $\lim_{x\to a} f(x) = L$, then $\lim_{x\to a} |f(x)| = |L|$.*

Proof. Since ${}^*f(a+\varepsilon) \approx L$,

$$||{}^*f(a+\varepsilon)| - |L|| \leq |{}^*f(a+\varepsilon) - L|.$$

Therefore $|{}^*f(a+\varepsilon)| \approx |L|$. Hence, by the Infinitesimal Characterization of Limits,

$$\lim_{x\to a} |f(x)| = |L|.$$

$\square$

Theorem 15.1.6 (Composition of Limits)**.** *Let $\lim_{x\to a} g(x) = b$ and $\lim_{y\to b} f(y) = L$. Suppose that $g(x) \neq b$ whenever x is sufficiently close to a and $x \neq a$. Then $\lim_{x\to a} f(g(x)) = L$.*

Proof. Let ε be a nonzero infinitesimal such that $a + \varepsilon \in {}^*A$. Since $\lim_{x\to a} g(x) = b$,

$${}^*g(a+\varepsilon) \approx b.$$

By the Transfer Principle and the assumption on g, ${}^*g(a+\varepsilon) \neq b$. Therefore, by the Infinitesimal Characterization of $\lim_{y\to b} f(y) = L$, ${}^*f({}^*g(a+\varepsilon)) \approx L$. Since ${}^*f({}^*g(a+\varepsilon)) = {}^*(f\circ g)(a+\varepsilon)$, we have

$${}^*(f\circ g)(a+\varepsilon) \approx L.$$

Hence, by the Infinitesimal Characterization of Limits,

$$\lim_{x\to a} f(g(x)) = L.$$

$\square$

Theorem 15.1.7 (Polynomial Limit)**.** *Let $p(x) = a_0 + a_1x + \cdots + a_nx^n$. Then $\lim_{x\to a} p(x) = p(a)$.*

Proof. By the Limit Laws, $\lim_{x\to a} x^k = a^k$ for every $k \in \mathbb{N}$. Therefore,

$$\lim_{x\to a} p(x) = a_0 + a_1 a + \cdots + a_n a^n = p(a).$$

□

Theorem 15.1.8 (Rational Function Limit)**.** *Let $r(x) = \frac{p(x)}{q(x)}$ where p and q are polynomials. If $q(a) \neq 0$, then*

$$\lim_{x\to a} r(x) = \frac{p(a)}{q(a)}.$$

Proof. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in {}^*A$. By the Polynomial Limit Theorem,

$$\operatorname{st}\bigl({}^*p(a+\varepsilon)\bigr) = p(a), \qquad \operatorname{st}\bigl({}^*q(a+\varepsilon)\bigr) = q(a) \neq 0.$$

Hence ${}^*q(a+\varepsilon) \neq 0$, and by the Standard Part Theorem,

$$\operatorname{st}\left(\frac{{}^*p(a+\varepsilon)}{{}^*q(a+\varepsilon)}\right) = \frac{p(a)}{q(a)}.$$

Therefore, by the Infinitesimal Characterization of Limits,

$$\lim_{x\to a} r(x) = \frac{p(a)}{q(a)}.$$

□

definition 15.1.2. *Let $f : A \to \mathbb{R}$ and let a be a limit point of A. We say that f has the* *****Right Limit***** *L at a if*

$$\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\,(a < x < a+\delta \to |f(x) - L| < \varepsilon)\,.$$

We write $\lim_{x\to a^+} f(x) = L$. Similarly, we say that f has the *****Left Limit***** *L at a if*

$$\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\,(a - \delta < x < a \to |f(x) - L| < \varepsilon)\,.$$

We write $\lim_{x\to a^-} f(x) = L$.

Theorem 15.1.9 (Infinitesimal Characterization of One-Sided Limits)**.** *We have $\lim_{x\to a^+} f(x) = L$ if and only if ${}^*f(a+\varepsilon) \approx L$ for every positive infinitesimal ε such that $a+\varepsilon \in {}^*A$. Similarly, $\lim_{x\to a^-} f(x) = L$ if and only if ${}^*f(a+\varepsilon) \approx L$ for every negative infinitesimal ε such that $a+\varepsilon \in {}^*A$.*

Theorem 15.1.10 (One-Sided Limits)**.** *The following are equivalent:*

$$\lim_{x\to a} f(x) = L \quad\leftrightarrow\quad \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L.$$

Proof. If $\lim_{x\to a} f(x) = L$, then every nonzero infinitesimal is either positive or negative. Therefore ${}^*f(a+\varepsilon) \approx L$ for both positive and negative infinitesimal ε. Hence $\lim_{x\to a^-} f(x) = L$ and $\lim_{x\to a^+} f(x) = L$.

Conversely, suppose $\lim_{x\to a^-} f(x) = L$ and $\lim_{x\to a^+} f(x) = L$. Every nonzero infinitesimal ε is either positive or negative, so ${}^*f(a+\varepsilon) \approx L$. Hence $\lim_{x\to a} f(x) = L$.

□

definition 15.1.3. *We say* $\lim_{x\to a} f(x) = +\infty$ *if*

$$\forall M > 0\ \exists \delta > 0\ \forall x \in A\,(0 < |x-a| < \delta \to f(x) > M)$$

. *Similarly,* $\lim_{x\to a} f(x) = -\infty$ *if*

$$\forall M > 0\ \exists \delta > 0\ \forall x \in A\,(0 < |x-a| < \delta \to f(x) < -M)$$

Theorem 15.1.11 (Infinitesimal Characterization of Infinite Limits)**.** *We have* $\lim_{x\to a} f(x) = +\infty$ *if and only if* ${}^*f(a+\varepsilon)$ *is positive infinite for every nonzero infinitesimal* ε *such that* $a+\varepsilon \in {}^*A$*. Similarly,* $\lim_{x\to a} f(x) = -\infty$ *if and only if* ${}^*f(a+\varepsilon)$ *is negative infinite for every such* ε*.*

definition 15.1.4. *Let* $f : A \to \mathbb{R}$ *where* $A \subseteq \mathbb{R}$ *is unbounded above. We say that* $\lim_{x\to\infty} f(x) = L$ *if* $\forall \epsilon > 0\ \exists M > 0\ \forall x \in A\,(x > M \to |f(x) - L| < \epsilon)$*.*

Theorem 15.1.12 (Infinitesimal Characterization of Limits at Infinity)**.** *We have* $\lim_{x\to\infty} f(x) = L$ *if and only if* ${}^*f(H) \approx L$ *for every positive infinite* $H \in {}^*\mathbb{R}$ *such that* $H \in {}^*A$*. Equivalently,*

$$\lim_{x\to\infty} f(x) = L \leftrightarrow \mathrm{st}({}^*f(H)) = L.$$

Proof. Suppose $\lim_{x\to\infty} f(x) = L$. Let H be positive infinite. If ${}^*f(H) \not\approx L$, then for some $\epsilon > 0$, $|{}^*f(H) - L| \geq \epsilon$. By the definition of the limit, there exists $M > 0$ such that $x > M$ implies $|f(x) - L| < \epsilon$. By Transfer, $H > M$ implies $|{}^*f(H) - L| < \epsilon$, a contradiction. Hence ${}^*f(H) \approx L$.

Conversely, suppose that ${}^*f(H) \approx L$ for every positive infinite $H \in {}^*A$. If $\lim_{x\to\infty} f(x) \neq L$, then for some $\epsilon > 0$, for every $M > 0$ there exists $x \in A$ such that $x > M$ and $|f(x) - L| \geq \epsilon$. Thus, for every $n \in \mathbb{N}$, choose $x_n \in A$ such that $x_n > n$ and $|f(x_n) - L| \geq \epsilon$. Let $H = [(x_n)]$. Then H is positive infinite, while by Transfer $|{}^*f(H) - L| \geq \epsilon$, so ${}^*f(H) \not\approx L$, a contradiction. Therefore $\lim_{x\to\infty} f(x) = L$.

□

definition 15.1.5. *We say that* $\lim_{x\to\infty} f(x) = +\infty$ *if*

$$\forall M > 0\ \exists N > 0\ \forall x \in A\,(x > N \to f(x) > M)$$

.

Similarly,$\lim_{x\to\infty} f(x) = -\infty$ *if*

$$\forall M > 0\ \exists N > 0\ \forall x \in A\,(x > N \to f(x) < -M)$$

.

Theorem 15.1.13 (Infinitesimal Characterization of Infinite Limits at Infinity)**.** *We have* $\lim_{x\to\infty} f(x) = +\infty$ *if and only if* ${}^*f(H)$ *is positive infinite for every positive infinite* $H \in {}^*A$*. Similarly,* $\lim_{x\to\infty} f(x) = -\infty$ *if and only if* ${}^*f(H)$ *is negative infinite for every positive infinite* $H \in {}^*A$*.*

Theorem 15.1.14 (Sequential Characterization of Limits)**.** *Let $f : A \to \mathbb{R}$ and let a be a limit point of A. Then $\lim_{x\to a} f(x) = L$ if and only if, for every sequence $(x_n) * n \in \mathbb{N}$ in A such that $x_n \neq a$ and $\lim *n \to \infty x_n = a$, we have $\lim_{n\to\infty} f(x_n) = L$.*
Theorem 15.1.15 (Limit of a Monotone Function)**.** *Suppose that f is monotone on an interval. Then the one-sided limits of f exist at every interior point of the interval, possibly taking the values $+\infty$ or $-\infty$.*
Theorem 15.1.16 (Limit of Powers)**.** *If $\lim_{x\to a} f(x) = L$, then for every $n \in \mathbb{N}$,*

$$\lim_{x\to a} f(x)^n = L^n.$$

Proof. This follows by repeated application of the Limit Laws.

□

The most important conceptual result of this section is the chain

$$\lim_{x\to a} f(x) = L \leftrightarrow {}^*f(a+\varepsilon) \approx L \leftrightarrow \mathrm{st}({}^*f(a+\varepsilon)) = L.$$

which will be used directly in the definitions of continuity, derivatives, and the Riemann integral.
Theorem 15.1.17 (Infinitesimal Characterization of Limits)**.** *Let $f : A \to \mathbb{R}$ and let a be a limit point of A. Then $\lim_{x\to a} f(x) = L$ if and only if, for every nonzero infinitesimal $\varepsilon \in {}^*\mathbb{R}$ such that $a + \varepsilon \in {}^*A$, we have ${}^*f(a + \varepsilon) \approx L$. Equivalently,*

$$\lim_{x\to a} f(x) = L \leftrightarrow \mathrm{st}\big({}^*f(a+\varepsilon)\big) = L.$$

Proof. Suppose first that $\lim_{x\to a} f(x) = L$. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in {}^*A$. If ${}^*f(a+\varepsilon) \not\approx L$, then for some $\epsilon > 0$ we have $|{}^*f(a+\varepsilon) - L| \geq \epsilon$. By the definition of the limit, there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - L| < \epsilon$. By Transfer, $0 < |\varepsilon| < \delta$ implies $|{}^*f(a+\varepsilon) - L| < \epsilon$, a contradiction. Hence ${}^*f(a+\varepsilon) \approx L$.

Conversely, suppose that ${}^*f(a + \varepsilon) \approx L$ for every nonzero infinitesimal ε such that $a + \varepsilon \in {}^*A$. Suppose that $\lim_{x\to a} f(x) \neq L$. Then for some $\epsilon > 0$, for every $n \in \mathbb{N}$ there exists $x_n \in A$ such that $0 < |x_n - a| < 1/n$ and $|f(x_n) - L| \geq \epsilon$. Let $H = [(x_n)]$. Then $H - a$ is a nonzero infinitesimal, but $|{}^*f(H) - L| \geq \epsilon$, so ${}^*f(H) \not\approx L$, a contradiction. Therefore $\lim_{x\to a} f(x) = L$.

Finally, ${}^*f(a+\varepsilon) \approx L$ if and only if $\mathrm{st}({}^*f(a+\varepsilon)) = L$.

□

definition 15.1.6. *Let $f : A \to \mathbb{R}$ where $A \subseteq \mathbb{R}$ is unbounded above. We say that $\lim_{x\to\infty} f(x) = L$ if*

$$\forall \epsilon > 0 \; \exists M > 0 \; \forall x \in A \, (x > M \to |f(x) - L| < \epsilon)$$

.
definition 15.1.7. *Let $x \in {}^*\mathbb{R}$. We say that x is* *****Infinite***** *if $|x| > n$ for every $n \in \mathbb{N}$. If $x > n$ for every $n \in \mathbb{N}$, we say that x is a* *****Positive Infinite***** *number.*

Theorem 15.1.18 (Infinitesimal Characterization of the Limit at Infinity)**.** *Let $f : A \to \mathbb{R}$ where $A \subseteq \mathbb{R}$ is unbounded above. Then $\lim_{x\to\infty} f(x) = L$ if and only if, for every positive infinite $H \in {}^*A$, we have ${}^*f(H) \approx L$. Equivalently,*

$$\lim_{x\to\infty} f(x) = L \leftrightarrow \mathrm{st}({}^*f(H)) = L.$$

Proof. Suppose first that $\lim_{x\to\infty} f(x) = L$. Let $H \in {}^*A$ be positive infinite. If ${}^*f(H) \not\approx L$, then for some $\epsilon > 0$, $|{}^*f(H) - L| \geq \epsilon$. By the definition of the limit, there exists $M > 0$ such that $x > M$ implies $|f(x) - L| < \epsilon$. By Transfer, $H > M$ implies $|{}^*f(H) - L| < \epsilon$, a contradiction. Hence ${}^*f(H) \approx L$.

Conversely, suppose that ${}^*f(H) \approx L$ for every positive infinite $H \in {}^*A$. If $\lim_{x\to\infty} f(x) \neq L$, then for some $\epsilon > 0$, for every $n \in \mathbb{N}$ there exists $x_n \in A$ such that $x_n > n$ and $|f(x_n) - L| \geq \epsilon$. Let $H = [(x_n)]$. Then H is positive infinite, while $|{}^*f(H) - L| \geq \epsilon$, so ${}^*f(H) \not\approx L$, a contradiction. Therefore $\lim_{x\to\infty} f(x) = L$.

□

15.2 Continuity

First we fix a “Main Point”. It is almost all the time a.
definition 15.2.1. *Let $f : A \to \mathbb{R}$, where $A \subseteq \mathbb{R}$, and let $a \in A$. We say that f is* *****Continuous***** *at a if*

$$\forall \epsilon > 0\ \exists \delta > 0\ \forall x \in A\, (|x - a| < \delta \to |f(x) - f(a)| < \epsilon)$$

.
Theorem 15.2.1 (Continuity and Limits)**.** *Let $f : A \to \mathbb{R}$ and let $a \in A$ be a limit point of A. Then f is continuous at a if and only if $\lim_{x\to a} f(x) = f(a)$.*

Proof. Suppose first that f is continuous at a. Then, by definition, for every $\epsilon > 0$ there exists $\delta > 0$ such that $|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$. In particular, this holds whenever $0 < |x - a| < \delta$. Hence $\lim_{x\to a} f(x) = f(a)$.

Conversely, suppose that $\lim_{x\to a} f(x) = f(a)$. Then for every $\epsilon > 0$ there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$. If $x = a$, then $|f(x) - f(a)| = 0 < \epsilon$. Therefore $|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$, and hence f is continuous at a.

□

Theorem 15.2.2 (Infinitesimal Characterization of Continuity)**.** *Let $f : A \to \mathbb{R}$ and let $a \in A$. Then f is continuous at a if and only if, for every infinitesimal $\varepsilon \in \mathbb{R}(\mathrm{Ord})$ such that $a + \varepsilon \in A$, ${}^*f(a + \varepsilon) \approx f(a)$.*

Equivalently,

$$f \text{ is continuous at } a \leftrightarrow \mathrm{st}\big({}^*f(a + \varepsilon)\big) = f(a)$$

for every infinitesimal ε such that $a + \varepsilon \in A$.

Proof. Suppose first that f is continuous at a. Let ε be an infinitesimal such that $a+\varepsilon \in A$. If ${}^*f(a+\varepsilon) \not\approx f(a)$, then there exists a positive real ϵ such that $|{}^*f(a+\varepsilon)-f(a)| \geq \epsilon$. By continuity, there exists $\delta > 0$ such that $|x-a| < \delta$ implies $|f(x)-f(a)| < \epsilon$. Since ε is infinitesimal, $|\varepsilon| < \delta$. By the Transfer Principle,

$|{}^*f(a+\varepsilon) - f(a)| < \epsilon$, a contradiction. Hence ${}^*f(a+\varepsilon) \approx f(a)$.

Conversely, suppose that ${}^*f(a+\varepsilon) \approx f(a)$ for every infinitesimal ε such that $a+\varepsilon \in A$. Suppose that f is not continuous at a. Then there exists $\epsilon > 0$ such that for every $\delta > 0$ there is $x \in A$ with $|x-a| < \delta$ and $|f(x)-f(a)| \geq \epsilon$.

For every $n \in \mathbb{N}$, choose $x_n \in A$ such that $|x_n - a| < 1/n$ and $|f(x_n) - f(a)| \geq \epsilon$. Taking an infinite index N gives $|x_N - a| < 1/N$, so $\varepsilon := x_N - a$ is infinitesimal. But $|{}^*f(a+\varepsilon)-f(a)| \geq \epsilon$, contradicting ${}^*f(a+\varepsilon) \approx f(a)$. Therefore f is continuous at a.

Finally, ${}^*f(a+\varepsilon) \approx f(a)$ is equivalent to $\mathrm{st}({}^*f(a+\varepsilon)) = f(a)$ by the Standard Part Theorem.

□

Theorem 15.2.3 (Algebra of Continuous Functions)**.** *Let f and g be continuous at a. Then $f+g$, $f-g$, and fg are continuous at a. If $g(a) \neq 0$, then f/g is continuous at a.*

Proof. For every infinitesimal ε, continuity gives ${}^*f(a+\varepsilon) \approx f(a)$ and ${}^*g(a+\varepsilon) \approx g(a)$. By the arithmetic of hyperreals,

${}^*(f+g)(a+\varepsilon) \approx f(a)+g(a)$, ${}^*(f-g)(a+\varepsilon) \approx f(a)-g(a)$, and ${}^*(fg)(a+\varepsilon) \approx f(a)g(a)$. If $g(a) \neq 0$, then ${}^*g(a+\varepsilon) \approx g(a) \neq 0$, and therefore ${}^*(f/g)(a+\varepsilon) \approx f(a)/g(a)$. Hence the stated functions are continuous at a.

□

Theorem 15.2.4 (Composition of Continuous Functions)**.** *Let $g : A \to \mathbb{R}$ be continuous at a, and let $f : B \to \mathbb{R}$ be continuous at $g(a)$, where $g(A) \subseteq B$. Then $f \circ g$ is continuous at a.*

Proof. Let ε be infinitesimal such that $a+\varepsilon \in A$. Since g is continuous at a, ${}^*g(a+\varepsilon) \approx g(a)$. Since f is continuous at $g(a)$, ${}^*f({}^*g(a+\varepsilon)) \approx f(g(a))$. By the composition property of nonstandard extensions,

${}^*(f \circ g)(a+\varepsilon) = {}^*f({}^*g(a+\varepsilon)) \approx f(g(a)) = (f \circ g)(a)$.

Therefore $f \circ g$ is continuous at a.

□

Theorem 15.2.5 (Continuous Functions Have Limits)**.** *Let $f : A \to \mathbb{R}$ be continuous at a, where a is a limit point of A. Then*

$$\lim_{x \to a} f(x) = f(a).$$

Proof. This follows immediately from the Continuity and Limits Theorem.

□

15.3 Derivatives

first we have to fix a "Main variable" it's almost all the time "x". (sometimes it's "t").
definition 15.3.1. *assume $f : {}^*\mathbb{R}_{Ord} \to {}^*\mathbb{R}_{Ord}$ we define **Difrential** $(d_\varepsilon(f))$*

$$d_\varepsilon(f) := f(x+\varepsilon) - f(x)$$

Main reason we choose "x" in definition because it's "Main variable". you can prove $\forall x \in {}^*\mathbb{R}_{Ord} \quad d_\varepsilon(x) = \varepsilon$.
definition 15.3.2. *we define **Derivative***

Let $f : \mathbb{R} \to \mathbb{R}$ and let $\varepsilon \in {}^*\mathbb{R}_{Ord}$ be a nonzero infinitesimal:

$$\varepsilon \neq 0, \qquad |\varepsilon| < \frac{1}{n} \quad \forall n \in \mathbb{N}.$$

Define

$$\frac{df}{dx} = f'(x) = \mathrm{st}\left(\frac{d_\varepsilon(f)}{d_\varepsilon(x)}\right)$$

Equivalently,

$$\frac{df}{dx} = \mathrm{st}\left(\frac{f(x+\varepsilon) - f(x)}{\varepsilon}.\right)$$

More generally,

$$\frac{d}{dx} := \{\big(f, st(\frac{d_\varepsilon(f)}{d_\varepsilon(x)})\big) \mid f : {}^*\mathbb{R}_{Ord} \to {}^*\mathbb{R}_{Ord}\}$$

there is a biggest set like A s.t. $\frac{d}{dx}A$ is a function (you can prove it as an exercise). we call that set "Derivatible funsions" and that function "Derivative".

15.4 Partial Derivatives

first we have to fix a "Main variable" it's almost all the time "x". (sometimes it's "t").
definition 15.4.1. *assume $f : {}^*\mathbb{R}^n_{Ord} \to {}^*\mathbb{R}_{Ord}$ we define **Partial Difrential** $(d_\varepsilon(f))$*

$$\partial_{\varepsilon,x_i}(f) := f(x_1, \ldots, x_i + \varepsilon, \ldots, x_n) - f(x_1, \ldots, x_i, \ldots, x_n)$$

Main reason we choose "x" in definition because it's "Main variable". you can prove $\forall x \in {}^*\mathbb{R}_{Ord} \quad \partial_{\varepsilon,x}(x) = \varepsilon$.
definition 15.4.2. *we define **Partial Derivative***

Let$f : \mathbb{R}^n \to \mathbb{R}, \qquad f = f(x_1, \ldots, x_n)$. and let $\varepsilon \in {}^*\mathbb{R}_{Ord}$ be a nonzero infinitesimal:

$$\varepsilon \neq 0, \qquad |\varepsilon| < \frac{1}{n} \quad \forall n \in \mathbb{N}.$$

define

$$\frac{\partial f}{\partial x_i} = \mathrm{st}\left(\frac{\partial_{\varepsilon,x_i}(f)}{\partial_{\varepsilon,x_i}(x_i)}\right)$$

Equivalently,

$$\frac{\partial f}{\partial x_i} = \mathrm{st}\left(\frac{f(x_1,\ldots,x_i+\varepsilon,\ldots,x_n) - f(x_1,\ldots,x_i,\ldots,x_n)}{\varepsilon}\right)$$

More generally,

$$\frac{\partial}{\partial x_i} := \{\big(f, st(\frac{\partial_{\varepsilon,x_i}(f)}{\partial_{\varepsilon,x_i}(x_i)})\big) \mid f : {}^*\mathbb{R}_{Ord} \to {}^*\mathbb{R}_{Ord}\}$$

15.5 Riemann Integral

definition 15.5.1. *Let $f : [a,b] \to \mathbb{R}$. A **Partition** of $[a,b]$ is a finite ordered set $P = \{x_0, x_1, \ldots, x_n\}$ such that*

$$a = x_0 < x_1 < \cdots < x_n = b.$$

*We define the **Mesh** of P by*

$$\|P\| := \max_{1\le i\le n}(x_i - x_{i-1}).$$

*For every partition P, define $M_i := \sup\{f(x) : x \in [x_{i-1}, x_i]\}$ and $m_i := \inf\{f(x) : x \in [x_{i-1}, x_i]\}$. The **Upper Darboux Sum** of f with respect to P is*

$$U(f,P) := \sum_{i=1}^{n} M_i(x_i - x_{i-1}),$$

*and the **Lower Darboux Sum** of f with respect to P is*

$$L(f,P) := \sum_{i=1}^{n} m_i(x_i - x_{i-1}).$$

*We say that f is **Riemann Integrable** on $[a,b]$ if f is bounded and*

$$\inf_P U(f,P) = \sup_P L(f,P).$$

In this case, we define

$$\int_a^b f(x)\,dx := \inf_P U(f,P) = \sup_P L(f,P).$$

Theorem 15.5.1 (Convergence of Riemann Sums)**.** *Suppose that $f : [a,b] \to \mathbb{R}$ is Riemann integrable. For each $n \in \mathbb{N}$, let*

$$x_i = a + \frac{i(b-a)}{n}, \qquad 0 \le i \le n.$$

Then

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} f(x_i)\frac{b-a}{n} = \int_a^b f(x)\,dx.$$

Proof. Let $P_n = \{x_0, x_1, \dots, x_n\}$. Since f is Riemann integrable,

$$\lim_{n\to\infty} U(f,P_n) = \lim_{n\to\infty} L(f,P_n) = \int_a^b f(x)\,dx,$$

because $\|P_n\| = (b-a)/n \to 0$.

Since $f(x_i)$ lies between the infimum and supremum of f on $[x_i, x_{i+1}]$,

$$L(f,P_n) \le \sum_{i=0}^{n-1} f(x_i)\frac{b-a}{n} \le U(f,P_n).$$

Therefore, by the Squeeze Theorem,

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=0}^{n-1} f(x_i)\frac{b-a}{n}$$

□

Since the Riemann sum converges to $\int_a^b f(x)\,dx$, by the Standard Part Theorem, for an infinite N and $\varepsilon = \frac{b-a}{N}$,

$$\int_a^b f(x)\,dx. = \mathrm{st}\left(\sum_{i=0}^{N-1} f(x_i)\varepsilon\right)$$

let $\varepsilon = \frac{b-a}{N}$ and $x_i = a + i\varepsilon$. we can prove for every choice of sample points $\xi_n(i) \in [x_i, x_{i+1}]$,

$$\int_a^b f(x)\,dx. = \mathrm{st}\left(\sum_{i=0}^{N-1} f(\xi_n(i))\varepsilon\right)$$

More generally,

$$\int_a^b := \left\{ \left(f,\, \mathrm{st}\left(\sum_{i=0}^{N-1} f(\xi_i)\varepsilon\right)\right) \middle| f \text{ is Riemann integrable on } [a,b] \right\}.$$

There is a biggest set A such that $\int_a^b A$ is a function (you can prove it as an exercise). We call that set "Riemann Integrable Functions" and that function "Riemann Integral".
definition 15.5.2 (More Generalisation for ξ). *Let $\varepsilon = \frac{b-a}{N}$ and*

$$x_i = a + i\varepsilon, \qquad i \in \{0, \ldots, N-1\}.$$

Let K be an index set. For each $k \in K$, let $\xi_N^k : \{0, \ldots, N-1\} \to [a, b]$ be a choice of sample points, such that there is a permutation $\sigma : \{0, \ldots, N-1\} \to \{0, \ldots, N-1\}$ such that

$$\xi_N^k(i) \in [x_{\sigma(i)}, x_{\sigma(i)+1}].$$

then for every Riemann integrable function $f : [a, b] \to \mathbb{R}$,

$$\int_a^b f(x)\,dx = \operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\varepsilon\right), \qquad k \in K.$$

Lemma 15.5.1 (Riemann Integrability and Sample Points). *Let $f : [a, b] \to \mathbb{R}$ be bounded. Then f is Riemann integrable if and only if for every $k, l \in K$,*

$$\operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\varepsilon\right) = \operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^l(i))\varepsilon\right).$$

Proof. Suppose first that f is Riemann integrable and let

$$I = \int_a^b f(x)\,dx.$$

For every $k \in K$, since σ is a permutation, the points $\xi_N^k(i)$ give one sample point in each interval of the partition $P_N = x_0, \ldots, x_N$. Hence, if

$$m_i = \inf_{x \in [x_i, x_{i+1}]} f(x), \qquad M_i = \sup_{x \in [x_i, x_{i+1}]} f(x),$$

then

$$\sum_{i=0}^{N-1} m_i\varepsilon \le \sum_{i=0}^{N-1} f(\xi_N^k(i))\varepsilon \le \sum_{i=0}^{N-1} M_i\varepsilon.$$

Taking standard parts and using the Riemann integrability of f, both the lower and upper sums have standard part I. Therefore

$$\operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\varepsilon\right) = I$$

for every $k \in K$, and hence the required equality holds for every $k, l \in K$.

Conversely, suppose that

$$\mathrm{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\varepsilon\right) = \mathrm{st}\left(\sum_{i=0}^{N-1} f(\xi_N^l(i))\varepsilon\right)$$

for every $k, l \in K$. Then the standard part of the Riemann sum is independent of the choice of sample points. Hence the upper and lower sums have the same standard part, so

$$\mathrm{st}(U(f, P_N)) = \mathrm{st}(L(f, P_N)).$$

Thus

$$\mathrm{st}\big(U(f, P_N) - L(f, P_N)\big) = 0,$$

and therefore f is Riemann integrable. $\square$

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Chapter 16

Well-Ordered Integral (more powerful than Lebesgue Integral)

definition 16.0.1 (Admissible Sample Points)**.** *Let* $\varepsilon = \dfrac{b-a}{N}$ *and* $x_i = a + i\varepsilon$ *for* $i \in \{0, \dots, N-1\} \subseteq {}^*\mathbb{R}_{Ord}$*. For each* $k \in K$ *let*

$$\xi_N^k : \{0, \dots, N-1\} \to [a, b]$$

be a choice of sample points. We say ξ_N^k *is **admissible** if:*

1. *For every* $i \in \{0, \dots, N-1\}$ *that is an ordinal,* $\xi_N^k(i) \in \mathbb{R}$ *is a standard real number.*
2. *For every* $i, j \in \{0, \dots, N-1\}$ *such that* $\xi_N^k(i)$ *and* $\xi_N^k(j)$ *lie in the same halo, we have*

$$\xi_N^k(j) = \xi_N^k(i) \pm \varepsilon_N$$

   for some infinitesimal ε_N*.*

3. *For every* $i \in \{0, \dots, N-1\}$ *that is not an ordinal, there exists an ordinal* $j < i$*,* $j \in \{0, \dots, N-1\}$*, such that*

$$\xi_N^k(i) \approx \xi_N^k(j),$$

   where $\approx$ *denotes infinite closeness.*

Let $K_{\mathrm{adm}} \subseteq K$ *denote the set of indices* k *for which* ξ_N^k *is admissible.*

definition 16.0.2 (Well-Ordered Integral)**.** *Let* $f : [a, b] \to \mathbb{R}$ *be bounded. We say* f *is **well-ordered integrable** on* $[a, b]$ *if for every* $k, l \in K_{\mathrm{adm}}$*,*

$$\mathrm{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\,\varepsilon\right) = \mathrm{st}\left(\sum_{i=0}^{N-1} f(\xi_N^l(i))\,\varepsilon\right).$$

In this case we define

$$^{\circledast}\!\!\int_a^b f(x)\,dx := \mathrm{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\,\varepsilon\right), \qquad k \in K_{\mathrm{adm}}.$$

Remark 16.0.1. *Since $K_{\text{adm}} \subseteq K$, agreement over K_{adm} alone is a weaker requirement than agreement over all of K. Hence every Riemann integrable function is well-ordered integrable, and*

$$\circledast\int_a^b f(x)\,dx = \int_a^b f(x)\,dx$$

whenever f is Riemann integrable.

Chapter 17

New beginning for Probability theory